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ABSTRACT

In this thesis we shall investigate the kinds of information which the Human Sentence Processing Mechanism employs during the resolution of local syntactic ambiguity in written texts.

The thesis is in three parts. In Part I we consider some current models of syntactic ambiguity resolution. On the one hand, we consider the structural approaches, in which the processor considers only syntactic information when choosing between alternatives. On the other, we consider the interactive approaches, in which different kinds of information are brought to bear during the resolution process.

In Part II, we describe a number of experiments which contrast the predictions of these two approaches. In particular, we investigate the processing of sentences which are locally ambiguous between a simple noun phrase analysis and a complex noun phrase analysis. Frazier (1979) predicts that the simple noun phrase analysis is chosen because it utilizes fewest phrasal nodes in its associated phrase marker. Crain and Steedman (1985), however, predict that the interpretation of the noun phrase is determined by referential factors, such as whether or not a unique referent can be identified for the noun phrase. The results support Crain and Steedman's interactive hypothesis.

In Part III, we consider some theoretical issues concerning the timing of the processor's decisions. Crain and Steedman's original model is modified in the light of such considerations. We explore the implications of the modification for the status of syntax and semantics within our model of sentence comprehension.

In the final chapter, we attempt to explain the existence of parsing preferences in sentences which are presented in isolation, and for which no explicit contextual information has been provided.

We conclude that contextual considerations, such as the distinction between what is and what is not already known to the hearer, are of fundamental importance during the resolution of local syntactic ambiguity by the Human Sentence Processing Mechanism.

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Part I: The Literature

"Il faut n'appeler Science que l'ensemble des recettes qui réussissent toujours. - Tout le reste est littérature"

The term Science should not be given to anything but the aggregate of the recipes that are always successful. All the rest is literature.

P. Valéry.

CHAPTER 1

Introduction

A fundamental problem for theories of natural language processing is the pervasiveness of ambiguity. Ambiguity can be found at all the major levels of description of a language, and a requirement of processing theories is that they should characterise both the information which the processor uses in order to decide between alternative analyses, and the timing with which this information is invoked.

The present study will ask, in particular, whether human sentence processors resolve local syntactic ambiguity in written texts on the basis of semantic or pragmatic information, or on the basis of syntactic information alone (cf. discussions concerning the "interactive" nature of language processing: e.g. Forster, 1979; Marslen-Wilson & Tyler, 1980; Crain & Steedman, 1985).

An often noticed feature of the Human Sentence Processing Mechanism (HSPM) is that ambiguities seem to pose little or no problem for it. In fact, when faced with a choice, the HSPM will often exhibit a preference for some one analysis, to the point of making the hearer unaware that there is an alternative analysis which could have been pursued (e.g. Fodor, Bever, & Garrett, 1974). In extreme cases, these preferences give rise to "garden path" effects, like the one illustrated by (1.1).

(1.1) The oil tycoon sold the off-shore oil tracts for a lot of money wanted to kill J.R. (Adapted from Bever, 1970).

The problem here is that the word "sold" is ambiguous: it can be interpreted either as a simple active, as in (1.2)

(1.2) The oil tycoon sold the off-shore oil tracts for a lot of money.

or it can be interpreted as a past participle, in a reduced passive construction, analagous to (1.3)

(1.3) The oil tycoon given the off-shore oil tracts for a lot of money wanted to kill J.R.

In example (1.1), the only way to make the whole string into a sentence is to interpret "sold" as a reduced form of the passive. But what seems to happen in this and similar examples is that people tend to interpret the word "sold" as an active main verb.

Example (1.1) gives rise to the garden path phenomenon because it is locally ambiguous: that is, the ambiguity can in principle be resolved on the basis of subsequent syntactic information. What is particularly interesting about such examples is that the HSPM makes what turns out to be the wrong choice of analysis: the one which is not compatible with subsequent syntactic information. Similar effects are experienced with

(1.4) My brother ordered the woman from Italy a spaghetti dinner had been spilled on accidently a large steak and a salad. (From Frazier, 1979).

In this case, the noun phrase "a spaghetti dinner" is interpreted as

the direct object of the verb "ordered" and not as the surface structure subject of the reduced passive, as made explicit in (1.5).

(1.5) My brother ordered the woman from Italy whom a spaghetti dinner had been spilled on ...

And in (1.6) below, the string "..that he was worried about.." tends to be interpreted as part of a complement clause, and not as a relative clause.

(1.6) The psychologist told the woman that he was worried about to visit him again. (Adapted from Crain, 1980).

For most readers, these examples demonstrate very clearly that the HSPM exhibits preferences when faced with a choice of analysis. But on what basis do these preferences exist? Why is it that in these examples the wrong analysis is consistently chosen?

Certainly these examples seem to show that, when faced with a choice, the HSPM commits itself to pursuing just one analysis. Garden path effects would be unlikely to occur if the HSPM pursued all analyses simultaneously. Furthermore, the nature of these effects suggests that the HSPM selects analyses not at random, but in some principled way. During the course of this study we shall attempt to determine what these principles might be, and the information over which these principles operate.

There are already a number of accounts of local ambiguity resolution. For the most part, these accounts are "structural". They maintain that syntactic ambiguities are resolved on the basis of

syntactic information alone. In other words, the principles which guide the parser's decisions are based on properties of the syntactic structure(s) associated with the ambiguous fragment (e.g. Kimball, 1973; Kimball, 1975; Fodor & Frazier, 1978; Frazier, 1979; Wanner, 1980; Frazier & Fodor, 1980; Ford, Bresnan, & Kaplan, 1982; Rayner, Carlson, & Frazier, 1983; Holmes, 1984). There are a number of alternative accounts, however, which maintain that such ambiguities are resolved on the basis of pragmatic information.[1] These accounts propose that the principles which guide the resolution process are based on notions such as "plausibility": an analysis is chosen which is most "likely" given the reader/hearer's knowledge of the world and/or discourse within which the target fragment has been uttered. This notion of plausibility requires that parsing decisions be sensitive to the context in which an utterance is produced (e.g. Tyler & Marslen-Wilson, 1977; Crain & Steedman, 1985). The experimental sections of this thesis (in Part II) will describe a number of experiments which contrast these two opposing views of the resolution process.

But models of the ambiguity resolution process should do more than simply describe the information that is used by the processor.

[1] For present purposes, we take "pragmatic" information to be knowledge about the world in general (which includes knowledge of the discourse as a whole). In contrast, we take "semantic" information to refer to knowledge about the language (or "linguistic knowledge"). As has been pointed out (e.g. Rayner, Carlson, & Frazier, 1983), it is often impossible to make a precise distinction between the two terms.

They should also specify just when this information is used. Computational models of sentence processing constitute the most explicit attempts to achieve these two aims. Although this thesis does not constitute a computational model, it is the intention of this work to provide this dual specification. To the extent that there exists this overlap of aims, the thesis has a contribution to make to that branch of Computational Linguistics which is concerned with modelling human linguistic behaviour.

The remainder of the thesis proceeds as follows: Part I consists of an overview of theories to date which have addressed the issue of syntactic ambiguity resolution. In Chapter 2 we review the structurally-based theories. The aim here will be to demonstrate that although a number of distinct approaches exist, they all share certain common principles. More specifically, we shall conclude that there exist just two structural strategies on which these various approaches converge: Minimal Attachment (Frazier, 1979) and Right Association (Kimball, 1973). In Chapter 3 we review the more interactive accounts of the resolution process. This chapter will consider sentence processing in the wider context of discourse comprehension. It ends with a detailed review of an interactive account based on a very specific definition of plausibility (Crain & Steedman, 1985).

Part II contains descriptions of a number of experiments which investigate the empirical adequacy of the Minimal Attachment strategy (one of the two strategies to be discussed in Chapter 2). The

experiments are designed to contrast this structural strategy with the Crain & Steedman account based on plausibility. It will be concluded that the evidence favours the latter approach to ambiguity resolution. Within Part II, Chapter 4 will be concerned with the processing of prepositional phrases, and Chapter 5 with the processing of relative and complement clauses.

Part III consists of a theoretical discussion which extends the ideas presented in Crain & Steedman. In Chapter 6, we consider the question of when parsing decisions are made. We conclude that the principles provided by Crain & Steedman require some revisions in order to specify both which information is used, and when it is used. We shall also consider the issue of autonomy in sentence processing. That is, we shall consider the status of the relationship between syntax and semantics within the model, and the effects that different kinds of relationship may have on the timing of the HSPM's decisions.

Chapter 7 concludes the study by exploring some implications which follow from findings in earlier chapters. We shall return to the original data which the structural strategies were designed to explain, data drawn from studies which have investigated the processing of isolated sentences. It will be argued throughout the thesis that during normal discourse, sentences are not processed in isolation. Rather the process of sentence comprehension is subsumed by the more general processes responsible for the comprehension of whole discourses. We shall see that this view is not incompatible with the existing data on isolated sentences. A number of

suggestions will be offered with regard to this data which follow naturally from the contextually-based account of ambiguity resolution developed here. We shall also consider the implications of these suggestions for the Right Association strategy introduced in Chapter 2. We shall consider the role that context may play in overcoming the preferences predicted by this strategy. We shall conclude, contrary to earlier studies on syntactic ambiguity, that syntactic information alone does <u>not</u> exclusively guide the resolution of local syntactic ambiguity.

CHAPTER 2

Structure and Syntactic Ambiguity Resolution

In Chapter 1, it was suggested that there are essentially two distinct approaches to syntactic ambiguity resolution. On the one hand are those studies which argue that local syntactic ambiguities are resolved on the basis of syntactic/lexical information alone. On the other, are those studies which argue for more "interactive" models of the resolution process, in which different kinds of knowledge (not just syntactic knowledge) are brought to bear (and therefore interact) during the resolution process.

The ambiguous examples presented in Chapter 1 were all structurally ambiguous: the alternative readings or interpretations could all be distinguished on the grounds of their respective syntactic structures. Figures 2.1 and 2.2 illustrate the differences between the structures which can be associated with the ambiguous sentence fragments (2.1) and (2.2).

(2.1) The tycoon sold the oil tracts .

(2.2) Blake thought Stephen died yesterday. (Adapted from Wanner, 1980)

The crucial issue is whether it is only these differences in structure that form the basis for the HSPM's decisions. It is this possibility that we shall consider in this chapter, leaving until

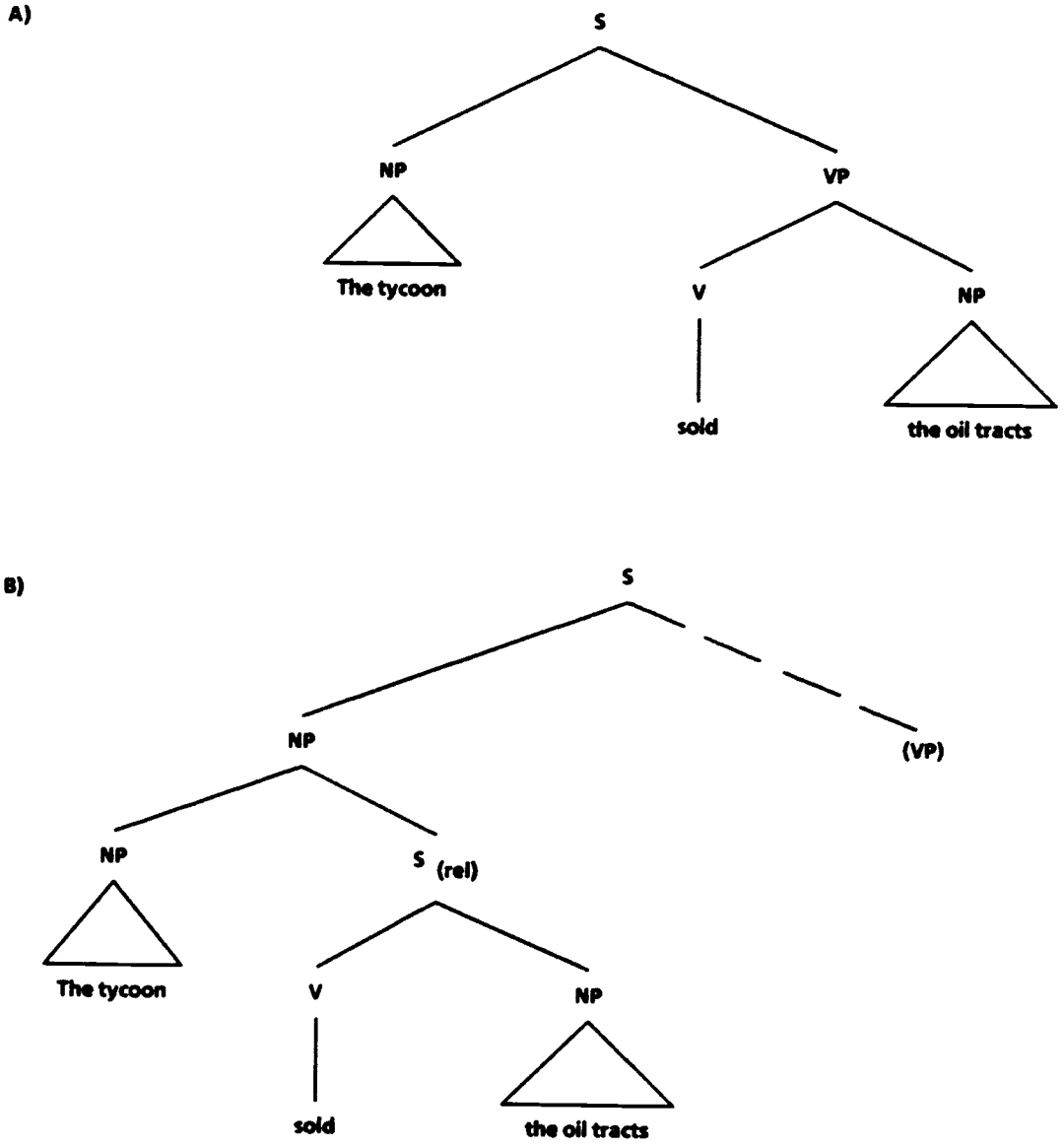


FIGURE 2.1

Phrase markers for the string "The tycoon sold the oil tracts ..." under (a) a main clause interpretation of "sold ...", and (b) under a subordinate interpretation of "sold ...".

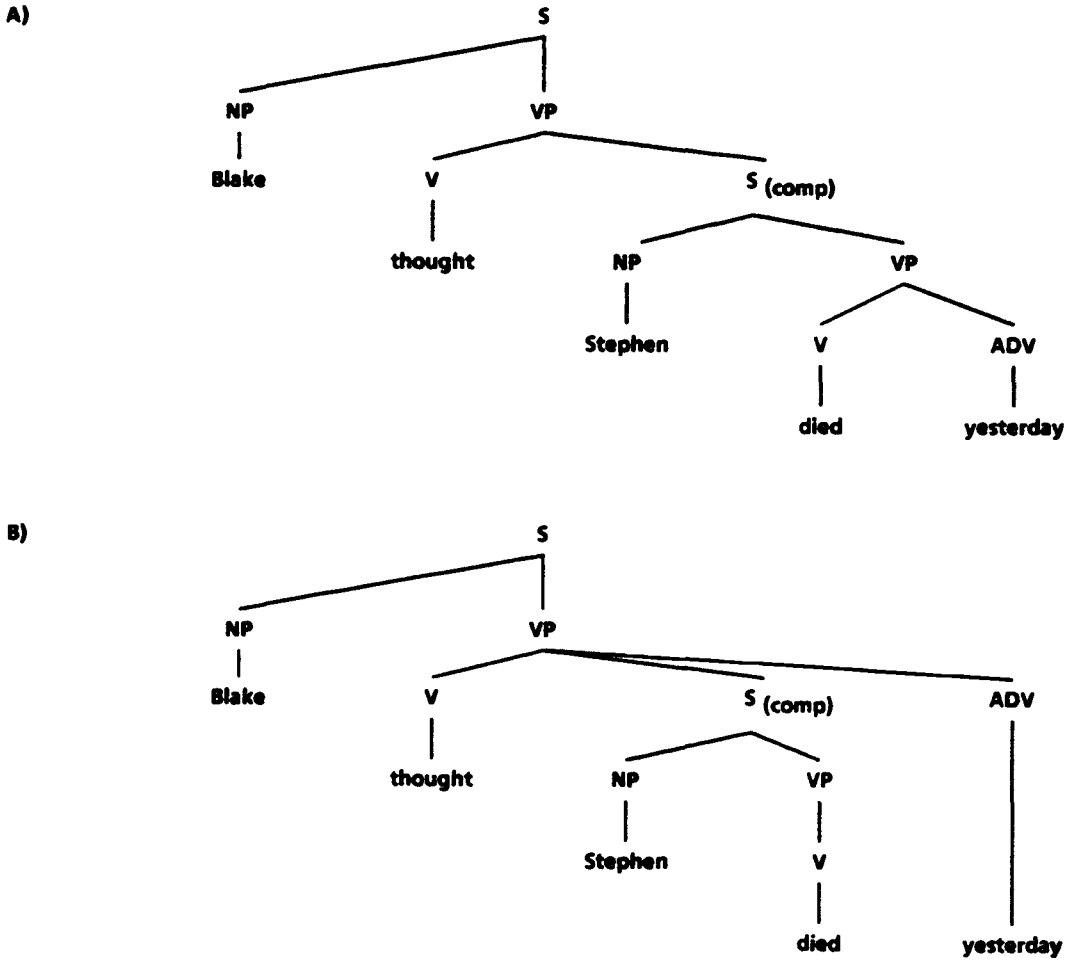


FIGURE 2.2

Phrase markers for the string "Blake thought Stephen died yesterday" with (a) the attachment of "yesterday" such as to modify the verb "died", and (b) the attachment of "yesterday" such as to modify the verb "said".

Chapter 3 consideration of the interactive approach to ambiguity resolution.

The chapter is organised into three main sections. In Section 2.1 we shall consider perhaps the earliest attempt to explain parsing preferences and the garden path phenomena. This attempt, by Bever (1970), differs from the other structural accounts in that Bever was concerned more with what he called "internal" structure than with what he called "external" (or "syntactic") structure. This distinction seems to correspond to that between "deep" structure, which describes the content of a sentence, and "surface" structure (cf. Figures 2.1 and 2.2) which describes the form of a sentence (Chomsky, 1965). Bever attributes responsibility for parsing preferences to the processes by which deep structures are created. This is in contrast to later theories (e.g. Kimball, 1973; Frazier, 1979) which attribute preferences to the processes involved in the construction of surface structures.[1] These theories will be discussed in Section 2.2.

In Section 2.3 we shall consider a number of alternative models of human sentence processing within which a variety of principles of surface structure parsing have been implemented. Although these

[1] The distinction between deep structure and surface structure is maintained here primarily for historical reasons. It allows the work of Bever (1970), and the work that followed, to be set in a historical context. For the purposes of the present discussion, however, little rests on the distinction. In later chapters we shall consider a view of sentence processing in which intermediate levels of representation, such as surface structures, are not constructed.

structurally-based models appear to differ markedly, they all share the same two parsing principles, namely Minimal Attachment (Frazier, 1979), and Right Association (Kimball, 1973). Of particular interest in this section will be the way in which the linguistic behaviours predicted by these principles can be taken to reflect the internal workings of the HSPM.

2.1 Linguistic Behaviour and Perceptual Parsing Strategies

Chomsky (1965) distinguished between a language user's knowledge of a language ("competence") and that user's use of the language ("performance"). In an influential article, Bever (1970) makes this same distinction: between linguistic knowledge on the one hand (from which tree structures such as those in Figures 2.1 and 2.2 are derived), and linguistic behaviour on the other. Bever added that there thus follows a distinction to be made between sentences which are "grammatical", as defined by linguistic knowledge, and sentences which are "acceptable", as defined by linguistic behaviour. He proposed that whereas sentence grammaticality might be defined by specifically linguistic rules, sentence acceptability (and thus linguistic behaviour) is constrained by the operation of a number of general perceptual heuristics, or "strategies". Thus the explanation for the effects illustrated with the examples in the preceding chapter lies not with any properties unique to the parsing mechanism itself, but rather with properties found specific to the perceptual/cognitive system as a whole.

The non-linguistic nature of these strategies is stressed by Bever when he describes them as processes

> "..in which the knowledge of linguistically defined syntactic structure is not utilized actively in actual perception." (Bever, 1970. p.298)

Bever proposed that the role of these strategies is to map "external sequences" (i.e. strings of words) onto "internal structures". In essence, Bever claims that it is not the linguistic representations themselves which govern the HSPM's choices, but rather the mapping process by which the internal structures are derived. Underlying this line of argument was Bever's concern to show that adult language users make use of perceptual strategies which resemble those used by children during the acquisition of the language. Moreover, he suggested that

> "..linguistic structure is itself partially determined by the learning and behavioural processes that are involved in acquiring and implementing that structure." (Bever, 1970. p.280)

For the purposes of the present discussion we shall consider just three of the twelve strategies proposed by Bever.

> Strategy A: Segment together any sequence X...Y, in which the members could be related by primary internal structural relations, "actor action object....modifier." (Bever, 1970. p.290)

> Strategy B: The first N...V...(N)...clause (isolated by strategy A) is the main clause, unless the verb is marked as subordinate. (Bever, 1970. p.294)

> Strategy D: Any Noun-Verb-Noun (NVN) sequence within a potential internal unit in the surface structure corresponds to "actor-action-object." (Bever, 1970. p.298)

These three strategies "explain", according to Bever, the unacceptability of (1.1) and (1.4):

(1.1) The oil tycoon sold the off-shore oil tracts for a lot of money wanted to kill J.R.

(1.4) My brother ordered the woman from Italy a spaghetti dinner had been spilled on accidently a large steak and a salad.

As suggested earlier, the difficulty with these two examples stems from the analysis of the noun phrases "the off-shore oil tracts" and "a spaghetti dinner" as main clause objects. Such analyses are ensured, Bever would claim, by the operation of Strategies B and D.

As corroboration for the operation of these strategies (and in particular Strategy B), Bever cites the evidence concerning relative pronoun deletion.

(2.3) The oil-tycoon who had financial problems was desperate.

(2.4) *The oil-tycoon had financial problems was desperate.

(2.5) The oil tycoon having financial problems was desperate.

(2.4) is clearly ungrammatical. Bever claims that this is because of Strategy B: the lexical item "had" is interpreted as the main verb. Unless the verb following the first "N" is clearly marked as

subordinate (as in 2.5), it is not possible to delete the relative pronoun. Relative pronoun deletion is possible in examples (2.6) and (2.7), because Strategy B is not violated.

(2.6) The oil tycoon whom Jamie married hated J.R.

(2.7) The oil tycoon Jamie married hated J.R.

Although Bever postulated the existence of these parsing strategies in order to account for perceptual preferences, he did notice that there exist occasions when his strategies do not appear to apply. For instance, (2.9) is more acceptable than (2.8).

(2.8) The man pushed past the queue crashed into the post.

(2.9) The light airplane pushed past the barn crashed into the post. (From Bever, 1970. p.316)

Example (2.9) prompted Bever to write:

> "Thus Strategy A and B are not offered as grammatical rules but as constraints to which otherwise optional rules may respond when the speaker/listener's knowledge of individual exceptions does not make them irrelevant." (Bever, 1970. p. 317)

Bever thus accounts for those occasions when his strategies are violated yet with no perceptual cost. As he says elsewhere (footnote 2, p.294), there are exceptions to every strategy: the strategies capture generalizations which, like all generalizations, are not necessarily always true. Bever does not expand on the notion of

"knowledge of individual exceptions". Clearly though, such knowledge lies within the domain of semantic/pragmatic information, which we shall return to in the next chapter.

Frazier (1979) points out there is a certain indeterminacy with regard to the relative ordering of Bever's strategies. It is unclear just which of the three strategies given above is or are actually responsible for the garden path effects in (1.1) and (1.4). Bever attributes the difficulty experienced with these examples to Strategy B, although Valian (1979) attributes it to Strategy D. One might even suppose that the difficulty could be attributed to Strategy A. The same indeterminacy applies to the evidence concerning relative pronoun deletion: Bever attributes these effects to Strategy B, but they could equally well be attributed to the operation of Strategy A. The point here is that Bever is unable to specify both the <u>source</u> of the information by which ambiguities are resolved, and the <u>timing</u> of that information's use.

Associated with this indeterminacy is a further indeterminacy concerning the status of Bever's strategies with respect to grammars. Valian (1979) argues that Bever cannot explain why some of his strategies affect "linguistic structure" (i.e. lead to certain strings being rejected by the grammar), whilst others do not. She cites Strategy B as an example of a strategy which does affect the grammar and Strategy D as a strategy which does not. She takes the former to account for the data discussed above concerning the permissible deletions of relative pronouns (an effect that manifests

itself in the grammar), whilst she claims that the latter accounts for the garden-path data (which is not an effect that manifests itself in the grammar: the garden-path examples are certainly not ungrammatical). Bever might counter Valian's criticism with the following:

> "..perceptual strategies affect grammatical structures in those cases in which the child acquires the strategies before he acquires certain grammatical structures; grammatical structures acquired after he learns the strategies will be affected by them." (Bever, 1970. p.325)

Bever would thus have to claim that Strategy B is acquired before the grammatical structures associated with the pronoun deletion data, whilst Strategy D is acquired after the grammatical structures associated with the garden path examples.

It is possible, however, that the situation is even worse than Valian suggests, and therefore not countered by Bever's remark concerning the order of acquisition of the different strategies with respect to the acquisition of grammatical structures. Because of the indeterminacy with which linguistic behaviours can be attributed to the different strategies, we might attribute to Strategy A the responsibility for both the garden-path effects and the pronoun deletion effects. One can then extend Valian's argument as follows: why should the same strategy (A) affect the grammar in one instance (e.g. data concerning relative pronoun deletion) but not in another (e.g. data concerning reduced passive constructions such as in 1.1)? So just what is the status of Bever's strategies? The answer to this question will remain unclear so long as the various indeterminacies

concerning Bever's theory remain.

In the following section we shall consider a number of principles of surface structure parsing which largely overcome these indeterminacies.

2.2 Linguistic Behaviour and Syntactic Parsing Strategies

In the preceding section, we saw that the status of Bever's strategies is questionable: at times they seem to operate like rules of grammar, and at others, like constraints on performance (which can be violated). As rules they are too imprecisely defined; no ordering is supplied, and thus they cannot be considered to constitute a parsing algorithm. As constraints they appear to have arbitrary effects on the grammatical rules they constrain. It is unclear therefore whether Bever actually succeeded in his goal of explaining various aspects of linguistic behaviour. Perhaps he has merely succeeded in describing these behaviours. It is unsatisfactory to say that an "N V N" sequence is interpreted as "actor action object" simply because there is a strategy which interprets each "N V N" sequence as "actor action object". Strategies like this merely mirror linguistic behaviour, they cannot explain it.

Bever attempted to explain certain linguistic behaviours according to a set of operating principles which govern the workings of the general perceptual system. An alternative approach, taken by Kimball (1973) and Frazier (1979), attempted to explain these behaviours according to a set of operating principles which govern

the workings of the parsing mechanism itself. Unlike Bever's, their principles of surface structure parsing do actively utilize linguistically defined syntactic structure. They propose that the resolution of local syntactic ambiguity does depend on the syntactic (or "structural") properties of a sentence.

Of the seven principles which Kimball claims govern the workings of the parser, three "account for" sentences like (1.1)

(1.1) The oil tycoon sold the off-shore oil tracts for a lot of money wanted to kill J.R.

They are

> "Principle Five (Closure): A phrase is closed as soon as possible, i.e. unless the next node parsed is an immediate constituent of that phrase." (Kimball, 1973, p.36)

> "Principle Six (Fixed Structure): When the last immediate constituent of a phrase has been formed, and the phrase ... closed, it is costly in terms of perceptual complexity ever to have to go back to reorganise the constituents of that phrase." (Kimball, 1973, p.37)

> "Principle Seven (Processing): When a phrase is closed, it is pushed down into a syntactic (possibly semantic) processing stage and cleared from short-term memory." (Kimball, 1973, p.38)

Implicit in Kimball's arguments is the notion that the parsing process proceeds in an essentially left-to-right fashion. Closure thus predicts the occurrence of a garden path in Example (1.1) because it stipulates that as soon as the potential S is reached

("the oil tycoon sold the off-shore oil tracts for a lot of money"), it is closed. Kimball claims that Bever's strategy B (the first "N V N" sequence is interpreted as a main clause), which Bever uses to account for this garden path, is just a particular case of Closure as applied to sentences. The Principles of Processing and Fixed Structure explain the difficulty encountered once a phrase is closed. Put simply, a "closed" phrase cannot be retrieved and opened again except at great perceptual cost

However, there is an important question which this explanation poses: until the potential S (in 1.1) is reached, does the parser pursue multiple analyses - here, both the active and passive analyses - or does it follow only one?[2] It would appear that Closure only describes the outcome of some decision process; it does not describe the process itself. It does not dictate what course of action the parser will take on encountering the ambiguity. To go back to example (1.1), Kimball's Closure does not explain what happens when the ambiguous lexical item "sold" is first encountered. However, there is an alternative explanation involving Closure which is not used by Kimball, and which largely overcomes this problem.

According to Closure, a phrase is closed as soon as possible. Clearly, the noun phrase "the oil tycoon" is subject to this same principle: as soon as a potential right-most boundary is found to

[2] See Mackay (1966) and Bever, Garrett, & Hurtig (1973) for experimental investigations which bear on this issue (which is discussed in Section 6.1).

this noun phrase (i.e. "tycoon"), it must, according to Closure, be closed, in which case the subsequent verb ("sold") can only be interpreted as the main verb.

On this account, the preferred interpretation of the that-clause in (1.6) as a complement clause is also explained.

(1.6) The psychologist told the woman that he was worried about to visit him again.

The noun phrase "the woman" is closed once the lexical item "woman" is encountered, and so there is no alternative but to interpret the that-clause as a complement clause.[3]

Despite this apparent "success" in capturing certain facts of linguistic behaviour, there are a number of problems with Closure. In (2.10), which we first encountered in Chapter 1, "with the revolver" is preferentially attached to "shot" (Rayner, Carlson, & Frazier, 1983).

(2.10) The criminal shot the policeman with the revolver.

Closure would predict two things: first, the noun phrase "the policeman" will be closed before "with the revolver" can be attached to it. This much seems to conform to findings. It also predicts,

[3] If noun phrases are closed as soon as possible, then any reduced relative construction (e.g. "the man given ..) should be problematic for the processor. This does not seem to be the case. It is interesting to note at this point that Frazier's (1979) Minimal Attachment strategy (to be discussed below) makes exactly the same prediction.

however, that the potential S ("the criminal shot the policeman") should also be closed. The principles of Processing and Fixed Structure predict that any attachment of the prepositional phrase "with the revolver" should be problematic. This is clearly not the case.

It could be argued, however, that the attachment of the prepositional phrase to the verb phrase does not necessitate the reorganisation of the constituents of the preceding clause, and hence is unproblematic. Such an explanation requires that retrieval of a phrase from the "syntactic (possibly semantic)" processing stage is not problematic. Rather, problems are only encountered if one subsequently has to reorganise the constituents of that phrase. But this line of argument threatens Kimball's original suggestion that Fixed Structure is merely a consequence of Processing.

Frazier (1979) points out a further problem with Closure: there are some examples in which Closure is empirically inadequate:

(2.11) While John was eating his toast became cold. (Adapted from Frazier, 1979)

The correct interpretation of this sentence requires that the phrase "while John was eating" be closed. Closure would predict this to happen. However, the tendency with such examples is not to close this phrase, but rather to close the phrase "while John was eating his toast". To account for (2.11), Frazier suggests that the parser does not abide by Kimball's principle of Closure (which could be called "Early Closure") but instead abides by a principle of Late

Closure:

> "Late Closure: When possible, attach incoming material into the clause or phrase currently being parsed." (Frazier, 1979. p.76)

Late Closure explains not only the behaviour associated with (2.11) but also that associated with a class of example (cf. 2.12 below) which differs slightly from earlier examples.

(2.12) Joe bought the book that I had been trying to obtain for Susan.

This example is ambiguous because the buying could be for Susan, or the trying to obtain could be for Susan. It appears to be the case that people interpret the "trying to obtain" as having been "for Susan". This is explained as follows: when the prepositional phrase "for Susan" is encountered, the phrase currently being parsed is the (infinitival) verb phrase "to obtain..". Late Closure thus predicts that "for Susan" will attach to this phrase and to no other.

Frazier suggests that Late Closure is just a more general formulation of Bever's Segmentation Strategy A; the relations actor-action-object are maintained only if each item is analysed as part of the same clause (i.e as being dominated by the same S-node). Late Closure is also very similar to one of Kimball's own principles, Right Association:[4]

[4] Frazier suggests, of Right Association and Late Closure, that "it may turn out that they are just alternative formulations of the same strategy." (Frazier, 1979, p.52)

"Principle Two (Right Association): Terminal symbols optimally associate to the lowest nonterminal node." (Kimball, 1973, p.24)

Figure 2.3 shows the phrase marker for the string "Joe bought the book that I had been trying to obtain" and the two attachments of the prepositional phrase "for Susan". The preferred reading of this sentence, following Right Association, corresponds to the attachment of the prepositional phrase to the lowest point in the phrase marker, that is, to the verb "obtain".

The question is, though, whether we are still in a position to explain the effects in the original examples (1.1, 1.4, and 1.6). Apparently we no longer can. Whilst (Early) Closure could explain these effects, it makes incorrect predictions concerning sentence complexity (cf. 2.10 and 2.11). Furthermore, neither Late Closure nor Right Association make any predictions concerning the original examples.[5]

[5] Strictly speaking, this is untrue. Frazier (1979) argues that Right Association would predict in Example (1.1) that the verb "sold" be interpreted as part of a reduced relative, as this is the only way in which it can be "associated" to the lowest nonterminal node in the phrase marker (there is a distinction to be made between the terms "associate" and "attach"). However, such an attachment involves the addition of extra nodes into the phrase marker (cf. Figure 2.5 below). She suggests that Right Association should therefore be constrained by the operation of a principle of Minimal Attachment (to be discussed below). This would, in effect, cause Right Association to read "...optimally attach to the lowest nonterminal node which already exists in the phrase marker". However, this restriction assumes that Closure applies only to sentential clauses and not to phrases (cf. the earlier discussion on Closure). If the principle is taken literally, and Closure does apply to phrases, then Frazier's restriction is arguably redundant, as Closure will operate in such cases before Right Association (if we assume a left-to-right parse).

Late Closure, on the other hand, is subject to the kind of

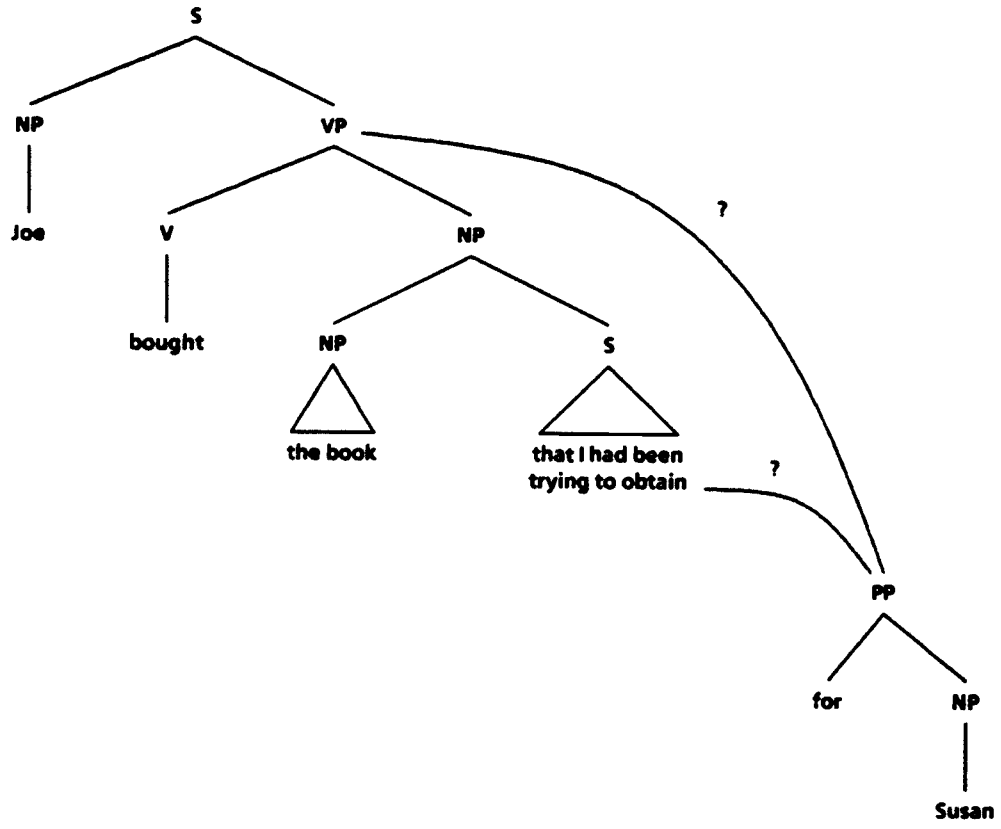


FIGURE 2.3

Phrase marker for the string "Joe bought the book that I had been trying to obtain for Susan", showing two possible attachment sites for the prepositional phrase "for Susan". The low right attachment seems to be preferred.

Kimball (1975) proposed a principle which seems to overcome some of the problems discussed above, and is apparently able to account for the garden path effects in the original examples.

> "Let us define an equivalence relation between paths, so that two paths are equivalent if they differ only in repetition of a given symbol. Thus, the paths Det NP S, Det NP NP S, and Det NP NP S S S are all equivalent. We can then pick a canonical representative of such an equivalence class to be the most collapsed string, in this case the first. ...let us assume that the machine will always pick ... the canonical representative of equivalence classes ..." (Kimball, 1975)

This suggests that of the two possible phrase markers in Figure 2.4, the "canonical" 2.4b. will always be the preferred structure.

Frazier (1979) refined this principle further as

> "Minimal Attachment: Attach incoming material into the phrase marker being constructed using the fewest nodes consistent with the well-formedness rules of the language." (Frazier, 1979. p.76)

She defends Minimal Attachment on a number of grounds. First, Minimal Attachment is a more general principle than Kimball's because

objection raised by Frazier: just as (Early) Closure can be applied to noun phrases, so too can Late Closure. Taken literally, Late Closure would demand that a noun phrase be closed as late as possible, in which case the string "the oil tycoon sold.." should be interpreted as part of a complex noun phrase, and not as the first N..V of a main clause. This requires the addition of extra nodes into the phrase marker (as compared to the main clause interpretation) and so can be constrained by the same constraint Frazier proposes for Right Association (given the similarity Frazier claims for Right Association and Late Closure, it is not surprising that she proposes to constrain them both using the same mechanism of minimal attachment).

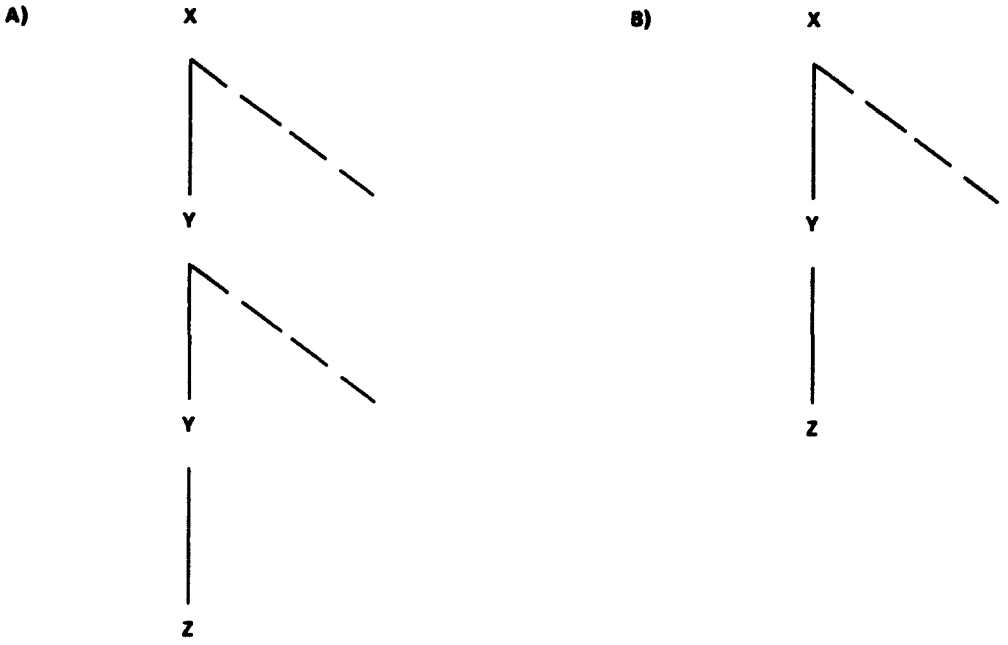


FIGURE 2.4

The "canonical" structure in (b) should be preferred, according to Kimball (1975).

while Kimball's version is dependent for its operation on the "repetition of a given symbol", Minimal Attachment depends only on the number of nodes in the phrase marker, irrespective of their actual identity. Minimal Attachment is also claimed to account for Bever's Strategy B: "The first N...V...(N)...clause (isolated by strategy A) is the main clause, unless the verb is marked as subordinate." (Bever, 1970. p.294). If phrase markers are constructed using the smallest possible number of nodes, then Strategy B follows as an automatic consequence of the operation of Minimal Attachment. Consider how Minimal Attachment would direct the parsing of (1.1).

(1.1) The oil tycoon sold the off-shore oil tracts for a lot of money wanted to kill J.R.

Figure 2.5 shows the two structures which can be constructed for the string "the oil-tycoon sold . The preferred interpretation of this string, with "sold" as the main verb, is represented by a phrase marker (Figure 2.5a) which requires fewer nodes than the phrase marker for the locally nonpreferred (but globally required) interpretation of the string, with "sold" as a subordinate passive verb (Figure 2.5b). The same is true of the other cases encountered earlier.

(1.4) My brother ordered the woman from Italy a spaghetti dinner had been spilled on accidently a large steak and a salad.

In (1.4), the determiner "a" (in "a spaghetti dinner") can either be interpreted as signaling the start of the main clause direct object,

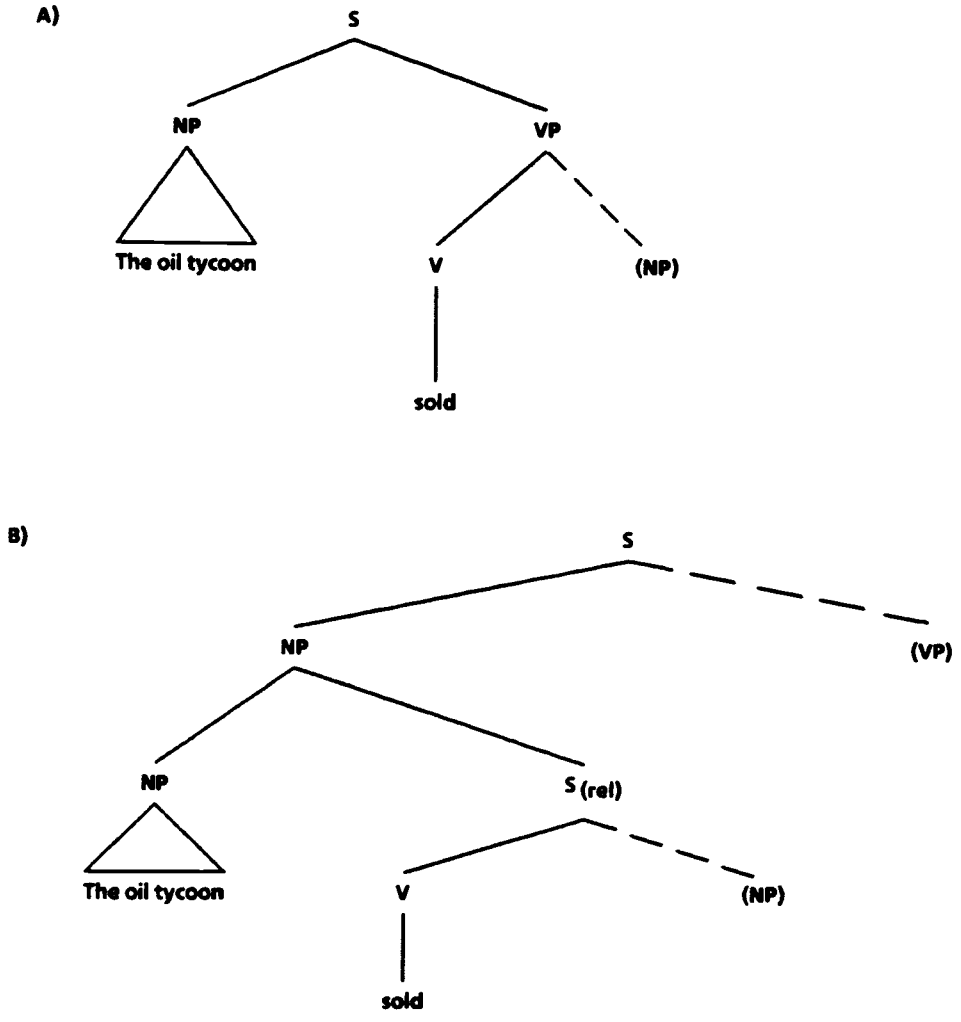


FIGURE 2.5

Phrase markers for the string "the oil tycoon sold ..." under (a) a main clause interpretation of "sold ...", and (b) under a subordinate clause interpretation of "sold ..."

or as signaling the start of a subordinate clause. The latter requires more nodes to be added to the phrase marker for the string "my brother ordered the woman from Italy" than does the former.

(1.6) The psychologist told the woman that he was worried about to visit him again.

In (1.6) the relative pronoun interpretation of "that" requires more nodes to be added to the phrase marker than does the former.

In fact, Frazier claims that in all the cases we have discussed, the resolution of local syntactic ambiguity is under the control of the principles of Late Closure and Minimal Attachment. As support for Late Closure, Frazier used a serial visual presentation technique to collect grammaticality judgements for examples like (2.13) which "demand" early closure and examples like (2.14) which demand late closure.

(2.13) Though George kept on reading the story still bothered him.

(2.14) Though George kept on reading the story Sue bothered him.

(From Frazier, 1979)

Kimball's (Early) Closure would predict that there should be more correct responses (i.e subjects should respond that the target was grammatical more often) to examples like (2.13) than to (2.14). Frazier's Late Closure predicts exactly the opposite. The results

supported Late Closure.[6] In support of Minimal Attachment, Frazier collected grammaticality judgements for the following two examples, both of which are grammatical.

(1.4) My brother ordered the woman from Italy a spaghetti dinner had been spilled on accidently a large steak and a salad.

(2.15) My brother ordered the woman from Italy a spaghetti dinner which she was enjoying until he spilled it all over her.

(From Frazier, 1979)

In (2.15), the NP "a spaghetti dinner" is "minimally attached". As we saw earlier, the direct object interpretation requires one less NP node than the interpretation which makes it the surface structure subject of a reduced passive. Minimal Attachment predicts that (2.15) should be judged grammatical on more occasions than (1.4). This is exactly what was found. Frazier (1979) thus claims to provide convincing evidence for the operation of both Late Closure and Minimal Attachment.

Frazier claims, then, that the HSPM abides by the principles of Late Closure and Minimal Attachment. But why should such principles exist? Why should the parser abide by them? In the following section we shall consider some "implementations" of syntactic parsing strategies which attempt to provide tentative answers to these questions. The motivation here will be that these principles actually reflect some underlying properties of the parser; they are a

[6] See also Frazier & Rayner (1982).

natural consequence of the way the parser operates.

2.3 Syntactic Parsing Strategies, the Architecture of the HSPM, and Rule-Ordering

The present section is concerned with what could be called the "realization problem". Right Association/Late Closure and Minimal Attachment are principles which, according to Frazier (1979) and Rayner et al. (1983), govern the operations of the processor. However, just how they govern it is another matter. There is a sense in which there is a distinction between, on the one hand, principles which describe the outcome of some processor's operation, and on the other, the implementation of these principles in a model of that processor.

2.3.1 The Sausage Machine: Frazier & Fodor (1978), Fodor & Frazier (1980)

In this subsection, we shall consider Frazier's two-stage model of the HSPM.[7] Frazier & Fodor (1978) proposed a 2-stage model, called the "Sausage Machine" (or sometimes just "SM"). The first stage is the Preliminary Phrase Packager (PPP) which assigns lexical and phrasal nodes to the words it has access to in the sentence. Access to the sentence is through a limited view window (about 6 words long) that shifts continuously through the sentence. The second stage, the

[7] The basic model was outlined in Frazier's Ph.D thesis (Frazier, 1979), but is most widely known through Frazier & Fodor (1978).

Sentence Structure Supervisor (SSS) takes the packages produced by the PPP and combines them to form a complete phrase marker. Right Association[8] arises out of a local attachment principle: any item to which some lexical node has been assigned cannot, if it has not already been linked to any other words in the window, be attached to any item on its left which may have passed out of the PPP's window. So Right Association is maintained in this model by virtue of the architecture of the HSPM. Frazier & Fodor propose that Minimal Attachment (sometimes referred to simply as "MA") governs the construction of phrase markers within both the PPP and the SSS, but only in Fodor & Frazier (1980) do they say how MA is to be implemented.

> "In the SM , the how question is answered by the assumption that the rules .. are not extrinsically ordered, but are accessed in parallel and selected in terms of the outcome of a 'race' - the first rule or combination of rules that successfully relates the current lexical item to the phrase marker dominates subsequent processing." (Fodor & Frazier, 1980. p.434)

However, they say nothing more about this, and do not attempt to explain why minimal attachments will always "win" this "race". Only if one assumes that processing time is directly proportional to the number of nodes in a phrase marker can one explain why minimal attachments will always be fastest.[9]

[8] Although in Frazier (1979) Late Closure is preferred over Right Association (given their apparent equivalence), in Frazier & Fodor (1978) the term "Right Association" is preferred.

[9] We shall return to this point in Section 2.3.2 in connection with a recent proposal by Wanner (1985) concerning race models.

Right Association and Minimal Attachment are implemented in this model in different ways. It is because of this that Frazier & Fodor can explain why, although there is a tendency to right associate (in 2.12), there is no such tendency in (2.16).

(2.12) Joe bought the book that I had been trying to obtain for Susan

(2.16) Joe read the letter to Susan

In (2.12) the tendency for the prepositional phrase "for Susan" to attach to "to obtain" arises because of the local attachment principle induced by the architecture of the HSPM (which is split between the SSS and PPP). This architecture does not permit attachment of "for Susan" to any other node. In (2.16), however, the entire sentence falls within the window of the PPP, and so the prepositional phrase "to Susan" can attach to either the verb phrase dominating "read" or to an NP node dominating the noun phrase "the letter". However, the latter attachment is prevented by Minimal Attachment.

Wanner (1980) points out a serious flaw in this argument. The difficulty with (2.17), which should be interpreted as (2.18), stems from the tendency to right associate (or locally attach, depending on whether one is talking about the principle or the implementation).

(2.17) Joe said that Martha expects that it will rain yesterday

(2.18) Joe said yesterday that Martha expects that it will rain

(From Wanner, 1980)

If Frazier & Fodor's account of Right Association is correct, and if the entire sentence were to fit within the PPP's viewing window, then this tendency to right associate should disappear. As (2.19) demonstrates, this does not happen, rather the tendency persists.

(2.19) Joe said Martha died yesterday

(Adapted from Wanner, 1980)

In this example, Minimal Attachment is not applicable and cannot explain the preferred interpretation. As a result of this observation, Fodor & Frazier (1980) concede the tendency for right association within the PPP. In other words, they concede that this tendency does not arise directly from the model's architecture. Given this concession, and given that a race model for Minimal Attachment does not actually explain why minimal attachments should <u>always</u> "win", Frazier & Fodor are left with a model of the HSPM which does little to "explain" the existence of these putative principles.

This lack of explanation is most clearly seen when we consider the following remark of Fodor & Frazier (1980).

> "..there seems to be no need to postulate any differences between the PPP and the SSS with respect to the kinds of operations they can perform or how they perform them.." (Fodor & Frazier, 1980, p.446)

Fodor & Frazier thus make clear that Right Association and Minimal Attachment are features of their 2-stage model which are <u>independent</u>

of the architecture of the model. This apparently leaves just one phenomenon which is explained by the architecture of the Sausage Machine: in Frazier & Fodor (1978), the distinction between the PPP and SSS accounted for the difficulty experienced with (2.20)

(2.20) The boy the dog the cat scratched bit cried

The problem here is that only the first three noun phrases fit within the limited-view PPP. Consequently, by the time the string of verbs is encountered, there is nothing in the PPP's window to which they can be attached. The whole of (2.21) below fits within the PPP, and consequently is perfectly acceptable.

(2.21) The boy the dog bit cried

However, Wanner (1980) proposes (2.22) to counter the Sausage Machine explanation of (2.20) and (2.21).

(2.22) Boys dogs cats scratch bite cry

Although this fits within the limited view of the PPP it is still unacceptable. The distinction, therefore, between the PPP and SSS cannot account for the unacceptibility of (2.22), and so cannot explain the status of (2.20). So it appears that there are no phenomena which are exclusively explained on the basis of the limited view of the PPP. There thus appears to be no motivation for the distinction between the PPP and the SSS, and hence no motivation for the Sausage Machine as a model of the HSPM.

2.3.2 Limited Parallelism in the ATN: a race model (Wanner, 1985)

An alternative proposal on how to "implement" principles such as Minimal Attachment and Late Closure or Right Association is provided by Wanner (1985). Wanner argued that a model based on the Augmented Transition Network (henceforth "ATN") could capture these principles in a more elegant and general way than does the Sausage Machine.

With an ATN the parsing process consists of attempting to traverse a network. A network consists of a series of "states" and "arcs" joining these states. On these arcs is a specification of the "material" which is needed in order to traverse the arc (this specification may be augmented by including a specification of some action which must be performed, such as placing the "material" in a particular register). An arc may also specify a whole network. In this case to traverse the arc one first needs to traverse the specified network successfully (hence the potential for recursion). After traversing this "sub"-network the processor then has to return to the network which called this subnetwork.

In an ATN, there are a number of different arcs, each of which has a particular function. There are WORD arcs which specify particular words which must appear in the input in order to traverse that arc; there are CAT arcs which are like the WORD arcs except that they specify a category of word which must be found; there are SEEK arcs which specify a subnetwork which must be traversed; there are SEND arcs which terminate a network and in effect return the ATN processor from a subnetwork to the network which called this

subnetwork; and there are JUMP arcs which enable the processor to jump from one state to the next without requiring anything in the input (hence allowing for the optionality of certain constituents).

Wanner proposes a model which exhibits "limited parallelism":

> "At each state, all transitions leaving that state are attempted simultaneously. If any of these transitions invoke sub-networks via SEEK operations, these derivative transitions will also be attempted in the usual way. But as soon as any one of these analysis paths successfully analyzes the current input word and attaches it to the parse tree, all other analyses are abandoned. Control then passes to the state at the end of the analysis path that won the race and the same limited parallelism resumes at that state." (Wanner, 1985, p.24)

Given the limited parallelism of the model, the analysis path which involves making fewest transitions will always win the race to attach the current word into the phrase marker. Such a model will always induce minimal attachment simply because nonminimal attachments will always involve an extra SEEK arc (as well as an extra SEND arc). Figure 2.6 illustrates this.

So Fodor & Frazier's (1980) suggestion that Minimal Attachment arise as a consequence of a race model is explained within the framework provided by Wanner. Of course, the assumption underlying Wanner's model is that processing time is directly proportional to the number of transitions. We shall return to this assumption below.

Interestingly enough, this same model will also induce right associations. As Wanner puts it:

"In a top-down, left-to-right parser, like the ATN, the

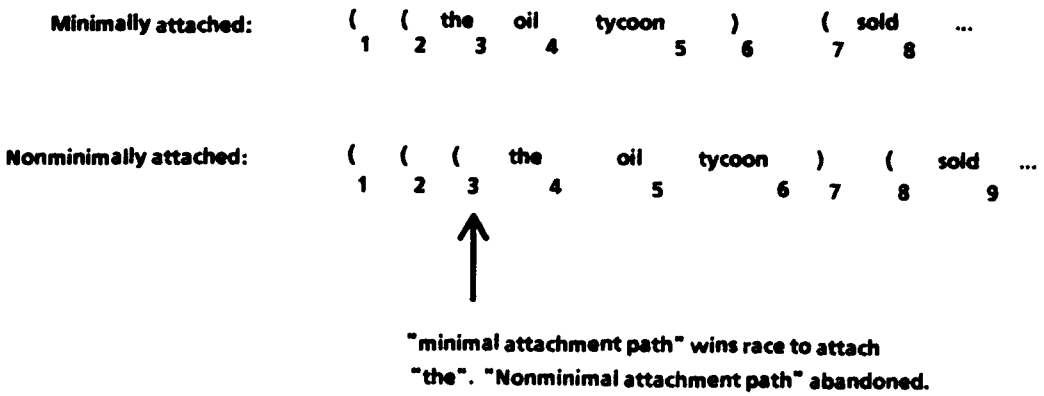


FIGURE 2.6

In this figure, left parentheses correspond to SEEK arcs, and right parentheses to SEND arcs. The lexical item "the" is reached sooner (in terms of the number of arcs traversed) in the minimal attachment path than in the nonminimal attachment path.

> lowest right constituent is always the current constituent under construction. And in the ATN, it will always require more transitions to close up the current constituent and make a higher attachment than it will to make an attachment inside the current constituent." (Wanner, 1985, p.25)

This is illustrated in Figure 2.7.

As Wanner points out, his explanation of Minimal Attachment and Right Association has advantages over that of Fodor & Frazier's simply because the two strategies fall out of the same architectural claim. Fodor & Frazier have to postulate two separate mechanisms to account for the two different strategies.

Despite the elegance of this implementation, there is one possible flaw in the argument. Implicit in Wanner's model is the notion that the analysis path with fewest arcs (i.e. fewest transitions) will always be traversed faster than any other analysis path which involves more transitions. It is by no means obvious, however, why this should necessarily be the case. Associated with each arc in the network will be an action, or possibly several actions, each of which must be successfully executed in order for that arc to be traversed. There is no a priori reason to suppose that all actions will take the same time to execute. And even if they did, it seems intuitively unlikely that an arc which has several actions associated with it will take the same time to "traverse" as an arc involving just one action.

Wanner's race model based on limited parallelism has advantages over the Fodor & Frazier (1980) model to the extent that it is more

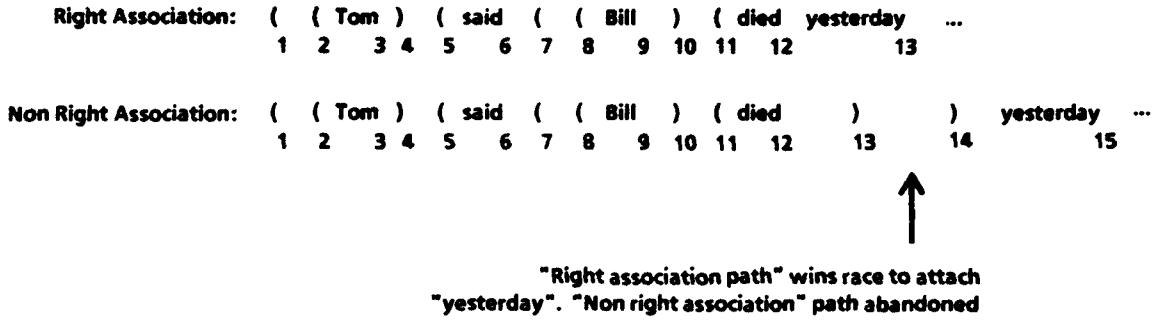


FIGURE 2.7

In this figure, left parentheses correspond to SEEK arcs, and right parenthese to SEND arcs. The lexical item "yesterday" is encountered sooner (in terms of the number of arcs traversed) in the Right Association path than in the Non Right Association path.

explanatorily "complete". We saw, however, that there are reasons for treating the assumption on which Wanner's model is based with a certain degree of caution.

2.3.3 Arc-Ordering in the ATN (Wanner, 1980)

In an earlier paper concerning the implementation of Minimal Attachment and Right Association within the ATN framework, Wanner (1980) proposed to capture these principles by ordering the arcs leaving each state in the ATN network.

Frazier & Fodor (1978) suggest that if one is to capture the principle of Minimal Attachment in an ATN then the ordering of the arcs leaving a state (i.e. the ordering of the rules in the grammar) has to be independently specified for each state in each network. However, rather than use such an ad hoc ordering, Wanner (1980) showed that there is in fact a principled account of arc-ordering in an ATN.

Wanner (1980) proposes to implement Minimal Attachment in an ATN by simply scheduling all WORD arcs and CAT arcs before all SEEK arcs. This has the effect of forcing the processor to explore all the possibilities available within the one network before attempting to explore those in any subnetwork. In terms of constructing phrase markers this means that one will always attempt to attach incoming material with the fewest possible intervening nodes.

Wanner implements Right Association in this model by simply scheduling all SEND and JUMP arcs after all the other arcs. Before

following a SEND or JUMP arc, the effect of which would be to terminate the current constituent, the parser would attempt to follow a CAT or SEEK arc. This has the effect of ensuring that the parser incorporates a new item or node into the current constituent being parsed (cf. Late Closure).

This ATN model adequately accounts for the interaction, discussed earlier, between Right Association and Minimal Attachment in (2.16).

(2.16) Joe read the letter to Susan

In Wanner's model, Minimal Attachment has priority over Right Association simply because the arcs whose properties (i.e. ordering) induce minimal attachments are ordered before those arcs whose properties induce right associations.[10]

The model can also account for the tendency, again discussed earlier, for Right Association in (2.19).

(2.19) Joe said Martha died yesterday

This example is problematic for Frazier & Fodor's 1978 version of the Sausage Machine simply because it exhibits right association. In their model, this should not be the case because the whole string

[10] In Wanner's 1985 model discussed earlier, the priority of Minimal Attachment over Right Association arises because the number of SEEK arcs traversed will always be kept to a minimum, irrespective of how many SEND arcs are subsequently required.

should fall within the limited-view window of the PPP, and right associations are predicted only if the string does not all fit within this window. The ATN model induces right associations irrespective of sentence length.

The advantage of this ATN model over the later version (Wanner, 1985) is that it is not based on any (possibly untenable) assumptions concerning the relationship between processing time and "distance" through the ATN network. Instead, it is based on the notion that, in effect, the rules of the grammar are ordered: there exists a preference for expanding one rule over another. A similar approach is taken by Ford, Bresnan, & Kaplan (1982).

2.3.4 Lexical and Syntactic Preferences: Ford, Bresnan, & Kaplan (1982)

In the preceding subsection, we considered a model in which Minimal Attachment and Right Association are explained on the basis of arc ordering in an ATN. In this subsection we shall consider what can be thought of as an extension to the arc or rule ordering approach.

Ford, Bresnan, & Kaplan (1982) provide a model in which, as in Wanner's (1980) ATN implementation of Minimal Attachment and Right Association, syntactic rules are ordered in the grammar. If two rules in the grammar apply (which is the case when a local syntactic ambiguity is encountered), then that rule which is ordered "highest" will be attempted first. In addition, Ford et al. attempt to

provide a mechanism whereby lexically-derived information, associated with the verb, can modify this ordering. They claim that their proposals do not constitute a simple reformulation of Minimal Attachment and Right Association. We shall argue, however, that this is not the case.

Ford et al. (1982) point out that there are some occasions in which parsing preferences are influenced by the particular lexical items used. As examples, they give the following:

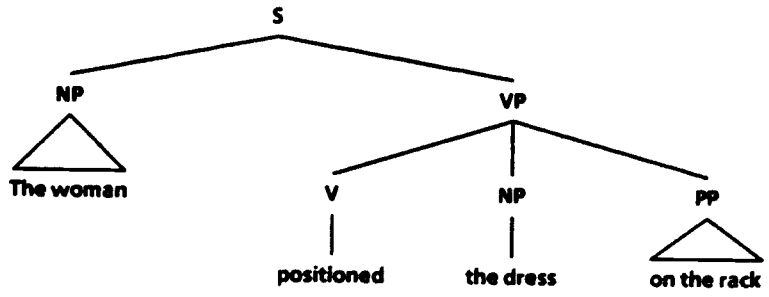
(2.23) The woman positioned the dress on the rack

(2.24) The woman wanted the dress on the rack

They claim that the verb "positioned" in (2.23) causes attachment of the prepositional phrase "on the rack" to the VP node dominating this main verb (see Figure 2.8a). This corresponds to a "minimal" attachment. In (2.24), however, they claim that the verb "want" causes attachment of the prepositional phrase to a (complex) NP node dominating the noun phrase "the dress" (see Figure 2.8b). This corresponds to the non-minimal attachment.

To explain these effects, they propose a model in which each verb has associated with it a number of "lexical forms" which may differ in terms of "strength" or "salience". For instance, the strongest lexical form of "position" is ⟨(subj),(obj),(loc)⟩ : it requires a subject, an object, and a location. The strongest lexical form of "want" is simply ⟨(subj),(obj)⟩, which requires just a

A)



B)

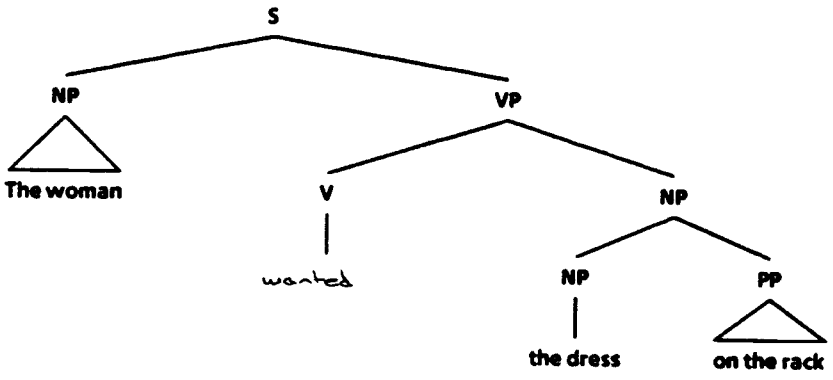


FIGURE 2.8

Phrase markers for (a) "the woman positioned the dress on the rack", and (b) "the woman wanted the dress on the rack".

subject and an object.

In addition to the notion of "strongest lexical form", Ford et al. propose a number of principles which govern the parsing process. The first of these is the principle of Lexical Preference.

"Lexical Preference

> If a set of alternatives has been reached in the expansion of a phrase structure rule, give priority to the alternatives that are coherent with the strongest lexical form of the predicate." (Ford et al., 1982. p.747)

Lexical Preference dictates that in (2.23) a VP node will be expanded which dominates the verb ("positioned"), a noun phrase (to become <obj>) and a prepositional phrase (to become <loc>). In (2.24), a VP node will be expanded which only dominates the verb ("wanted") and a noun phrase (<obj>). In order to explain the effects in (2.23) and (2.24) fully, two further principles are required.

Of the four principles postulated by Ford et al., two are concerned with the expansion of rules in the grammar (of which Lexical Preference is one), and two are concerned with the attachment of phrases into the phrase marker. One of these is the principle of Final Arguments.

"Final Arguments

> Give low priority to attaching to a phrase the final argument of the strongest lexical form of that phrase and to attaching any elements subsequent to the final argument. Low priority is defined here with respect to other options that arise at the end position of the element whose attachment is to be delayed" (Ford et al., 1982. p.752)

As can be seen from its definition, Final Arguments ensures that the Late Closure principle is maintained. This principle, together with Lexical Preference, "explains" (2.24). If the noun phrase "the dress" were attached to the NP node introduced by the lexically preferred expansion of VP, then this noun phrase would constitute the final argument of the strongest lexical form of the verb "want". An alternative option is to attach the noun phrase "the dress" as the first constituent in a complex NP. Because of the principle of Final Arguments, this latter option is preferred.[11]

Associated with Final Arguments is a default principle, which "explains" (2.23).

"Invoked Attachment

> When there are alternative options for attaching a phrase into a structure, give (default) priority to the option for attaching the phrase to the partial constituent that caused the phrase to be hypothesised." (Ford et al., 1982. p.754)

When, in (2.23), the noun phrase "the dress" is encountered, two options arise. One of these is to attach it to the NP node created during the lexically preferred expansion of VP (which in this case creates the nodes V, NP, and PP). Another is to attach it as the first constituent to a complex NP. Final Arguments does not apply here because "the dress" could not be the final argument of the strongest lexical form of the verb "positioned". Instead, the

[11] There is a possible inconsistency in this explanation to which we shall return below.

default principle of Invoked Attachment applies. According to Ford et al., the VP constituent caused the NP to be hypothesised, and so the first of these two options should be chosen.[12]

This same explanation can be used to account for the preference in (1.6).

(1.6) The psychologist told the woman that he was worried about to visit him again.

The strongest lexical form of the verb "told" is ⟨(subj),(obj),(comp)⟩. According to Lexical Preference, a VP node will be created which dominates a V, an NP (to become ⟨obj⟩), and an S´ (to become ⟨comp⟩). Because of Invoked Attachment, the noun phrase "the woman" will be attached to this VP node, and will not be attached as the first constituent in a complex NP.

The principles described thus far are claimed to account for lexical effects on parsing preferences. These lexical effects are "triggered" by the particular verbs used. However, in (1.1), parsing preferences are seen to occur <u>before</u> a verb is encountered.

(1.1) The oil tycoon sold the off-shore oil tracts for a lot of money wanted to kill J.R.

In this example, it is impossible for the verb "sold" to exert any influence over the interpretation of the noun phrase "the oil

[12] Again, this explanation gives rise to a possible inconsistency which will be returned to below.

tycoon". To account for examples like this one, Ford et al. suggest that, as in Wanner's ATN model, rules in the grammar are so ordered that the first rule to be accessed is determined by this ordering:

> "Syntactic Preference
>
> The (default) order of priority for alternative categories in the expansion of a phrase structure rule is the order of the strengths of the alternative categories." (Ford et al., 1982, p.749)

This principle is the default for Lexical Preference: The first rule to be expanded is, if Lexical Preference does not apply, the rule which is ordered first in the grammar. To explain (1.1), the rule which expands a noun phrase as a simple NP must be ordered before that which expands a noun phrase as a complex NP.[13]

At this point there arises what appears to be a slight inconsistency in the model proposed by Ford et al. They explain the complex NP preference in (2.24) according to their principles of Lexical Preference and Final Arguments.

(2.24) The woman wanted the dress on the rack

Their explanation relies on there being two options for attaching the string "the dress" into the phrase marker. One option is to attach it to a VP node dominating V and NP (Figure 2.8a), and the other is to attach it as the first constituent in a complex NP (as in Figure

[13] This was accomplished in the ATN model by ordering WORD and CAT arcs before SEEK arcs (Wanner, 1980).



2.8b). Final Arguments favours the latter option (in which "the dress" is not the final argument of the strongest lexical form). If, however, the simple NP expansion is ordered before the complex NP expansion (as must be the case, cf. the explanation given for 1.1 above), then the simple NP expansion, and not the complex NP expansion, will have been proposed first. In other words, there would not be any complex NP node to which the string "the dress" could be attached.

This would constitute an inconsistency in the Ford et al. model if it were the case that the processor had access to only one rule expansion at a time. However, if <u>all</u> the options available to the processor (i.e. all the rule expansions) are stored in an "option list", then a principle like Final Arguments will simply re-order the options within this list. So although the simple NP alternative may be ordered before the complex NP expansion, this ordering may be changed by Final Arguments. Such a proposal is in fact made by Ford et al.[14] Their account for (2.23) and (1.6) must now be questioned, however.

(2.23) The woman positioned the dress on the rack

(1.6) The psychologist told the woman that he was worried about to visit him again.

[14] For more detail, refer to Ford et al. (1982), Appendix B, p. 794.

Invoked Attachment (reprinted below) requires that the strings "the dress" and "the woman" be attached to a VP node and not to a (complex) NP node:

> "When there are alternative options for attaching a phrase into a structure, give (default) priority to the option for attaching the phrase to the partial constituent that caused the phrase to be hypothesised." (Ford et al., 1982. p.754)

According to Ford et al. only the partial VP constituent caused the noun phrases "the dress" and "the woman" to be hypothesised. However, if all the possible alternatives are put onto an "option list", then both the partial VP constituent and the partial (complex) NP constituent will appear on this list. Furthermore, both these constituents hypothesise the existence of the noun phrases: the partial VP constituent in (2.23) hypothesises the existence of a noun phrase at "position 4"[15] as does the partial complex NP. Similarly for (1.6) at position 4. Invoked Attachment is therefore still faced with a choice: there is no single option to which Invoked Attachment can assign priority. However, the essential difference between these two alternatives is that one of them (the partial VP alternative) introduces fewer nodes into the phrase marker than does the other (the complex NP alternative). Any procedure which could decide between these two would be equivalent to Frazier's (1979) Minimal Attachment. Invoked Attachment thus appears to be effectively little

[15] Ford et al. propose that partial constituents are labelled according to the serial position within the sentence of the lexical items they span.

different from Minimal Attachment.

To summarise this subsection: Ford et al. (1982) claim to provide a model of sentence processing which does not abide by Minimal Attachment or Right Association, but which can nonetheless account for the observed parsing preferences. They propose that syntactic rules are ordered in the grammar, and that this ordering can be superceded on the basis of lexical information. It appears, however, that their model is equivalent to one incorporating Minimal Attachment and Right Association: their principle of Final Arguments maintains Right Association, whilst their use of the default principle of Invoked Attachment suggests that their model abides elsewhere by a principle of Minimal Attachment. In Part II, we shall question the empirical adequacy of Minimal Attachment.

2.4 Summary

In this chapter we have considered a number of studies which take the view that local syntactic ambiguities are resolved on the basis of syntactic, and not semantic/pragmatic, information. In Section 2.2 we considered the proposals of Kimball (1973; 1975) and Frazier (1979). They put forward what amount to two principles of surface structure parsing: Right Association (Kimball, 1973) or the equivalent Late Closure (Frazier, 1979), and Minimal Attachment (Frazier, 1979; and see also Kimball, 1975). In Section 2.3, we considered the implementation of these principles within models of sentence processing. At issue was whether these principles explicitly guide the actions of the parser, or whether they are

merely derivative on the workings of the parser. In other words, are these principles merely arbitrary rules to which the parser must respond, or do they reflect some underlying property of the parsing mechanism?

Fodor & Frazier (1978)'s model "accounted" for these principles only with the proposal that Right Association is brought about by the architecture of their 2-stage model, the Sausage Machine. No "explanation" was given for Minimal Attachment. In the later version of the Sausage Machine (Fodor & Frazier, 1980), they proposed a race model to account for Minimal Attachment. No explanation was given for Right Association (which now became a principle which governed both stages of the 2-stage Sausage Machine and could not be explained by their system's architecture).

Wanner (1985) proposed a general race model for ATNs which would predict both minimal attachments and right associations. His model is, unfortunately, based on the assumption that processing time is dependent only on the number of arcs that must be traversed during a parse, and not on the kinds of action that must be executed in order to traverse each arc. Reasons were given for treating this assumption with extreme caution.

In an alternative ATN-based model, Wanner (1980) demonstrated that both minimal attachments and right associations could be made to arise by using a very general arc-ordering in an ATN.

Finally, Ford et al. (1982) continued with the rule-ordering approach, but deviated slightly from the two preceding positions in that they attempted to allow lexically-derived information to influence the parsing process and override the preferences that would otherwise be induced by Minimal Attachment and Late Closure. Their proposals differ from the others in that they do not make any claims about the architecture of the HSPM but rather merely elaborate on the principles which must govern the workings of the parser.

It can be concluded on the basis of these studies that there are just two principles of surface structure parsing which must be postulated if one is to adopt a structural approach to syntactic ambiguity resolution. These are Minimal Attachment (Frazier, 1979) and Right Association (Kimball, 1973).

It is certainly interesting to speculate on the properties of the HSPM which principles such as Minimal Attachment and Right Association might reflect. The studies discussed above demonstrate that there are a variety of directions in which such speculation might lead. There is a sense, however, in which speculation of this sort is premature: it should first be demonstrated that the HSPM does actually abide by these principles (whatever the reasons for their existence). The models discussed in this section would be of less interest if it were the case that the HSPM did not. <u>always</u> abide by these principles.

Bever (1970) observed that there are occasions when parsing preferences can be overridden. What Bever did not do was attempt to

characterise those occasions. In the chapter that follows, we shall attempt such a characterisation, and shall review the studies which take the view that local syntactic ambiguities can be resolved on the basis of semantic/pragmatic information.

CHAPTER 3

Context and Syntactic Ambiguity Resolution

The main emphasis of Chapter 2 was with surface structure parsing. That is, we were primarily concerned with the construction of syntactic tree structures, or phrase markers. These structures were considered important because different structures could be associated with the alternative analyses of an ambiguous sentence fragment. It was suggested that the HSPM might choose between the alternative analyses on the basis of some property or other of these different syntactic structures. Much of Chapter 2 was taken up with detailed consideration of two principles of surface structure parsing: Minimal Attachment (Frazier, 1979), and Late Closure/Right Association (Frazier, 1979; Kimball, 1973). But there must be something more to sentence processing than the construction of syntactic phrase markers. In this chapter we shall consider some aspects of sentence processing which, unlike surface structure parsing, demand that we consider more than just the processing of single isolated sentences.

The chapter is organised as follows: In Section 3.1 we shall briefly consider some issues concerning sentence processing within the wider scope of discourse processing. Central to this discussion will be the notion of "context", which we shall define in this section. We shall illustrate a number of principles of discourse processing using as an example Winograd's (1972) language

understanding program SHRDLU.

In Section 3.2 we shall review some early studies which purport to show that the HSPM's decisions can be influenced by contextual (i.e. semantic/pragmatic) information.

Section 3.3 consists of a detailed review of some proposals made by Crain (1980) and Crain & Steedman (1985). They claim that parsing preferences can be induced or overturned depending on the context within which a sentence is to be comprehended. More importantly, and unlike the earlier studies, they actually characterise that aspect of the context which is responsible for influencing the parser's decisions. They discuss in detail one specific class of ambiguity, corresponding to those examples otherwise accounted for by Minimal Attachment. Crain (1980) provides experimental evidence in support of the thesis that Minimal Attachment does not always govern the choices of the HSPM. The investigation of these claims will form the basis of the remainder of this thesis (Part II onwards).

We conclude Part I of this thesis in Section 3.4.

3.1 Sentence Processing and Discourse Comprehension

Language has evolved as a means of conveying some aspect of a speaker's knowledge or beliefs to a hearer. This transaction, between speaker and hearer, takes place by way of discourse.[1]

[1] For the purposes of this discussion, we shall tend to deal with spoken discourses, i.e. with speakers and hearers. The same arguments will, in general, apply also to written texts.

Discourses can be thought of as a series of sentences, each of which contributes to this transaction. Within the communicative act, the speaker attempts to convey information, via each sentence, to the hearer. The hearer must then attempt to integrate this new information with information he or she already has about the world, or even about the preceding sentences. It is this act of integration that makes discourse comprehension different from isolated sentence processing: each sentence must be interpreted within the context of the discourse. Sanford & Garrod (1981) comment on this interpretive process as follows:

> "... the sentences only have significance because of knowledge in the mind of the reader which was there before the text was encountered. Rather than simply describe a text as a linguistic object, it therefore seems more appropriate to describe it as a series of instructions which tell the reader how to utilize the knowledge he already has, and contingently modify this knowledge in the light of the literal content of the discourse itself." (Sanford & Garrod, 1981, p.8).

All sentences, whether ambiguous or not, are produced, and processed, in a context. But what constitutes a sentence's context? The context of a sentence can be thought of as including the physical environment, the state of the speaker's knowledge or beliefs, the state of the hearer's (or reader's) knowledge or beliefs, the content of any preceding sentences, the form of these sentences, and even the content and form of the sentence whose context we are attempting to define. Following Brown & Yule (1982), we shall distinguish between "context", which is a general term to describe the overall situation within which the sentence is being processed, and "co-text", which is

a more specific term which refers only to the set of sentences, or text, which precedes the "target" sentence whose context concerns us. We may also distinguish between intra-sentential (or sentence-internal) context and extra-sentential (or sentence external) context. The former pertains to information conveyed by the target sentence itself, whilst the latter pertains to information which is "external" to the sentence (and which of course includes that sentence's co-text[2]).

In order to consider in more detail the manner in which contextual information may aid sentence comprehension, we shall take as an example a simple language understanding system developed by Winograd (1972; 1973). Although the domain of discourse is highly constrained, and the manner in which the program interprets the discourse does not necessarily reflect human behaviours, the example will illustrate a number of different kinds of information which are available during discourse processing. The example is not intended to provide an exhaustive list of such kinds of information, but a limited number which we shall have cause to return to in later sections, when we shall conclude that aspects of the program do reflect aspects of human discourse processing. As such, the example provides a historical setting for some of the later work to be reported here.

Winograd's system consisted of a simulated robot (called "SHRDLU") which could manipulate "toy blocks" in its simulated blocks world. The robot made use of a "model" of its world. This model contained information which described the current state of the blocks world, and information about procedures for changing the state of the world, or for making deductions about it. The robot could respond to simple commands or questions. The following dialogue illustrates this. Sentences in lower case would be typed in by the user, and sentences in upper case would be the robot's response. The blocks world consisted of three pyramids and five blocks (all of different

[2] For the remainder of the thesis, the term "context" will be used in its most general sense, and will thus be taken to subsume co-textual information.

sizes and colours) and a box.

(3.1) a. Pick up a big red block.
b. OK
c. Grasp the pyramid.
d. I DON'T UNDERSTAND WHICH PYRAMID YOU MEAN.
e. Find a block which is taller than the one you are holding.
f. OK.
g. Put it into the box.
h. BY "IT" I ASSUME YOU MEAN THE BLOCK WHICH IS TALLER THAN THE ONE I AM HOLDING. OK.
i. What does the box contain?
j. THE BLUE PYRAMID AND THE BLUE BLOCK.
k. What is the pyramid supported by?
l. THE BOX.

(Adapted from Winograd, 1973, pp. 155-157)

There are a number of features of this dialogue which illustrate the interpretive nature of discourse comprehension. The first feature is seen in examples (3.1c) and (3.1d).

(3.1) c. Grasp the pyramid.
d. I DON'T UNDERSTAND WHICH PYRAMID YOU MEAN.

Here, SHRDLU cannot establish the referent of the expression "the pyramid". That is, SHRDLU cannot determine which particular pyramid this expression is being used to refer to. The definite article "the" suggests that this should be a specific pyramid which SHRDLU should know about and be able to identify. In the blocks world, however, there were three pyramids, and the expression "the pyramid" was not sufficiently precise to enable SHRDLU to pick just one pyramid from amongst the three candidates.

Sentence (3.1e) demonstrates how sentence-internal information can be used to decide on what is meant by the expression "the one".

(3.1) e. Find a block which is taller than the one you are holding.

SHRDLU must work out that the referent of the expression "the one" is the "big red block".

Similarly, in (3.1g)

(3.1) g. Put it into the box.

SHRDLU must work out that the referent of "it" is the block which is taller than the red block (in this example there was only one block which was taller than the red one), as opposed to the red block itself. To do this, SHRDLU makes use of information it has stored about the preceding dialogue.

The definite referring expression "the box" in (3.1g) causes SHRDLU no problem because there is only one box in the blocks world. SHRDLU uses its knowledge about its world in order to determine this fact. In (3.1k), the expression "the pyramid" also causes SHRDLU no problem, even though there was a problem the first time this expression was encountered.

(3.1) k. What is the pyramid supported by?

In (3.1c) the expression "the pyramid" was rejected because there was no particular pyramid that could be distinguished. In (3.1k), however, there is: namely the pyramid that was mentioned in answer to question (3.1i).

(3.1) i. What does the box contain?
j. THE BLUE PYRAMID AND THE BLUE BLOCK.

As in the case of the interpretation of "it" in (3.1g), SHRDLU now makes use of its knowledge of the preceding discourse in order to determine just which object is being referred to.

This simple dialogue with SHRDLU shows that decisions concerning the intended interpretation of a sentence must often be made in the light of information which is external to the sentence being processed (and which is therefore contained within that sentence's context). For instance, we saw that the processing of an expression such as "the pyramid" is dependent on the context within which the expression occurs. The hearer/reader presumably has access to this information via his or her "mental model".

SHRDLU had available both a model of the toy blocks world (which contained "real world knowledge") as well as a model of the dialogue within which each sentence was to be interpreted. SHRDLU's "world model" provided the information which led to "the pyramid" being uninterpretable the first time it was mentioned (in 3.1c), even though "the box" was interpretable (in 3.1g). SHRDLU's "discourse model" provided the information which led to "the pyramid" being interpretable the second time it was mentioned (in 3.1k). Of course, the distinction between world model and discourse model is by no means clearcut, and the latter may simply be a subset of the former. The structure and organisation of such mental models, however, is of less concern to us here than is the fact that such internalised

information must exist.[3]

The sample dialogue in (3.1) not only demonstrates the need for internal representations of both the world and the discourse, but also illustrates one kind of process which draws heavily on sentence-external context; namely the process of identifying the referents for referring expressions. In Section 3.3 we shall return to this process, and consider the way in which such information might form the basis for resolving certain kinds of ambiguity. In the following section, we shall consider some of the early studies which investigated the effects of contextual information on ambiguity resolution.

3.2 The Role of Context during Syntactic Ambiguity Resolution

Frazier (1979) acknowledges that contextual information may play an important role during the resolution of local syntactic ambiguity. With regard to Minimal Attachment and the role of context she says

> "When a [suitably ambiguous] sentence ... is preceded by contextual information which specifies that the NON-minimal analysis of the sentence is appropriate, the parser might: (a) ONLY compute the appropriate non-minimal analysis of the sentence, if contextual information may be used as a basis for FORMING syntactic hypotheses, or (b) construct the appropriate non-minimal analysis only after a minimal analysis was formed and (when necessary) revised. That is, even though contextual information probably determines which analysis ... will eventually be selected, it is not known how the parser ARRIVES at this analysis." (Frazier, 1979.

[3] For a discussion of issues concerning the structure and organisation of mental models, see for example Sanford & Garrod (1981); Reichgelt (1982); Johnson-Laird (1983); and Shadbolt (1983).

pp.59-60)

Frazier's own experiments, reported in Section 2.2, manipulated only syntactic structure, not semantic/pragmatic plausibility. Test sentences which were "minimally-attached" were judged to be grammatical more often than "nonminimally-attached" counterparts. Similarly for materials which "demanded" late closure as compared with early closure. It is impossible on the basis of these data alone to determine whether semantic/pragmatic information can guide the resolution process. Rayner, Carlson, & Frazier (1983) attempt to address exactly this issue.

The Rayner et al. study, which is primarily concerned with the operation of Minimal Attachment, consists of two experiments. The first purports to show that the initial syntactic analysis pursued is determined on the basis of Minimal Attachment, and the second purports to show that the final analysis arrived at is determined by a principle of "Thematic Selection". These two principles can be summarised as follows: if more than one analysis of a sentence is possible then the HSPM will initially pursue whichever analysis requires the phrase marker with fewest nodes. If this chosen analysis subsequently proves "implausible", then the alternative analysis is then, and only then, attempted.

Their first experiment is based on consideration of the following two examples:[4]

[4] Structurally equivalent to examples 1.1 and 2.8.

(3.2) The florist sent the flowers was very pleased.

(3.3) The performer sent the flowers was very pleased.
(From Rayner et al., 1983)

Rayner et al. claim that if pragmatic plausibility determines the initial analysis then (3.3) should be "easier" than (3.2), because (3.3) describes a more plausible state of affairs than (3.2). Knowledge of what florists tend to do would make the active interpretation of "sent" most likely in (3.2), whilst knowledge of what performers tend to do would make the passive interpretation of "sent" most likely in (3.3). Only the passive interpretation of "sent" (i.e. only 3.3) avoids a garden path. Minimal Attachment, on the other hand, predicts that there should be no difference between (3.2) and (3.3). The initial analysis pursued will be determined not on grounds of pragmatic plausibility, but on grounds of syntactic structure, and this will interpret "sent" as an active verb, rather than as a past participle. Since in both examples this analysis is incompatible with subsequent material, subjects should garden path in both cases. To test these two approaches, Rayner et al. collected reading time and eye movement data for these examples. As predicted by Minimal Attachment, they found no differences between sentence pairs of this type.

Having demonstrated that the initial analysis of a sentence is determined by Minimal Attachment, Rayner et al. propose that the final interpretation is chosen via Thematic Selection. A verb like

"see" will have, they claim, two "thematic structures" associated with its lexical entry. One of these reflects the fact that the verb may occur just with an "experiencer" and a "theme" (as in "John saw the lunar eclipse"), whilst the other reflects occurrences with a "theme, an "experiencer", and an "instrument" (as in "John saw the lunar eclipse with a telescope").[5] As Rayner et al. put it,

> "These thematic structures could be used to delimit the set of relations whose pragmatic plausibility will be considered. Given the phrases 'John', 'the lunar eclipse', and 'with a telescope', the processor could consider whether the set of relations including just an experiencer and a theme, or the set of relations including an experiencer, a theme, and an instrument is more plausible on the basis of pragmatic factors concerning real world knowledge and knowledge of the preceding discourse." (Rayner et al., 1983. p.367)

To test this hypothesis, Rayner et al. ran a second experiment with examples similar to (2.10) above.

(3.4) The spy saw the cop with binoculars but the cop didn't see him.

(3.5) The spy saw the cop with a revolver but the cop didn't see him.

(From Rayner et al., 1983)

Rayner et al. predicted that (3.5) would take longer than (3.4) because Minimal attachment would favour attachment of the "with-clause" to the verb "saw" rather than to the noun phrase "the cop".

[5] There is an obvious similarity, noted by Rayner et al., between their proposal and that of Ford, Bresnan, & Kaplan (1982), discussed earlier.

However, in (3.5) subjects would "realise" their mistake, and on the basis of Thematic Selection reanalyse the sentence attaching "with the revolver" to "the cop". It is this process of reanalysis which should increase reading times to (3.5). This is exactly what they found.

In short, the Rayner et al. study is claimed to provide conclusive evidence in support of the notion that Minimal Attachment, when applicable, determines which is the initial analysis to be pursued in cases of local syntactic ambiguity. They claim that semantic/pragmatic information has no part to play in this initial selection process.

Rayner et al.'s materials consisted only of single isolated sentences. Although they manipulated intra-sentential factors, it is not the case that they determined whether or not extra-sentential contextual factors determine which is the first analysis to be pursued.

Few studies have investigated the effects of extra-sentential context on ambiguity resolution. Much of the early data on the processing of ambiguous material was gleaned from studies concerned with lexical ambiguity. Some of these studies were concerned with the effect of ambiguity on processing load. For instance, a number of studies demonstrated that the presence of a lexically ambiguous item (i.e. a word with two or more possible meanings) increased the perceptual complexity of the carrier sentence (e.g. Mackay, 1966; Foss, 1970; Bever, Garrett, & Hurtig, 1973). Others demonstrated

that disambiguating contextual information did not offset this increase in complexity (e.g. Foss & Jenkins, 1973; and see also Mehler, Segui, & Carey, 1978). Later studies on lexical ambiguity were concerned more with whether or not multiple entries for lexically ambiguous words are accessed (cf. Hogaboam & Perfetti, 1975; Schvanavelt, Maier, & Becker, 1976; Holmes, 1979; Swinney, 1979). The issue here was whether just one meaning of the ambiguous word was accessed, or whether more than one sense was accessed. Also of interest was whether prior contextual information had any effect on this. Swinney (1979) demonstrated that more than one meaning <u>is</u> accessed, regardless of whether the prior context contains any disambiguating information. Contextual disambiguation seems to occur only <u>after</u> the alternative senses are accessed.

None of these studies, however, have been concerned with the effects of contextual information on <u>syntactic</u> ambiguity. Early studies in this area (e.g. Carey, Mehler, & Bever, 1970; and Lackner & Garrett, 1972) suggested that information which is external to the ambiguous sentence can influence the interpretation of that sentence. For instance, Lackner & Garrett, who investigated lexical, surface structure, and "deep-structure" ambiguities, used a dichotic listening task in which an ambiguous sentence was presented to one ear, and a disambiguating context to the other (cf. also Mackay, 1973). Subjects were instructed to attend to one particular channel (which carried the ambiguous sentence). Using a paraphrase task, they found that syntactically ambiguous items such as (3.6) were interpreted differently depending on which of two biasing sentences

was presented to the unattended ear (this was true even for subjects who reported being unaware of the ambiguity).

(3.6) John left with a dog he found last Saturday. (From Lackner & Garrett, 1972).

If the structural strategies discussed in the preceding chapter were in operation, then we would always expect there to be a preferred interpretation for sentence (3.6). Late Closure/Right Association would predict that the first analysis to be attempted will always be that in which "last Saturday" modifies "found" and not "left".

But Lackner & Garrett's results suggested that extra-sentential information could influence the selection of one or other analysis in cases of ambiguity. This finding must, however, be treated with caution, because of the nature of the task that was used. Lackner & Garrett found that the biasing sentence was equally effective when the disambiguating portion temporally preceded the ambiguous portion of the test sentence, when it coincided with the ambiguous portion, or when it followed the ambiguous portion. The "mechanism" by which these effects occurred is thus far from clear. Although this study demonstrates contextual effects on the resolution of syntactic ambiguity, we are little closer to being able to specify just which information is used when.

Carroll, Tanenhaus, & Bever (1978) describe an experiment which also demonstrates a biasing effect of context. They presented subjects with two-sentence discourses as in (3.7) and (3.8).

(3.7) Why did Smith resign?
Smith who Ford asked to remain in Vermont, for political reasons, decided to resign his post.

(3.8) Why did Ford want Smith to remain in Vermont?
Smith who Ford asked to remain in Vermont, for political reasons, decided to resign his post.

(From Carroll et al., 1978)

Again, Late Closure/Right Association has a single prediction to make in both instances: that the first analysis to be attempted should be that in which "for political reasons" attaches to its left ("remain in Vermont"), and not to its right ("decided to resign"). Contrary to any such predictions, subjects' paraphrases differed "overwhelmingly" (p. 207) depending on the context question. However, this experiment does not indicate whether contextual information was used on-line during the processing of the target sentences, it indicates only that it affected the ultimate interpretation of the sentence.

A more convincing demonstration of the on-line effects of contextual information on ambiguity resolution was provided by Tyler & Marslen-Wilson (1977). They considered the following sentence fragments.

(3.9) Although it doesn't require much work, cooking apples ..(is/are)

(3.10) Although they may be very tart, cooking apples ..(are/is)

(From Tyler & Marslen-Wilson, 1977)

The ambiguity arises in these examples because "drying" can be interpreted either as a gerund (3.9), or as an adjective (3.10). Tyler & Marslen-Wilson claim that in (3.9) the gerund interpretation is the more plausible, and in (3.10) the adjectival interpretation is the more plausible. This in turn means that in (3.9), use of the singular copula ("is") is pragmatically[6] more plausible a continuation than is the plural form ("are"), whilst in (3.10), use of the plural form is pragmatically more plausible than is use of the singular. Using a naming latency task, Tyler & Marslen-Wilson showed that subjects responded faster to the pragmatically more plausible form of the copula than they did to the less plausible form. They interpret this to suggest that contextually derived information ("if you enjoy it" vs "if they take up a lot of space") is used in resolving the local ambiguity encountered with the string "drying clothes".

Despite the convincing demonstration provided by Tyler & Marslen-Wilson, there is other evidence which is claimed to show that structurally-based parsing preferences cannot be overturned by the provision of prior contexts. Holmes (1984) preceded (3.12) below by either (3.11a) or (3.11b). (3.12) is structurally equivalent to the examples used by Rayner et al. (1983).[7]

[6] The term "pragmatic" is used here as introduced in Chapter 1: something is pragmatically plausible if it is plausible on the grounds of knowledge about the world in general or the discourse as a whole.

[7] Holmes advocates a theory of parsing preferences which resembles the proposals of Ford, Bresnan, & Kaplan (1982) more than those of

(3.11) a. The parents discussed with the math teacher what could be done about the difficult exam, but no solution was reached.

b. The parents discussed among themselves what could be done about the math teacher, but no general agreement was reached.

(3.12) The parents talked about the problem with the mathematician.

(From Holmes, 1984)

Holmes claims that (3.11a) is "felicitous" with the minimally-attached reading of (3.12), in which the prepositional phrase "with the mathematician" is used to modify the action denoted by the verb "talked", whilst (3.11b) is felicitous with the nonminimally-attached reading of (3.12), in which the prepositional phrase modifies "the problem".[8] Given this assumption (which we shall return to), Holmes formed two predictions: if contextual information can determine the outcome of the resolution process, then (3.12) should be judged to be consistent with (3.11a) as often as with (3.11b). The alternative is that the outcome of the resolution process is determined by structural considerations (which favour the minimally-attached interpretation of 3.12), in which case (3.12) will be judged to be consistent with (3.11a) more often than with (3.11b). Her results

Frazier (1979). However, all these theories make the same predictions of these examples (see Rayner et al., 1983, for discussion).

[8] For present purposes, we can define the term "felicity" as follows: two texts are felicitous with one another if the second is a pragmatically plausible continuation for the first. Clearly, much rests here on the notion of pragmatic plausibility. We shall return to this below.

favoured the structural approach.

Holmes claims to show that there is no effect of "felicity" on the outcome of the resolution process. However, she offers no explanation for what constitutes "felicity" in these examples. Presumably (3.11b) is claimed to be felicitous with the nonminimally-attached reading of (3.12) because it implies that there is "a problem with the mathematician", whereas (3.11a) is felicitous with the minimally-attached reading of (3.12) because it implies that the "talking" was "to the mathematician". For her claim to be valid, her criteria for what constitutes pragmatic plausibility (and hence "felicity") must be correct. If, on the other hand, her manipulations do not constitute manipulations of felicity, then her claims concerning the experiment's outcome cannot be valid.

The Holmes study can be criticised because it is unclear whether her notion of felicity (as applied to the specific examples she investigates) is correct. The Tyler & Marslen-Wilson study has been criticised on essentially similar grounds: it fails to characterise the nature of the information which is present in the preceding sentential context and which is responsible for the observed effects on the resolution process. We shall now consider a study which, like the Tyler & Marslen-Wilson study, demonstrates the effect of contextual information on the resolution process, and which in addition provides an explicit characterisation of the nature of this contextual information. Furthermore, we shall see that this characterisation is compatible with the contextual manipulations of

Tyler & Marslen-Wilson but incompatible with those of Holmes.

3.3 Reference and the Resolution of Local Syntactic Ambiguity

Crain (1980) and Crain & Steedman (1985)[9] propose an approach to ambiguity resolution in which the functional roles of the particular constructions involved are taken into account. In (3.1a), the functional role of the adjective "red" is to provide extra information for the hearer which can then be used as a basis for identifying just which "block" is intended.

(3.1) a. Pick up a big red block.

In particular, Crain & Steedman were concerned with the presuppositions which are implicated by each alternative analysis of an ambiguous sentence fragment. The term "presupposition" tends to be used with a number of different senses (for discussion see Givon, 1979; Brown & Yule, 1982; Crain & Steedman, 1985). We shall take a presupposition to be anything that the speaker believes to be true of the hearer's mental model of the universe. In (3.1a), the speaker presupposes (amongst other things) that without the adjective "red", SHRDLU would not be able to identify any particular block. For present purposes this definition of presupposition will suffice. In Chapter 7 we shall see, however, that this definition needs to be expanded in order to adequately account for certain phenomena.

[9] The relevant material from Crain (1980) is reported in Crain & Steedman (1985).

Crain & Steedman's discussion is based around the presuppositions which are implicated by the use of restrictive relative clauses, as in (3.13)

(3.13) The psychologist told the wife whom he was having trouble with to leave her husband.

The function of restrictive relatives is to provide additional information as to who or what is being referred to. This additional information is essential, to the extent that without it, the hearer would not be able to identify uniquely in his or her model of the discourse the object being referred to. By using a restrictive relative, the speaker presupposes that the information conveyed within the relative clause is already known to the hearer, for if the hearer did not have this information available, then the relative clause would not help in the task of identifying the intended referents. The information conveyed by the relative must therefore be "Given" information. Following Clark & Haviland (1977) we shall take Given information to be information already known about by the hearer, and New information to be information which is not already known to the hearer.[10]

The function of restrictive relatives is illustrated by the following two discourse fragments.

[10] Brown & Yule (1982) point out that this "psycholinguistic" definition is perhaps more similar to the notion of presupposition (cf. Givon, 1979) than it is to the original distinction as formulated by Halliday (1967).

(3.14) a. A psychologist was counseling a married couple.
One member of the pair was fighting with him but the other one was nice to him.
The psychologist told the wife .

b. A psychologist was counseling two married couples.
One couple was fighting with him but the other one was nice to him.
The psychologist told the wife .

(From Crain, 1980)

The critical difference between these two discourse fragments is that (3.14a) introduces just one couple into the discourse (and hence just one "wife"), whilst (3.14b) introduces two couples (and hence two "wives") into the discourse. In the last sentence of (3.14a), it is possible, on the basis of the description "the wife", to identify uniquely just which individual is intended. In the case of (3.14b) this is not the case. The preceding co-text has introduced more than one possible referent. If the last sentence of (3.14b) were replaced by

(3.15) The psychologist told the wife whom he was having trouble with

then it would be possible to identify just which "wife" was being referred to: the relative clause enables one to determine which member of the set of possible referents is intended. Indeed, its use presupposes (amongst other things) the existence of such a set, and the hearer's inability to determine which particular member of it is intended when only the simple referring expression "the wife" is used.

But how can information concerning the functional role of restrictive relatives help during the resolution of local syntactic ambiguity? To go back to Winograd's (1972) SHRDLU program, one of the commands which the robot could respond to was

(3.16) Put the blue pyramid on the block in the box. (From Winograd, 1973).

This command is of course ambiguous (see Figure 3.1): the pyramid could already be on the block or the block could be in the box. In this example, the prepositional phrase "on the block" could function in the same way as a restrictive relative clause modifying the referring expression "the blue pyramid" (cf. Figure 3.1a). Alternatively, the phrase "in the box" could be restrictive on the referring expression "the block" (cf. Figure 3.1b). Winograd's SHRDLU resolved the ambiguity as follows: on finding a definite noun phrase, SHRDLU would search the blocks world (and also a representation of the preceding discourse) for a referent (or antecedent) to this referring expression. If a unique referent could be found to the referring expression "the blue pyramid", SHRDLU would then look for "the block in the box". If no unique referent could be found to "the blue pyramid", SHRDLU would then look for "the blue pyramid on the block

SHRDLU resolves the ambiguity in (3.16) in a very different way to that advocated by Frazier (1979) or Rayner et al. (1983). According to Minimal Attachment, the string "the block in the box"

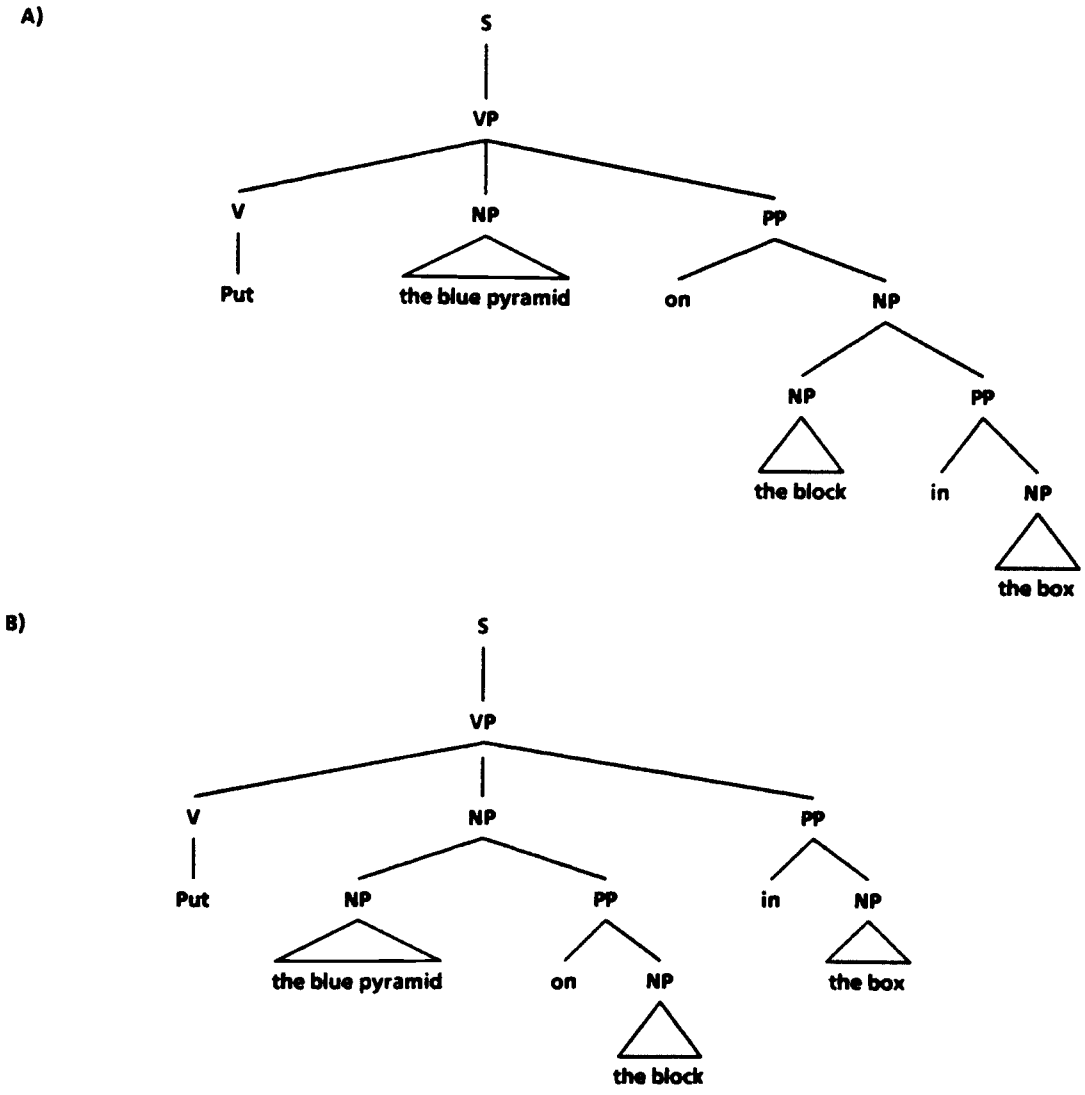


FIGURE 3.1

Phrase markers for the string "Put the blue pyramid on the block in the box". In (a) the target location for the pyramid is "the block", and in (b) the target location is "the box".

would always be interpreted as a complex noun phrase: "the blue pyramid" would be minimally attached as the direct object of "put", and not as the head NP of a complex noun phrase. According to Winograd's system, this would not always be the case. It would only be interpreted as a complex noun phrase if no unique referent could be identified for the expression "the blue pyramid".

Crain & Steedman formalise the kind of approach taken by Winograd (1982) with the following principle:

> "The Principle of Referential Success: If there is a reading which succeeds in referring to an entity already established in the hearer's mental model of the domain of discourse, then it is favoured over one that does not." (Crain & Steedman, 1985, p. 331)

If the Principle of Referential Success is correct, then the garden path effect in (3.17) should be overcome when (3.17) is preceded by a context which introduces more than one "wife" into the "discourse" (cf. 3.14b).

(3.17) The psychologist told the wife that he was having trouble with to leave her husband.

Furthermore, this same context should induce a garden path in (3.18).

(3.18) The psychologist told the wife that he was having trouble with her husband.

Similarly, a context which introduces just one "woman" into the discourse (cf. 3.14a) should induce a garden path in (3.17) but not in (3.18).

Crain & Steedman propose, then, that it should be possible to overcome garden path effects such as those found with (3.17) if a context can be provided which is felicitous (as defined by the Principle of Referential Success) with the particular interpretation required of the example (cf. one of the alternative theories which Holmes, 1984, set out to test). According to a purely structural principle such as Minimal Attachment, (3.17) should <u>always</u> induce a garden path, even if felicitous semantic/pragmatic information is present.

Using an incremental grammaticality judgement task (cf. Frazier, 1979[11]), in which he preceded each one of the targets (3.17) and (3.18) with either a "relative-inducing" context (cf. 3.14b) or a "complement-inducing" context (cf. 3.14a), Crain showed that garden path effects could be overcome <u>or</u> induced for <u>both</u> targets depending on which co-text preceded the target. Had Minimal Attachment been correct, then subjects should always have garden pathed with (3.17), regardless of the preceding context. Furthermore, it should not have been possible to induce garden path effects in (3.18). The results Crain obtained are presented in Table 3.1.[12]

[11] The incremental grammaticality judgement task involves presenting each target one word at a time. As soon as the sentence ceases to be "grammatical", subjects are asked to press a response button.

[12] Crain (1980, and in Crain & Steedman, 1985) reports the results of a number of Chi-square tests which confirm the difference between the two contexts. All the values of Chi-square reported were significant at $p<0.001$.

TABLE 3.1

RESPONSES INDICATING UNGRAMMATICALITY (PERCENT)
(Crain & Steedman, 1985)

	Context	
Target	complement-inducing	relative-inducing
complement	12	54
relative	50	22

These results provide clear support for the notion that the resolution of local syntactic ambiguity is influenced by referential considerations, such as whether or not a referring expression is found to be "referentially successful" in its discourse context. This is particularly important because all the examples which have so far been "explained" by the purely structural Minimal Attachment have one feature in common: of the two (locally) possible interpretations, one is functionally identical to a restrictive relative clause. This includes the reduced passive constructions such as (1.1) and (1.4)

(1.1) The oil tycoon sold the off-shore oil tracts for a lot of money wanted to kill J.R.

(1.4) My brother ordered the woman from Italy a spaghetti dinner had been spilled on accidently a large steak and a salad.

and the prepositional phrase examples of Rayner et al. (1983).

(3.4) The spy saw the cop with binoculars but the cop didn't see him.

(3.5) The spy saw the cop with a revolver but the cop didn't see him.

The success of Crain's experiment has implications for the manner in which the results of Tyler & Marslen-Wilson (1977) and Holmes (1984) should be interpreted. As the reader will recall, Tyler & Marslen-Wilson claimed to show via a naming latency task that contextual information can influence the ambiguity resolution process. Holmes claimed to show via a "plausibility judgement" task that contextual information cannot override structural preferences. Both these studies can, however, be criticised for failing to account for their (implicit) notions of felicity. The essential question is this: given Crain's results, can one now explain those of both the earlier experiments?

The majority of examples used by Tyler & Marslen-Wilson were of the form described earlier:

(3.9) Although <u>it</u> doesn't require much work, cooking apples ..(is/are)

(3.10) Although <u>they</u> may be very tart, cooking apples ..(are/is)

It is interesting to note the use of the cataphoric pronoun "they". This can be construed as creating a "referential context" which predisposes the processor to interpret anything of the form V-ing N as a (plural) noun. Similarly for those occasions when "it" was used: a referential context will have been created which predisposed the processor to interpret anything of the form V-ing N as a

(singular) gerund.[13] In other words, the results found by Tyler & Marslen-Wilson seem to be perfectly compatible with some notion of Referential Success.[14] But what of the Holmes study?

The following sentence (3.12), which was discussed earlier, is one of those used by Holmes.

(3.12) The parents talked about the problem with the mathematician.

This example is structurally ambiguous because there is more than one site on the phrase marker to which the prepositional phrase "with the mathematician" can attach. Holmes predicted, and found, that this target prepositional phrase would be attached to the verb even if the preceding co-text were felicitous with the attachment to the noun phrase.

[13] 25 of the 28 stimuli in this experiment had "they", or equivalent, in the version corresponding to (3.10), and 13 used "it" in the version corresponding to (3.9).

[14] Tyler & Marslen-Wilson (1985) report an analogous experiment using the following materials which do <u>not</u> include any cataphoric pronouns:

(3.19) If you have the time and energy, rowing boats is/are ...

(3.20) If you have a car and trailor, rowing boats are/is ...

They report finding a strong effect of context in the adjectival condition ("rowing boats are") but no effect of context in the verbal (gerund) condition ("rowing boats is"). They suggest that this may be because the gerund-inducing context ("If you have the time and energy") was not "strong" enough. The fact that in the earlier experiment, containing cataphoric expressions, there was a strong effect of context in the verbal condition suggests that referential context does at least contribute to the observed effects.

But this apparently less preferred attachment to the noun phrase is equivalent, in terms of referential function, to a restrictive relative clause on a noun phrase which begins with a definite article. The prepositional phrase here tells us which of several problems is at issue specifically "the problem with the mathematician". By the argument we have been using, a prior context felicitous with this attachment should have introduced more than one "problem" into the discourse. But in the "felicitous" context (3.11b) actually provided by Holmes, this has not occurred.

(3.11b) The parents discussed among themselves what could be done about the math teacher, but no general agreement was reached.

According to Crain & Steedman's Principle of Referential Success, (3.11b) is not felicitous with the nonminimal attachment of the target prepositional phrase, because a set of alternative candidate "problems" is not introduced. Holmes' finding, that this context did not induce this attachment, is therefore not surprising.

Crain & Steedman provide evidence that the resolution of (at least one kind of) local syntactic ambiguity is under the control of contextually-derived information. Furthermore, they provide a relatively explicit characterisation of this information in terms of the notion of "reference" and the hearer's "mental model of the domain of discourse". But how do they account for the observed preferences in the absence of any explicit prior context (what Crain & Steedman call the "null context" condition)

Crain & Steedman propose, of the null context, that an analysis will be adopted which requires fewest "alterations" or "extensions" to the hearer's model. Their discussion is based on examples similar to (3.21).

(3.21) The horse raced past the barn ... (fell.)

The main clause interpretation presupposes (or rather, the speaker, in intending a main clause interpretation, presupposes) that there is established in the model a uniquely identifiable "horse". It is also presupposed that there exists a uniquely identifiable "barn". The reduced passive interpretation presupposes not only the existence of a "horse" and a "barn", but also (i) that there exists a <u>set</u> of "horses", and (ii) that it is already established that the relative clause "applies" to a single unique member of this set. In the null context, therefore, the reduced passive analysis requires more extensions to a (newly created) model than does the main clause analysis. Hence the preference for the main clause in (3.21). Crain & Steedman encapsulate this notion with their Principle of Parsimony (which subsumes their Principle of Referential Success):

> <u>"The Principle of Parsimony:</u> If there is a reading which carries fewer unsatisfied but consistent presuppositions or entailments than another, then, other criteria of plausibility being equal, that reading will be adopted as most plausible by the hearer, and the entailments in question will be incorporated in his or her model." (Crain & Steedman, 1985. p. 333)

Parsimony favours a main clause interpretation for (3.21) because such an analysis carries fewer entailments than does the reduced

passive analysis.

So Crain & Steedman can apparently account for all the examples to which Frazier's (1979) Minimal Attachment has been applied. Moreover, they provide experimental evidence which supports the claim that (referential) context can influence the HSPM's decisions during the resolution of local syntactic ambiguity. But does this necessarily mean that Minimal Attachment cannot influence the HSPM's decisions? What would happen if, in the examples considered by Crain & Steedman, the co-texts were felicitous with both locally possible analyses of the locally ambiguous target sentence? Crain & Steedman suggest that

> "..in the absence of any difference in a priori plausibility between the rival analyses, any residual preference for one or other would seem to require a structural explanation." (Crain & Steedman, 1985. p.344)

They report an experiment by Crain (1980) in which co-texts like the following were devised which supposedly satisfied the presuppositions implicated by the use of both relatives and complements:

(3.22) Co-text:

Three new players joined the baseball team.
Several older players picked one of the new players to join them for the first practice.
The other new players were upset by this.

This co-text would be followed with either (3.23) or (3.24) below.

(3.23) Complement target:

The coach convinced the player that the old pros chose to practice by themselves for the rest of spring training.

(3.24) Relative target:

The coach convinced the player that the old pros chose to practice by himself for the rest of spring training.

(The critical disambiguating items in the target sentences are underlined.) Crain & Steedman claim that the context above is felicitous with both targets: either "the player" can be interpreted as that player introduced in the second sentence ("one of the new players"), which would lead to referential success and hence leave the way open to a complement clause interpretation of "that", or "the player" can be interpreted as referring to one of the set of all players introduced in the first two sentences, which would not lead to referential success and would hence require a restrictive relative clause interpretation of "that". Using an incremental grammaticality judgement task, they claim that

> "when the presuppositions of both the complement clause and relative clause analyses of the target sentences were fulfilled in this way subjects indicated no significant preference for either the complement clause or relative clause analysis of the target sentence." (Crain & Steedman, 1985. p.345)

Crain & Steedman claim, then, that once context has been controlled for, there are no residual effects of structure. In other words, they claim that referential context does influence the decisions of the HSPM, whereas Minimal Attachment does not.

It is unclear, however, whether Crain's experiment does in fact prove this point: in order to make both the relative and complement clause analyses felicitous with the co-texts, Crain designed co-texts which introduced a number of different sets of players. One set had just a single member (the "player" picked by the "old pros"), whilst another had more than one member (the set of "new" and "old" "players"). In doing this, Crain introduced a further degree of ambiguity into these examples (i.e. over and above the local syntactic ambiguity in the target sentences) to the extent that the expression "the player" could be taken as referring to a member of either one set or the other. Crain & Steedman's interpretation of Crain's (1980) results rests, however, on the premise that both the complement and relative clause analyses are possible (i.e. felicitous) given the preceding co-text. There is an alternative interpretation of their results, however. Perhaps half the subjects chose to interpret "the player" as belonging to one set, and half as belonging to the other. Once this choice is made, it is no longer the case that either of the two locally possible analyses (i.e. relative vs. complement) is possible. In other words, their results need not have arisen because of the lack of any residual effect of structure. They arose simply because of the lack of any preference for interpreting "the player" as belonging to one "set" of players or another.

Despite such criticisms, Crain & Steedman provide a convincing demonstration of the role that semantic/pragmatic information can play during the resolution of local syntactic ambiguity. In the

following section, we shall summarise the approach to syntactic ambiguity resolution which we have taken in the present chapter. In the following section, we shall consider the course that further research into the resolution process should take.

3.4 Summary and Conclusions

In this chapter, we have taken the view that sentence processing consists of more than just the construction of syntactic phrase markers. In Section 3.1, some examples were provided which demonstrated that there do exist occasions when a sentence has to be interpreted within the context of the discourse within which the sentence is produced. A number of examples were drawn from Winograd's (1972) natural language understanding program, SHRDLU. In particular, we considered the interpretation of definite referring expressions, and this interpretation's dependence on some form of model of the discourse.

In Section 3.2 we considered several studies which investigated whether or not the resolution of local syntactic ambiguity could be influenced by contextual information. Frazier's (1979) own experiments did not address issues concerning the role of semantics/pragmatics simply because semantic/pragmatic plausibility was not a variable in her experimental design. In the Rayner et al. study discussed in this section, it was, and no effect on the resolution process was found. There are several ways, however, in which semantic/pragmatic plausibility can be manipulated, and a number of other studies were presented which did report effects of

context on ambiguity resolution. None of these studies, however, specified exactly which information was responsible for these contextual effects. The section ended with a discussion of two studies: Tyler & Marslen-Wilson (1977) and Holmes (1984). The former demonstrated an effect of prior context on syntactic processing, whilst the latter study failed to find any such effects.

In Section 3.3, we considered the work of Crain (1980) and Crain & Steedman (1985). Their work is particularly important because, unlike other studies, it does characterise just which contextual information can be used by the HSPM during the resolution of certain kinds of syntactic ambiguity. The (null) result of Holmes (1984) was, according to this characterisation, to be expected. The Crain & Steedman account of ambiguity resolution is based on a Principle of Parsimony: a reading is chosen which leaves unsatisfied fewest presuppositions or entailments. A specific instance of this principle is their Principle of Referential Success: if there is a reading which enables the hearer/reader to establish the referent of a referring expression, then that reading is preferred over one that does not. It was suggested that this principle, together with Parsimony, could account for the garden path effects that had previously been attributed to the operation of the purely structural strategy Minimal Attachment (Frazier, 1979).

3.5 Part I: Summary and Conclusions

During the course of Part I of this thesis, we have encountered two distinct approaches to the study of local syntactic ambiguity resolution. On the one hand, is the structural approach, advocated most recently by Frazier and colleagues (eg. Frazier, 1979; Rayner et al., 1983), and Ford, Bresnan, & Kaplan (1982). On the other, is the contextual approach, advocated by Tyler & Marslen-Wilson (1977) and Crain & Steedman (1985).

In Chapter 2 we considered the structural approaches to syntactic ambiguity resolution. We saw that these approaches are all based on just two principles of surface structure parsing: Right Association (Kimball, 1973) and Minimal Attachment (Frazier, 1979). Wanner (1980) showed that these two principles could be captured in a simple Augmented Transition Network model in which the arcs leaving any particular state are scheduled according to a very general ordering. In a later implementation of these two principles, Wanner (1985) showed that they could be captured in an ATN which exhibited "limited parallelism". Ford, Bresnan, & Kaplan (1982) proposed a model which incorporates Minimal Attachment and Late Closure, and which, like Wanner's (1980) model, is based on a notion of rule-ordering. In addition, their model allows lexical information associated with verbs to override the preferences that would otherwise be induced by Minimal Attachment and Late Closure. Both Rayner et al. (1983) and Holmes (1984) - discussed in this chapter - claimed to provide experimental evidence in support of this

structural approach to ambiguity resolution.

In this chapter, an alternative approach was discussed in which contextual, and not structural, information guides the parsing process during the resolution of local syntactic ambiguity. The studies which had been discussed in Chapter 2 were all concerned with the construction of syntactic surface structures (or phrase markers). In this chapter we concluded that sentence processing must involve more than just this. We were specifically concerned with the interpretation of sentences within the context of the discourses within which they occurred. We took the view that the comprehension process should not be viewed when isolated from contextual information. With regard to syntactic ambiguity, Tyler & Marslen-Wilson (1977) and Crain (1980) provided evidence to suggest that contextual considerations do guide the resolution process. In addition, Crain & Steedman provided a rigorous characterisation of the contextual considerations at issue.

The implications of Crain & Steedman's work are far-reaching. Their account of syntactic ambiguity resolution explains the occurrence of parsing preferences not only in the presence of an explicit prior context (cf. Crain's experimental results), but also in the absence of any explicit prior context (in the "null context"). This would suggest that the results of Rayner et al. (1983) and Ford, Bresnan, & Kaplan (1982) may be artefactual on the fact that all the stimuli were presented in the null context. The one study which did not present sentences in isolation from an explicit context was the

Holmes (1984) study. She found no effects of context on the resolution process, even though she claimed to be manipulating the "felicity" of her texts. According to Crain & Steedman's account, however, none of her contextual conditions were in fact felicitous with anything but the minimally-attached interpretation of each target sentence.

Rayner et al. made a strong claim on the basis of their experimental results: they claimed that their study showed that semantic/pragmatic information cannot overturn the preferences that are induced by Minimal Attachment. Similarly for the Holmes study.[15] The Crain & Steedman account would predict that effects of semantic/pragmatic information would have been found in these studies had the critical information been supplied. Crain & Steedman's interactive account of syntactic ambiguity therefore makes a strong claim about the results of Rayner et al.'s and Holmes' experiments.

Part II of the thesis describes a number of experiments which explore further the effects of context on sentence processing the claims made by Crain & Steedman. In Chapter 4 we shall conclude that Crain's study alone is not sufficient to justify the rejection of the claims based on studies of ambiguity resolution which did not take into account referential function. Chapter 4 will explore Crain &

[15] As detailed in an earlier footnote, Holmes (1984) did not advocate Minimal Attachment as a parsing strategy. Nonetheless, her theory makes exactly the same predictions of her experimental stimuli as does Minimal Attachment (and Ford et al.'s, 1982, theory of Lexical and Syntactic Preferences).

Steedman's claims as they apply to the processing of materials which include multiple attachment sites for prepositional phrases. Chapter 5 explores them as they apply to the processing of relative and complement clauses. Both Crain's experiments and the experiments to be reported in these later chapters, deal exclusively with written texts. We might suppose that the HSPM will have available during the processing of spoken discourses at least the same information (if not more) than it has during the processing of written texts. Whether it uses the information in the same way, and whether referential considerations interact with other factors such as intonation in spoken discourses, will remain empirical questions.

An account of syntactic ambiguity resolution involves more, however, than simply a characterisation of just which information is used by the HSPM when a decision is to be made. As suggested in Chapter 1, such an account should also detail just when this information is used. We shall consider this point in Part III.

The emphasis in Chapters 4 and 5 of Part II and Chapter 6 of Part III will be on the operation of Minimal Attachment. Results will be described in the experimental chapters which seem incompatible with the operation of such a strategy. The theoretical discussion in Chapter 6 will focus on the Principles of Referential Success and Parsimony as a suitable alternative to Minimal Attachment. Although the Principle of Referential Success does characterise the nature of the information that is used by the HSPM when resolving certain ambiguities, we shall conclude that it does not successfully characterise the time course of the events surrounding the HSPM's decisions. An alternative principle, the Principle of Referential Failure, which does provide a successful characterisation of the time course of these decisions will be offered in its place.

In Chapter 7 the emphasis will return to a consideration of the original null context data with which this study started, and the <u>two</u>

structural strategies which we encountered in Chapters 2 and 3. We shall develop an account of the null context data which, although different to Crain & Steedman's, is still based on their Principle of Parsimony. We shall speculate about whether or not the Principle of Parsimony might also apply to the Right Association/Late Closure data. We shall conclude that an approach based on Parsimony is a suitable alternative to the structural approaches advocated elsewhere.

Part II: The Experiments

"Seal up the mouth of outrage for a while,
Till we can clear these ambiguities."

W. Shakespeare.

CHAPTER 4

Referential Context and Prepositional Phrases

In preceding chapters, we encountered a number of distinct positions concerning the resolution of local syntactic ambiguity. The discussion culminated with a review of an experiment reported in Crain (1980) and Crain & Steedman (1985) which showed strong effects of referential context on parsing decisions. This work is important in two respects. First, it provides a strong theoretical account of the factors which can influence the resolution of local syntactic ambiguity; and second, it makes use of an important experimental technique: because the same target is observed under two different contextual conditions, any differences observed are due to contextual factors; and because the contexts are "minimally different", any contextual differences can be accounted for.

Crain & Steedman view the comprehension process as being contextually determined: sentences are not processed in isolation from the contexts in which they are produced. Their work suggests that a major methodological shortcoming in the work of Frazier and her colleagues is the absence of any context within which the particular syntactic constructions they were studying could be allowed to "function". According to this contextually-based account, Rayner et al. (1983), whose experiments were discussed in Section 3.2, violated the presuppositions which are implicated by the use of restrictive constructions. Their results may thus be considered to

be artefactual on this violation. Perhaps Rayner et al.'s results would have been quite different had they not violated these presuppositions.

But would these criticisms be justified on the basis of Crain's experiments alone? In this chapter, we shall see that a number of methodological (as opposed to theoretical) difficulties in Crain's work prevent it from supporting the view just summarised. This chapter presents a contextually-based replication of the Rayner et al. study which is able to confirm the significance of Crain's work.

In Section 4.1 we consider a number of points concerning experimental design and their application to Crain's experiment. In Sections 4.2 and 4.3 we consider the form that a suitable replication of the Rayner et al. study should take. In Section 4.4 the experiment itself is described, and in Sections 4.5 and 4.6 its results are discussed and summarised. A number of points will arise which motivate a second experiment which is described in Sections 4.7 onwards.

4.1 Experimental Methodology

In this section, we shall consider a number of methodological points which arise out of consideration of Crain's experiment.

The degree to which an experiment either does or does not support an experimental hypothesis can be evaluated according to three criteria:

4.1.1 The Design

The first concerns the "design" of the experiment. There are two aspects to this issue. One of these concerns the relationships which hold between the experimental conditions, the test materials, and the subjects. Ideally each subject should see no item (whether in one version or another) more than once, but each subject should nonetheless be exposed to all experimental conditions and each item should be represented in all conditions. The design reported in Crain & Steedman (1985) seems to conform to this pattern. But such a pattern has important consequences for the number of experimental items (and also subjects) which must be used in an experiment. In this experiment there were just four basic sets of materials. Given that there were four experimental conditions (i.e. two levels of Context crossed with two levels of Target), this means that any one subject was exposed to only one test item in each condition. Because so few materials were used, we might want to question the generalisibility of the reported effects (see below, under Section 4.1.3). It may be the case that these effects are peculiar only to the specific examples used by Crain.

Under the heading "design", one might also include the construction of the test materials. In the case of an experiment like Crain's, the experimenter presents subjects with different versions of the same material. What is fundamentally important, however, is that there should exist a correspondence between how these versions differ in the theory and how they differ in practice.

Differences in the dependent variable which are seen to occur across experimental conditions may be due to any changes made to the materials in order to transform them from one version to another. For theory and practice to agree, the incidental changes to the materials across conditions should be kept to a minimum.

In the test materials used by Crain, however, more than just the one critical manipulation (e.g. "one girl" vs. "two girls") was made to transform the basic co-text from one version to the other. Examples (4.1) and (4.2) below illustrate this.

(4.1) Complement-inducing co-text

A psychologist was counseling a married couple. One member of the pair was fighting with him but the other one was nice to him.

Relative-inducing co-text

A psychologist was counseling two married couples. One couple was fighting with him but the other couple was nice to him.

Complement Target Sentence

The psychologist told the wife that he was having trouble with her husband.

Relative Target Sentence

The psychologist told the wife that he was having trouble with to leave her husband.

In this example, the co-text explicitly introduces either one wife or two wives into the context. In example (4.2) below the critical noun phrase in the target sentences is "the girl". The co-texts, however, do not merely introduce either one girl or two girls.

(4.2) Complement-inducing co-text

A teacher had three new girls in class. One of the new girls hated boys.

Relative-inducing co-text

A teacher had three new girls in class. All the boys in the class hated two of the three girls.

Complement Target Sentence

The teacher told the girl that the boys liked to help themselves to some paper.

Relative Target Sentence

The teacher told the girl that the boys liked to help herself to some paper.

In this example, the complement-inducing co-text introduces the same number of girls as does the relative-inducing co-text. The important differences occur in the second sentence: "One of the new girls hated boys" vs. "All the boys in the class hated two of the three girls". Amongst the differences between these are: first, the different roles of the boys and girls (i.e. who hated whom); second, the different numbers of individuals referred to (i.e. "one of" in the complement-inducing co-text vs. "all" and "two of the three" in the relative-inducing co-text); and third, the explicit reference in only one of the co-texts to "boys in the class".

A further difference between these two examples used by Crain is that in (4.1) we must infer on the basis of the relative target that the psychologist is having trouble with someone if he's fighting with them. In (4.2), however, we are forced to infer that if a girl isn't hated by a boy then she must be liked by him. These two inferences

are very different in character.

Because the four examples used by Crain (given in Crain & Steedman, 1984) were not all of the same format, there do appear to be grounds for questioning the correspondence between the manipulations which supposedly occur in the theory (manipulations concerning the cardinality of the referent set) and those which occurred in practice: the "minimal pairs of co-texts" were not as "minimal" as they could have been. There is thus no single contextual manipulation which can be held responsible for the observed changes in dependent variable across experimental conditions.

4.1.2 The Task

The second criterion by which an experiment can be judged concerns the experimental task. The serial visual presentation / grammaticality judgement task used by Crain (and also Frazier, 1979) is an example of a task in which there exists a delay between the fluctuation in processing to which the subject's response is somehow sensitive and the occurrence of the subject's response. An important consideration here is that subjects' responses may be sensitive not only to those aspects of sentential processing under the experimenter's control, but also to additional processes, not under the explicit control of the experimenter, which may intervene during the response delay.

A state of affairs must be avoided where any variation in the dependent variable (i.e. in subjects' responses) is largely artefactual on the experimental procedure. This might occur, for instance, where a task forced subjects into using particular processing strategies which would not be used under more "natural" circumstances.

With Crain's incremental grammaticality task, subjects took on average 1.84 seconds (or approximately 3.3 words) from the appearance of the critical disambiguating item to respond (i.e. to signal ungrammaticality) in those cases where garden pathing did occur. This surprisingly long response delay may have been partly a reflection of the delay between presentations of successive words: 550 msec. (compared to, for instance, 300 msec. used by Frazier, 1979). Thus, the very slow presentation rate coupled with the grammaticality judgement task may have induced subjects not only to respond slowly but also to use processing strategies which under more natural circumstances would not have been used.

Particularly worrying is the fact that the response delay of 1.84 seconds is also sufficiently long to have allowed subjects to attempt a reanalysis of the sentence in those cases where the first analysis led to a "garden path". If they arrived at the same analysis on this second pass, they would presumably respond (signalling ungrammaticality) whereas if they arrived at a different (but now grammatical) analysis they would not report the garden path at all. This possibility suggests that the frequencies found by

Crain (see Table 3.1) may be underestimates. Again, these criticisms seem to provide serious doubts with respect to the generalisibility of the reported effects.

4.1.3 The Analysis

The third criterion by which one can judge an experiment concerns the treatment of the results. The issues here are primarily ones concerned with the appropriateness of the statistical analyses with which the results are treated. This is not the place to describe statistical theory in detail. However, there are two points which are worth being reminded of: first, statistical tests should be used not just to confirm that the dependent variable is indeed dependent (i.e. that the results are not just due to chance). They should also be used to show the relationship between one experimental condition and another in terms of the effects each condition has on the dependent variable. There is an important difference between saying that there is some effect of the experimenter's manipulations on the dependent variable, and saying which manipulations are responsible for which aspects of this effect. This is especially important in those cases involving more than one independent variable and where one needs to know which, if any, are responsible for the observed effects. It follows that statistical analyses which can provide such information should be used.

The second point about statistical testing concerns "generalisibility". Clark (1973) points out that it is not enough merely to show (statistically) that a given result should generalise

to a situation involving new subjects but the same materials. Nor is it necessarily enough to demonstrate (again statistically) that the results should generalise to new materials but using the same subjects. Rather one should ideally show that the results will generalise to both new materials and new subjects. Clark (1973) goes on to suggest that an appropriate test which can provide the necessary information with which to evaluate generalisibility is the Analysis of Variance both "by subjects" (i.e. by averaging across materials), and "by materials" (i.e. by averaging across subjects). By combining the two F scores associated with each analysis (F_1 and F_2 respectively) to produce "Min F'", one can then evaluate the generalisibility of the observed results.[1]

In Crain's experiment, there were two independent variables, Context and Target. An analysis would have been appropriate which could distinguish between these two in terms of the effects they had on the dependent variable (frequency of the "ungrammatical" response). Chi-square, which Crain used, is unsuitable because it will merely indicate whether there is any difference in frequency across the experimental conditions: it will not indicate which differences are significant and which are not. It offers no information on the relative contributions of Context and Target to the response data. An Analysis of Variance could in principle have

[1] For more detail, the reader is referred to Clark (1973) - though see Wike & Church (1976) and Wickens & Keppel (1983) for further discussion on this.

provided the necessary information about the contributions of Context and Target, but Crain's very small set of materials violates assumptions concerning the distribution of the data which underly the use of Analysis of Variance. In short, the Analysis of Variance requires a larger set of data than that provided by Crain's experiment.

Thus, in the light of a number of criteria, Crain's experiment falls considerably short on those features which contribute to the generalisibility of the results. The fact that Crain used so few materials severely limits the generalisibility of his results: not only can one not generalise to other similar materials, but because he used only one class of local syntactic ambiguity, one certainly cannot generalise to other classes (cf. that used by Rayner et al., 1983, and Holmes, 1984).

It would seem, then, that there are grounds for approaching the results reported by Crain with caution. Nonetheless, the rationale behind the experiment is theoretically justified: not only has Crain provided a characterisation of that aspect of a context which can effect parsing decisions, but he has also suggested the means by which these effects can be explored, namely by preceding locally ambiguous target sentences with minimally differing co-texts in which only the "cardinality of the referent set" has been changed. Crain's experiments can therefore be interpreted as "pilots" on the basis of which other experiments can be designed and run.

In later sections just such experiments are described, but first, we shall consider the form that an appropriate replication of Crain's study should take.

4.2 Experiment 1: Introduction

To summarise the preceding section, there are a number of reasons for supposing that Crain's results do not provide sufficient justification for the outright rejection of the claims made by Rayner et al. (1983). First, there are a number of features concerning the design of his experiment which do not permit statistical generalisibility; and second, the critical experiment reported by Crain investigated just one kind of syntactic construction: his results do not, therefore, necessarily generalise to other constructions which can be described using Minimal Attachment.

The materials used by Rayner et al. (reproduced below) in their eye movement/reading time experiment display a class of ambiguity which differs from that used by Crain.

(3.4) The spy saw the cop with binoculars but the cop didn't see him. (Minimal VP-attachment)

(3.5) The spy saw the cop with a revolver but the cop didn't see him. (Nonminimal NP-attachment)

As we mentioned earlier (in Section 3.3), Crain & Steedman's Principle of Referential Success can nonetheless be applied to these examples (henceforth "PP-examples"). Crain's experimental results, however, need not necessarily generalise to these kinds of example.

This is important because the Rayner et al. study (like that of Holmes, 1984) makes very strong claims about the role (or lack of it) that semantic/pragmatic information has to play during the resolution of local syntactic ambiguity. What is needed is a replication of Rayner et al.'s experiment with the provision of adequate contextual information.

If Rayner et al. are correct in proposing Minimal Attachment and Thematic Selection, then VP-attached targets will always invoke reading times which are lower than those to NP-attached targets because the VP-attached target is "minimally-attached", and hence attempted first. But if Crain & Steedman are correct, this will not necessarily be the case: reading times will instead be determined by an interaction between target sentence and the number of potential referents to the simple NP "the cop"

4.3 Experiment 1: Methodological Preface

The examples used by Rayner et al. are qualitatively different to the relative/complement examples used by Crain (1980) because structurally speaking, they are globally ambiguous: there is no purely syntactic information which marks which of the two analyses is the correct one. The relative/complement examples, however are only locally ambiguous: they can be resolved on the basis of syntactic information alone. Clearly, this carries with it important implications for the kind of experimental task which can be used in order to explore the effects of referential context on these examples. Although the incremental grammaticality judgement task

used by Crain might be suited to cases in which a garden path leads the subject to recognise ungrammaticality, it is less well suited to examples in which the inappropriate analysis is merely pragmatically implausible. It would be difficult to provide instructions for the subjects which defined "plausibility" in anything other than terms defined by the theory which we would be attempting to test. Because Rayner et al. found strong effects using reading times, it seems sensible to abandon the judgement task in favour of reading times in the present study.[2]

The global reading time measure is, in its way, subject to problems. The measure used by Crain was a relatively direct measure of "garden pathing", and as such was arguably only sensitive to those manipulations of the context which did result in changes occurring to the cardinality of the referent set. Reading time, on the other hand, has been shown to be sensitive to many factors[3] , and thus it is essential that the members of each pair of co-texts truly be only "minimally" different (given the theory's definition of "minimal"). The materials described in the following section conform to this

[2] Reading times were chosen rather than eye movements because of the uncertainty which accompanies the interpretation of eye movement data. Differences in the mean number and/or duration of fixations or regressions for any particular region within the string could reflect a number of different processes. It is unclear, therefore, whether eye movement data would provide any useful information over and above that already provided by global reading times.

[3] See, for example, Sanford & Garrod (1981). The later experiments in this study will also illustrate some of the factors (other than garden pathing) which can influence reading times.

requirement. The importance of this fact will become increasingly apparent when we discuss the experiments making up the present study.

4.4 Experiment 1

The aim of Experiment 1 was to determine whether the contextual manipulations proposed by Crain & Steedman would have any effect on the reading time results reported by Rayner et al. (1983). The target sentences used in the Rayner et al. study (examples 3.4 and 3.5) are locally ambiguous between the attachment of a prepositional phrase to the immediately preceding noun phrase (the NP-attachment) or to the preceding verb (the VP-attachment). The experiment was designed to test the hypothesis that the NP-attachment is facilitated (in terms of the reading time measure) when the preceding co-text introduces more than one candidate referent to the NP. The VP-attachment should be facilitated if there is no need to modify the NP restrictively: that is, if just one candidate referent to the NP is introduced in the co-text.

4.4.1 Materials

Thirty target sentence pairs were constructed (see Appendix A). For each target pair, a pair of co-texts were constructed: in one co-text only one candidate referent for the post-verbal noun phrase was introduced (to induce VP-attachment in the target sentence[4]) and

[4] The terms "VP-inducing" and "NP-inducing" are used only to signify that a co-text is felicitous with one attachment or the other. As defined in Chapter 3, two texts are felicitous if the second is a pragmatically plausible continuation for the first.

in the other, two candidates were introduced (to induce NP-attachment). The following are examples of the co-texts and targets used in the experiment. The full set of materials is listed in Appendix A.

VP-inducing co-text ("1", i.e. a single candidate referent is introduced)

A burglar carrying some dynamite broke into an heiress's house.
Once inside he found a safe and a jewelry box.
One of them had some diamonds inside whilst
the other had several priceless emeralds.

NP-inducing co-text (">1", i.e. more than one candidate referent is introduced)

A burglar carrying some dynamite broke into an heiress's house.
Once inside he found two safes.
One of them had some diamonds inside whilst
the other had several priceless emeralds.

VP-attached target ("vp")

The burglar blew open the safe with the dynamite
and made off with the loot.

NP-attached target ("np")

The burglar blew open the safe with the diamonds
and made off with the loot.

The co-texts are "minimally-different" to the extent that the only change is from "a safe and a jewelry box" (one candidate referent and therefore VP-inducing) to "two safes" (two candidate referents and therefore NP-inducing). Thus any differences in reading time for the same target sentence can only be due to this one modification, the

effect of which, in the theory, is to change the cardinality of the set of "safes" introduced into the story from "1" to ">1".

Target sentences were presented on just one line of the screen, and were not split between two lines as shown here. Target sentence-pairs were matched as closely as possible for length (i.e. number of characters). A "Null" context condition (in which target sentences were not preceded by any co-text) was included in order to replicate the effects found by Rayner et al. (1983). There were six versions of each example (corresponding to the three levels of Context crossed with the two levels of Target). The thirty test items were embedded in fifty fillers which bore varying degrees of resemblance to the test items.

4.4.2 Design

The design was a factorial repeated measures design incorporating a Latin Square. This design was used for all subsequent experiments, and thus warrants a slightly extended description here:

Six sets of stimuli were constructed, with only one version of each of the thirty examples appearing in any one given set. Each stimulus set was presented to a different group of subjects. For instance, subjects in Group A would see basic examples 1 to 5 in condition ">1.NP", examples 6 to 10 in "1.NP", etc. Subjects in Group B, on the other hand, would see basic examples 1 to 5 in condition "1.NP", examples 6 to 10 in "1.VP", etc. Thus, while no

subject saw more than one version of the same example, each subject was exposed to all conditions (1.NP, 1.VP, >1.NP, >1.VP, 0.NP, 0.VP). Accordingly, each example was represented equally in each one of its six versions.

4.4.3 Apparatus and Procedure

The same apparatus and procedure were used for all the reading-time experiments reported here. What follows is an extended description which will apply to all subsequent experimental sections.

A BBC microcomputer was used both to present the materials and to time and record subjects' responses.

All materials were presented one sentence at a time on a VDU screen. In those cases where the length of a sentence exceeded 80 characters (the maximum permitted on one line of the screen) the sentence was presented on two lines. None of the target sentences ever exceeded 80 characters.

Subjects were instructed to press a button as soon as they had read and understood the sentence presented on the screen. This caused the next sentence in the sequence to appear. For Experiment 1 only, subjects were warned that some "stories" might only be one sentence long (the "null-context" condition). At the end of each sequence, subjects were presented with a simple yes/no question (cf. 4.3) in order to test, and hence encourage, comprehension.

(4.3) Did the burglar find a jewelry box?

4.4.4 Subjects

Forty-two members of the University of Edinburgh were paid to participate in the experiment. All were native speakers of English and naive as to the purpose of the experiment.

4.4.5 Results

Reaction times falling more than 1.96 standard deviations away from the mean were replaced by the critical cut-off value. Less than 4.4% of the data points were affected by this procedure. The results are summarised in Table 4.1 and Figures 4.1 and 4.2.

Two-way Analyses of Variance (henceforth ANOVAs) by subjects and by materials, were performed on the data (see Clark, 1973). In the following analyses, the Context dimension is transformed into one of Plausibility (defined in terms of felicity - cf. Figure 4.2): for the NP-attached targets, the NP-inducing co-text is felicitous

TABLE 4.1

READING TIME PER SENTENCE (IN CENTISECONDS)

	Context		
Target	1	>1	0
VP-attachment	293.1	300.9	321.7
NP-attachment	289.1	266.1	344.8

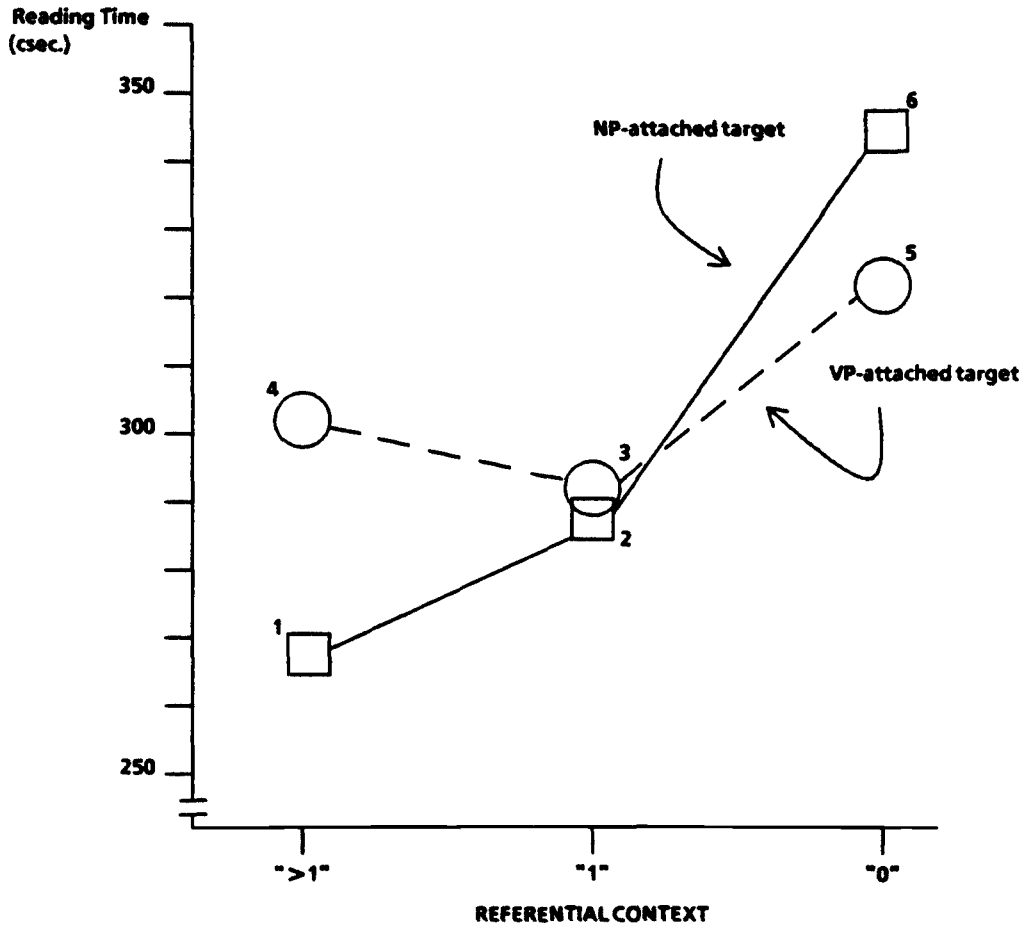


FIGURE 4.1

Mean reading times in centiseconds for Experiment 1 under three conditions of referential context: NP-inducing (">1"), VP-inducing ("1"), and null ("0").

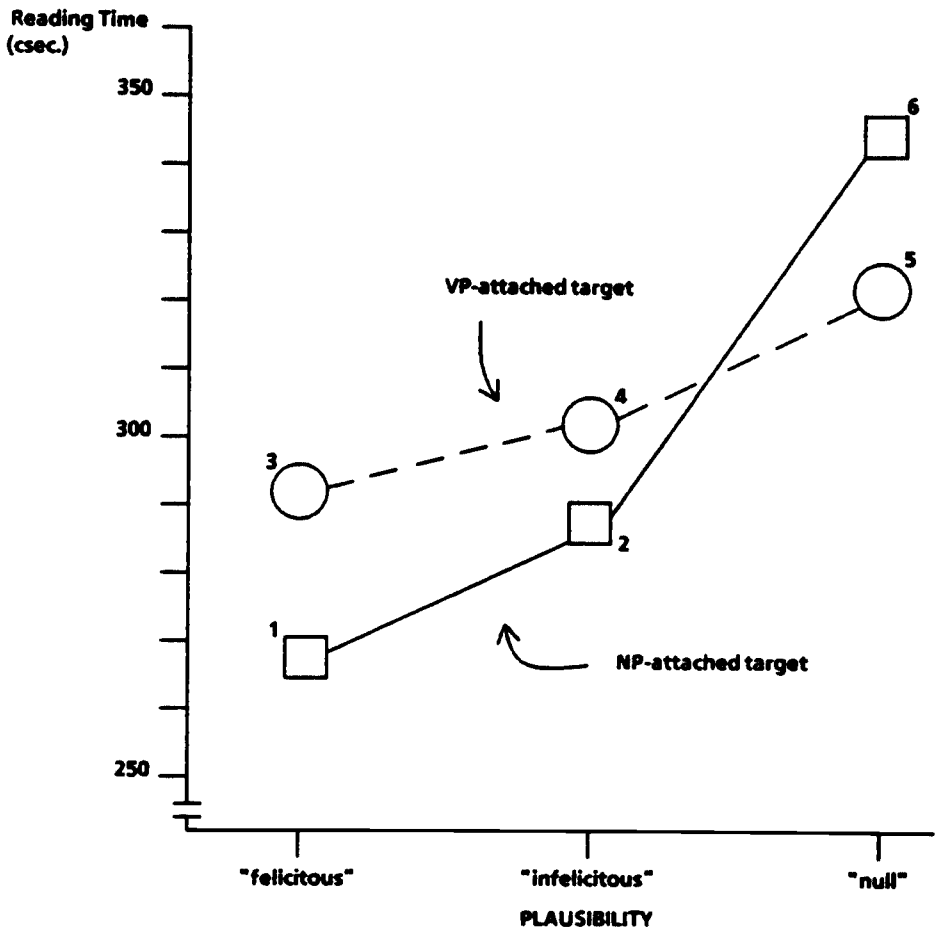


FIGURE 4.2

Mean reading times in centiseconds for Experiment 1 under three conditions Plausibility: "felicitous", "infelicitous", and "null".

and the VP-inducing co-text infelicitous; for the VP-attached targets, the VP-inducing co-text is felicitous and the NP-inducing co-text infelicitous. A main effect of Plausibility thus corresponds to what would have been an interaction between Context (as defined by the number of candidate referents introduced) and Target.

A strong main effect of Plausibility was found: $F_1(2,72)=56.56$, $p<0.0001$; $F_2(2,58)=26.15$, $p<0.0001$; Min $F'(2,106)=17.88$, $p<0.005$. There was also a strong Plausibility x Target interaction: $F_1(2,72)=11.33$, $p=0.0001$; $F_2(2,58)=8.26$, $p=0.0007$; Min $F'(2,121)=4.78$, $p<0.05$. There was no main effect of Target ($F_1(1,36)=1.42$, $p=0.2405$; $F_2(1,29)=0.79$, $p=0.3826$). Multiple Scheffé tests showed the following (where numbers in brackets refer to the numbered points on Figures 4.1 and 4.2: it is recommended that the reader refer to the Figures during the following description of the results).

1. (6)"0.NP" is significantly greater than (5)"0.VP" ($p<0.005$).

2. Both (6)"0.NP" and (5)"0.VP" are significantly greater than all other points ($p<0.005$).

3. (2)"1.NP" is greater than (1)">1.NP" ($P<0.005$).

4. (4)">1.VP" is not greater than (3)"1.VP" ($P>0.1$), though it is just greater than (2)"1.NP" ($p<0.05$).

4.5 Experiment 1: Discussion

There are three features of the results which warrant discussion. First, when target sentences were not preceded by any context (the "null" condition) reading times were significantly slower than when there was a preceding context. But apart from being slower, the basic effects reported by Rayner et al. (1983) were replicated in the null context: the minimal (VP) attachments were processed faster than the nonminimal (NP) attachments. Second, as Figures 4.1 and 4.2 illustrate, the cardinality of the candidate referent set does play a part in determining reading times to the (nonminimally) NP-attached target sentences. And third, there is no statistical effect of the cardinality of the referent set on the (minimally) VP-attached target sentences.

Discussion of these results will be in two parts: first, we shall consider their implications for the theory of Minimal Attachment, and second, we shall consider their implications for Referential Success.

4.5.1 Minimal Attachment in Experiment 1

Is the proposal that Minimal Attachment always governs the initial choice of analysis still tenable? It is certainly the case that in the null-context condition Minimal Attachment does seem to explain the differences which were found: the nonminimally-attached targets took longer to process than the minimally-attached targets. But how does one explain the fact that all reading times were

significantly reduced once a context was provided?[5] Could this finding be reconciled with a theory of Minimal Attachment?

Context as an overriding factor

How might we go about attempting such a reconciliation? We could start by supposing that the pattern of results is influenced only by the resolution process, and that Minimal Attachment does govern the initial choice of the HSPM but only in the null context: it is overridden when contextual information is present. This is implausible, however, if we consider the information on which parsing decisions would be based under the two conditions (null context vs. explicit context).

If Minimal Attachment only operates in the null-context condition, then the local ambiguity is resolved in this condition on the basis of structural information alone. In conditions where a context is provided, Minimal Attachment would not operate, and instead the resolution process would be based on information derived from knowledge about the preceding discourse. Such pragmatic information must surely be more complex than the structurally-derived information over which Minimal Attachment operates in the null context conditions. But how could it be that of two kinds of decision process (one for null contexts and one for explicit

[5] See also Holmes (1977) who reports a similar effect. She found that reading times to subject/object relatives were decreased when they were preceded by a co-text.

contexts), the one which requires less information (structurally derived) is found to take longer than the one which requires more information (pragmatically derived)?

So one argument against Minimal Attachment is that it cannot per se account for the relationship between the conditions in which no contextual information is explicitly provided and those in which such information is provided.

Other arguments against the operation of Minimal Attachment can be found by considering the relationship between the target sentences in the two context conditions: the minimally VP-attached targets took at least as long to process as their nonminimally NP-attached counterparts.[6] If Minimal Attachment were in operation, how could we explain these results? The only coherent explanation would be that the initial analysis is chosen according to structural criteria but that subsequent processing masks the predicted effects on reading times. In other words, perhaps subjects garden-pathed on the nonminimally NP-attached targets under both context conditions, but for some reason or other, reading times to the minimally VP-attached targets were still longer. But what might these subsequent masking processes be? There are at least two possibilities: first, they might be the interpretive processes which cause the implausible

[6] A separate ANOVA on just the four context conditions yielded a significant effect of Target (Min $F' < 0.05$). The minimally attached targets actually took longer to process than their nonminimally attached counterparts.

analysis to be abandoned and the more plausible analysis to be attempted (these would be the processes which are responsible for what Rayner et al. called Thematic Selection see Section 2.4). Second, they might be interpretive processes which occur after the final analysis has been chosen.

On the masking effects of Thematic Selection

Is is plausible to suggest that the initial analyses were selected according to Minimal Attachment, but that any predicted effects on reading time were in fact masked by the processes responsible for the selection of the final analyses? First of all, it is hard to see how such a proposal could explain why the minimally VP-attached targets, which required no reanalysis, took longer than the nonminimally NP-attached targets which did require reanalysis. And second, this suggestion cannot explain the difference in reading time found for the (nonminimally) NP-attached targets under the two context conditions. According to Minimal Attachment, the VP-attachment will be attempted first. The NP-attached target is reproduced below.

(4.4) NP-(nonminimally)-attached

The burglar blew open the safe
with the diamonds.

According to the Thematic Selection Hypothesis, the minimal (VP) attachment in (4.4) is tried first and then rejected on grounds of "implausibility" and "real world knowledge": one can't "blow things open with diamonds". But if Thematic Selection does operate in this

way, as Rayner et al. claim it does, then the VP-attachment should be rejected in these cases independently of whether there is one, or more than one, possible referent for the postverbal noun. The cardinality of the referent set should not affect the plausibility of using diamonds to blow open a safe. The results demonstrate, however, that the rejection of the VP-attachment was not independent of the cardinality of the referent set: it took longer to reject the VP-attachment when there was just one candidate referent to the postverbal noun than when there were two candidate referents. Could it instead be the case that to determine the plausibility of one attachment the processor must also consider the plausibility of the other? If plausibility is a relative measure, and if as a result we require that the processor be capable of evaluating the relative plausibilities of different attachments, then the motivation underlying the theory of Minimal Attachment must be questioned: why use Minimal Attachment to determine which analysis to pursue when the processor will in any case have to consider all analyses?

On the existence of post-syntactic masking effects

The alternative suggestion concerning masking effects was that the reading time differences in the four explicit context conditions were caused by "post-syntactic" interpretive processes. Such a view would, however, bring into question all the reading time evidence on which the theory of Minimal Attachment is based. After all, one could then explain the differences which Rayner et al. find in the null context (and which were replicated here) on the basis of these

processes alone. It would not be necessary to invoke Minimal Attachment in order to explain the observed effects.

Rayner et al. suggest that

> "..if pragmatic plausibility governs the initial syntactic analysis assigned to fully ambiguous sentences, then there is no reason to expect a difference in the complexity of ..[the NP and VP attached examples]." (Rayner et al., 1983, p.368)

In order to defend Minimal Attachment given the results of Experiment 1, however, they would have to argue that complexity is not determined only by whether or not the processor has to re-compute its initial analysis. They would have to argue that differences in complexity could also be brought about by differences incurred during the post-syntactic interpretive stage of processing. Thus, they would have to admit that even if pragmatic plausibility did govern the initial analysis assigned to the sentence there might still be differences in complexity between the two targets.[7] The predictions Rayner et al. would be forced to make of their experiment would not therefore distinguish between Minimal Attachment on the one hand, and the interactive Referential Success Hypothesis on the other. There would thus be no grounds on which to defend Minimal Attachment given the present results.

[7] Later experiments will shed more light on why such differences should exist.

4.5.2 Referential Success in Experiment 1

So Minimal Attachment per se seems unable to explain the effects observed in Experiment 1. In this subsection we shall consider whether the account based on Referential Success can satisfactorily account for the results. As in the case of our attempt with Minimal Attachment, we must account for a number of different patterns in the results: first, the null context data; second, the data in the explicit context conditions; and third, the relationship between these two sets of data (i.e. the fact that the null context reading times were significantly slower than those in the explicit contexts).

Referential Success and the null context

Crain & Steedman (1985) discussed the reasons why in the null context subjects should garden path to examples such as (4.5)

(4.5) The horse raced past the barn fell.

We considered their account in Section 3.3. Briefly, it relies on the notion of "extensions" to a model. The active main verb interpretation of the string "the horse raced past the barn" requires fewer extensions to the model than does the reduced passive interpretation. Their account can also be applied to the present class of examples, illustrated in (4.6) below.

(4.6) The burglar blew open the safe with .

The critical difference between the two possible attachments of the prepositional phrase is that the VP-attachment presupposes that there

exists in the hearer/reader's model of the discourse a uniquely identifiable "safe". The NP-attachment presupposes the existence of a set of such safes. Moreover, it presupposes that the prepositional phrase "applies" to a single member of this set (i.e. there is a member of this set which exhibits a certain property which can be described by the proposition conveyed by the prepositional phrase). Crain & Steedman would claim, then, that the VP-attachment violates fewer presuppositions than does the NP-attachment, and is therefore preferred (cf. Crain & Steedman's Principle of Parsimony, discussed in Section 3.3).

On this account, subjects will have garden pathed to the nonminimally NP-attached targets in the null context. Hence the increase in reading times for these targets as compared to the reading times for the minimally VP-attached targets. But why should the reading times in the null context be slower than the reading times in those conditions in which prior co-texts were present? It might be argued that the increased latencies in the absence of any context are due to experimental artefact. The first sentence in a sequence might be read more slowly than, say, the fifth simply because of an order or practice effect. However, several studies have demonstrated the dependence of reading time on, for instance, information (or indeed the lack of information) concerning the antecedents of noun phrases (e.g. Haviland & Clark, 1974; Sanford & Garrod, 1981; Garnham, 1984).

We can explain the increased reading times in the null context as follows: in the null context, an expression such as "the burglar" will require that the discourse model be extended to include an entity which could be described by this expression. In the conditions in which prior co-texts were present, such extensions would not have been required: an entity corresponding to "the burglar" will have been already introduced into the discourse model by the co-text. It would not therefore be necessary to extend the model on encountering this expression in the target sentence. So the target sentences will have required fewer extensions to the model when preceded by a co-text than when in the null context. We can therefore account for the difference in reading times between the null context and prior co-text conditions by assuming that processing time increases as more extensions are required of the discourse model.

So we can account for the relationship between the two targets in the null context, as well as for the relationship between the null context data and the prior co-text data.[8] All that now remains is that we account for the pattern of results in the four explicit context conditions.

[8] We shall modify this account of the null context in Part III when we consider in more detail some of the theoretical aspects of Crain & Steedman's work.

Referential Success and nonminimal (NP) attachments in context

As predicted by the Referential Success Hypothesis (i.e. the hypothesis which states that the Principle of Referential Success does govern the resolution process where applicable), reading times to the NP-attached targets were decreased if a prior context was provided which introduced more than one candidate referent for the postverbal noun phrase. This was expected because with more than one candidate referent, subjects should choose that analysis which provides restrictive information. With just the one candidate referent, however, subjects should need no further specification for the NP and should therefore choose an analysis which presupposes this very fact (i.e. a simple NP and the nonrestrictive (VP) attachment of the prepositional phrase). Such an analysis leads to an implausible state of affairs (concerning the suitability of diamonds as a tool for blowing open safes), and so the alternative analysis is then, and only then attempted. Hence the increase in reading times to the condition in which only one candidate referent was introduced.

But might there not be an alternative explanation for the pattern of results observed for the NP-attached targets? Could the increase in reading times in the VP-inducing co-text be attributable to something other than a garden path? A VP-inducing co-text and NP-attached target are reproduced below:

(4.7) VP-inducing co-text

A burglar carrying some dynamite broke into an heiress's house. Once inside he found a safe and a jewelry box. One of them had some diamonds inside whilst the other had several priceless emeralds.

NP-attached target

The burglar blew open the safe with the diamonds and made off with the loot.

In the co-text of example (4.7), a safe and a jewelry box are introduced, as are some diamonds and some emeralds. But nothing in the co-text tells us whether the safe contains the diamonds and the jewelry box the emeralds, or whether the safe contains the emeralds and the jewelry box the diamonds. We might expect, simply on the basis of the ordering of the items in the co-text, that the diamonds would be assigned to the safe and the emeralds to the jewelry box. If this were not the case, however, we would only be able to resolve this "indeterminacy" on encountering the target sentence. It is at this point in the text that we discover that "the burglar blew open the safe with the diamonds". The target sentence provides us with the new information that the safe contained the diamonds.

The following example does not share this indeterminacy:

(4.8) VP-inducing co-text

A burglar broke into a bank carrying some dynamite and some gelignite. He planned to blow open a safe. Once inside he saw that there was a safe with a new lock and a strong box with an old lock.

NP-attached target

The burglar blew open the safe with the new lock and made off with the loot.

In this case the last sentence in the text does not provide any new information about the safe. So one difference between (4.7) and (4.8) is that the target sentence in (4.7) resolves the "local indeterminacy" of the preceding co-text. The target sentence in (4.8) does not serve to resolve any such indeterminacy. Perhaps, then, the effect of context on the NP-attached targets in Experiment 1 can be attributed just to this difference.[9]

Stenning (1984), in a paper on the resolution of transient indeterminacy, describes an experiment in which he collected reading times for sentences in texts which, like (4.7), were locally indeterminate. He found that reading times for a sentence were significantly increased if that sentence resolved the indeterminacy in the preceding text. Stenning's results suggest that the increase in reading times to the NP-attached targets in the VP-inducing condition may have resulted from the fact that the target sentence

[9] Only the VP-inducing conditions are locally indeterminate. In the NP-inducing condition, two safes are introduced in the co-text, and there is thus no contrast, as in the VP-inducing condition, between whether the diamonds were contained by a safe or by something else (the jewelry box in the VP-inducing condition).

resolved the indeterminacy introduced in the co-text. In other words, the effect of context in Experiment 1 need not have been an effect of just referential context. There is a clear need for a replication of Experiment 1 in which this indeterminacy is removed. In Section 4.7 we shall explore whether or not the effect of referential context remains once the local indeterminacy of the texts used in Experiment 1 is removed.

Stenning's results suggest of Experiment 1 that the effect of context on reading times to the NP-attached targets did not necessarily stem from referential factors. The Referential Success Hypothesis is nonetheless compatible with these results in a way that the Minimal Attachment hypothesis is not. In the next subsection we shall consider whether the lack of any effect of context on the VP-attached targets is also compatible with the Referential Success Hypothesis.

Referential Success, VP-attachment and felicity conditions

Crain's results, found with the relative/complement examples, suggested that a context which was felicitous for one analysis was infelicitous for the other. It can be seen from Figure 4.2 that in Experiment 1 on the PP-examples, this was not the case: although there appears to be an effect of "felicity" (in terms of cardinality of the referent set) on the NP-attached examples, this same notion of felicity does not seem to apply to the VP-attached examples. What is infelicitous for NP-attachment is not felicitous for VP-attachment. What are the differences between the relative/complement examples of

Crain's experiments and the PP-examples here which could explain this asymmetry? One of the relative/complement examples is reproduced below.

(4.9) The psychologist told the woman that he was worried about her husband.

(4.10) The psychologist told the woman that he was worried about to leave her husband.

Why was it that with these examples what was felicitous for the relatives was infelicitous for the complements, and vice versa? As suggested earlier, the presence of a restrictive relative clause should presuppose the existence of more than one woman to whom the expression "the woman" can refer. The complement clause analysis, which contains an unrestricted noun phrase, presupposes just one woman. The propositional content of the complement clause itself (the "message" told to the unique woman) carries with it no presuppositions with respect to the contents of the hearer's mental model of the discourse. In other words, the relative clause target is felicitous with respect to a "2-women" context and infelicitous with respect to a "1-woman" context. The complement clause target, which contains a simple noun phrase, is felicitous with respect to a 1-woman context and infelicitous with respect to a 2-women context.

It should be clear, therefore, that with the relative/complement examples, what is felicitous for one target (where "felicity" is determined by the cardinality of the candidate referent set to "the woman") will be infelicitous for the other target, and vice versa.

The next question is why this was not the state of affairs with the PP-examples.

Use of the restrictive NP-attachment for a prepositional phrase presupposes, as does the relative clause target, that there is more than one candidate referent available for the referring expression "the safe". Use of the nonrestrictive VP-attachment presupposes, as does the complement clause target, that there is just one candidate referent for this expression. Is it the case, though, that as with the complement clause nothing else is presupposed with regard to information already contained within the model? In order to answer this question, we should consider what it "means" to modify a verb phrase with a prepositional phrase.

When we modify a verb phrase with a prepositional phrase we are providing extra information about the "action" denoted by the verb. In the example above, the burglar didn't just "blow open the safe", he "blew it open <u>with the dynamite</u>". In modifying the verb, the phrase "with the dynamite" is, in these examples, providing information about the <u>instrument</u> with which the action denoted by the verb was carried out. There is a sense in which it serves to "restrict" the class of action being referred to. The prepositional phrase is used to tell us something new about the action "blow open". This in turn presupposes that the hearer/reader already knows something about the blowing open. Use of the VP-attachment presupposes, therefore, that the action denoted by the verb "blew open" is Given.

As can be seen from the "VP-inducing" co-text used in Experiment 1, such presuppositions were not satisfied.

(4.11) VP-inducing co-text ("1")

A burglar carrying some dynamite broke into an heiress's house. Once inside he found a safe and a jewelry box. One of them had some diamonds inside whilst the other had several priceless emeralds.

Target:

The burglar blew open the safe with the dynamite and made off with the loot.

There is no mention of "blowing open" here prior to the final target sentence. If it is the case that the VP-attachment presupposes that the action denoted by the verb be Given, then the following slightly modified co-text should be felicitous with its target.

(4.12) Co-text:

A burglar carrying some dynamite broke into an heiress's house. He planned to blow open a safe. Once inside he found a safe and a jewelry box. One of them had some diamonds inside whilst the other had several priceless emeralds.

Target:

The burglar blew open the safe with the dynamite and made off with the loot.

There is just one difference between the co-text presented here and the earlier versions: the action "blow open" is known about in advance of the target sentence (i.e. is Given).

On the assumption that the VP-inducing co-texts used in Experiment 1 were not felicitous with VP-attachment, it is not

surprising that no effect of "felicity" was found. In the condition in which there was just one candidate referent to the postverbal noun (and which was therefore supposedly felicitous with the VP-attachment), any expected decrease in reading time to the VP-attached target may have been offset by an increase in reading time due to the actual infelicity of the VP-attachment. There are at least two accounts for this increase. One explanation is that it arose because of a garden-path effect: the NP-attachment may have been attempted first. The second, and more likely, is that the VP-attachment was attempted first, but that there then occurred a subsequent increase in processing which was required in order to overcome the infelicity of the attachment.

If the felicity conditions underlying the VP-attachment had been satisfied, then we could have expected to observe an effect of referential context on the VP-attached targets. Clearly, a second experiment is required in which these felicity conditions are manipulated. In Section 4.7 we shall consider such an experiment, and find that referential context can affect reading times to VP-attached targets.

4.6 Experiment 1: Summary

Experiment 1 aimed to contrast the claims of Crain & Steedman (1985) with those of Rayner et al. (1983). Rayner et al.'s effects were replicated in the null context conditions: nonminimally attached targets induced significantly longer reading times than did their minimally attached counterparts. Reading times for both targets in

this condition, however, were significantly longer than reading times to these targets when preceded by explicit contexts. Both these effects were explained within the framework provided by Crain & Steedman. Minimal Attachment per se could not account for the reduction in reading times when targets were preceded by co-texts.

Minimal Attachment was also shown to be unable to account for the observed effects of referential context on targets which were pragmatically biased (according to Rayner et al.'s criteria) towards the NP attachment. It was pointed out that this effect of context on the NP-attached targets did not necessarily stem from referential factors: the two texts (i.e. co-text and target) differed not only with respect to the cardinality of the target referent set, but also with respect to relations that held between the objects introduced prior to the target sentence (one text was "locally determinate" - there was a unique relationship between the objects - and the other text was "locally indeterminate" - there was no such unique relationship). This possibility is tested in the next experiment.

Contrary to initial expectations, no effects of referential context were found on the examples which were pragmatically biased (again, according to Rayner et al.'s criteria) towards the VP attachment. It was suggested that this lack of effect arose because the presuppositions implicated by the attachment of a prepositional phrase to a verb phrase had not been satisfied. Again, this possibility is tested in the next experiment.

4.7 Experiment 2: Introduction

There are two points to arise from the discussion of Experiment 1 which suggest that a further experiment is required: first, the effect of context on the NP-attached targets may have been confounded with the determinacy of the texts. Second, the lack of any effect of context on the VP-attached targets may have arisen because the felicity conditions underlying the VP attachment had not been fully satisfied. In this section we shall reconsider the materials which, following preceding discussion, could be used in a replication of Experiment 1.

As suggested earlier, the indeterminacy of the texts used in Experiment 1 can be removed as illustrated by the following examples.

(4.13) Once inside he found a safe and a jewelry box.
One of them had some diamonds inside whilst
the other had several priceless emeralds.

(4.14) Once inside he saw that there was a safe with
a new lock and a strong box with an old lock.

The text in (4.13) does not uniquely specify the relationship between the various objects (either the safe contains the emeralds and the jewelry box the diamonds, or vice versa). Example (4.14) does not share this same indeterminacy (the safe and the new lock are paired, as are the strong box and the old lock). The NP-inducing version of (4.14) would be

(4.15) Once inside he saw that there was a safe with a new lock and a safe with an old lock.

If an effect of referential context on the NP-attached targets can be found using materials similar to (4.14) and (4.15), then we cannot attribute the reading time differences to differences in the determinacy of the two sets of experimental texts.

With regard to effects of referential context on the VP-attached targets, we were left at the end of Experiment 1 with the following claim: the VP-attachment is felicitous if just a single candidate referent can be found for the postverbal noun phrase and if the action denoted by the verb is Given (i.e. known about by the subject in advance of the target sentence). In this regard, the following co-text/target pair should be felicitous:

(4.16) Co-text:

A burglar carrying some dynamite broke into an heiress's house. He planned to blow open a safe. Once inside he found a safe with a new lock and a strong box with an old lock.

Target:

The burglar blew open the safe with the dynamite and made off with the loot.

So the minimal change in co-text from "a safe .. and a strong box . (VP-inducing) to "a safe .. and a safe .." (NP-inducing) should only have an effect on the VP-attached targets if the co-texts contain the underlined sentence. This ensures that the action denoted by the verb "blew open" in the target sentence is Given. This in turn has the effect of ensuring that the felicity conditions associated with

the VP-attachment are satisfied when there is just one candidate referent to "the safe".

The preceding suggestion is based on the premise that the prepositional phrase in these examples provides New information about the Given action when it is used to modify the verb phrase. But in giving us new information, it might be providing "contrastive" information: the safe was blown open with the dynamite as opposed to with something else. Whether or not this requires that an explicit contrast be set up in the co-text is unclear. If it did, then the following co-text, and not (4.16) would be felicitous with its target.

(4.17) Co-text:

A burglar carrying some dynamite and some gelignite broke into an heiress's house. He planned to blow open a safe. Once inside he found a safe with a new lock and a strong box with an old lock.

Target:

The burglar blew open the safe with the dynamite and made off with the loot.

In order to explore this possibility, we should present co-texts which differ not just with respect to the cardinality of the set of safes but also with respect to whether or not there is an explicit contrast between the dynamite and anything else.

4.8 Experiment 2

The aim of Experiment 2 was to determine whether the contextual manipulations proposed by Crain & Steedman would have any effect on the reading times to VP-attached targets when the action denoted by the main verb in the target sentence had previously been introduced by the preceding co-text. It is expected that reading times to the VP-attached target should be facilitated if just one candidate referent to the post verbal NP is introduced in the co-text, as compared with the condition in which more than one referent is introduced (which should facilitate reading times to the NP-attached targets). In addition, the experiment examined whether or not the "contrastiveness" of the prepositional phrase had any effect: either one instrument (e.g. dynamite) was introduced in the co-text, or two instruments (e.g. dynamite and gelignite) were introduced. The local indeterminacy of the co-texts from Experiment 1 was removed in order to determine whether an effect of context could still be found on the NP-attached targets.

4.8.1 Materials

Thirty-two examples were constructed, each of which conformed to the format illustrated by the following example. The materials are listed in Appendix B.

NP-inducing co-text/1 instrument:
(2 "objects", 1 "instrument" - "02.I1").

A burglar broke into a bank carrying some dynamite.
He planned to blow open a safe.
Once inside he saw that there was a safe with
a new lock and a safe with an old lock.

VP-inducing co-text/2 instruments:
(1 "object", 2 "instruments" - "01.I2").

A burglar broke into a bank carrying
some dynamite and some gelignite.
He planned to blow open a safe.
Once inside he saw that there was a safe with
a new lock and a strong box with an old lock.

VP-inducing co-text/1 instrument:
(1 "object", 1 "instrument" - "01.I1").

A burglar broke into a bank carrying some dynamite.
He planned to blow open a safe.
Once inside he saw that there was a safe with
a new lock and a strong box with an old lock.

NP-inducing co-text/2 instruments:
(2 "objects, 2 "instruments" - "02.I2").

A burglar broke into a bank carrying
some dynamite and some gelignite.
He planned to blow open a safe.
Once inside he saw that there was a safe with
a new lock and a safe with an old lock.

Target sentences were constructed like the following:

NP-attached target ("np").

The burglar blew open the safe with the new lock
and made off with the loot.

VP-attached target ("vp").

The burglar blew open the safe with the dynamite
and made off with the loot.

Target sentences were presented on one line of the screen, and were not split between two lines as shown here. As in previous experiments, target sentence-pairs were matched as closely as possible for length. There were eight versions of each example (2 Instrument x 2 Object x 2 Target). The design was a factorial repeated measures design incorporating a Latin Square. The thirty-two test items were embedded in forty-eight fillers.

4.8.2 Apparatus and Procedure

The same Apparatus and Procedure were used as in Experiment 1.

4.8.3 Subjects

Sixty-four members of the University of Edinburgh were paid to participate in the experiment. All were native speakers of English and naive as to the purpose of the experiment.

4.8.4 Results

Reaction times falling more than 1.96 standard deviations away from the mean were replaced by the critical cut-off value. Less than 4.6% of the data points were affected by this procedure. The results are summarised in Table 4.2 and Figure 4.3. Three-way ANOVAs (Instrument (2) x Object (2) x Target (2)), by subjects and by materials, were performed on the data. While there were strong main

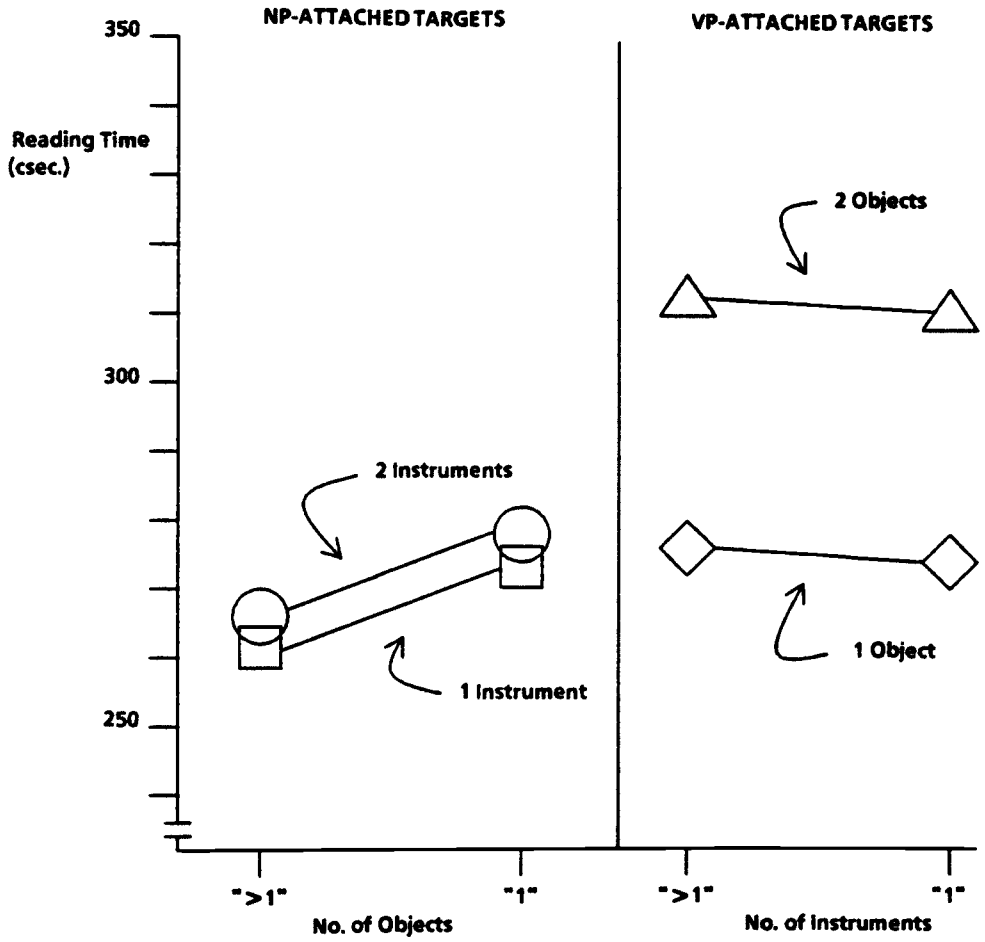


FIGURE 4.3

Mean reading times in centiseconds for Experiment 2. The data are separated according to Target (NP-attached versus VP-attached). Each graph represents four experimental conditions: 2 levels of Object crossed with 2 levels of Instrument.

TABLE 4.2

READING TIME PER SENTENCE (IN CENTISECONDS)

NP-attached target

	Instrument	
Object	2	1
2	265.0	262.0
1	276.8	272.7

VP-attached target

	Instrument	
Object	2	1
2	313.4	310.9
1	275.5	275.2

effects of Target ($F_1(1,56)=56.05$, $p<0.0001$; $F_2(1,31)=27.00$, $p<0.0001$; Min $F'(1,60)=18.22$, $p<0.005$) and Object ($F_1(1,56)=85.74$, $p<0.0001$; $F_2(1,31)=41.51$, $p<0.0001$; Min $F'(1,60)=27.97$, $p<0.005$), there was no main effect for Instrument ($F_1(1,56)=0.98$, $p=0.3254$; $F_2(1,31)=0.90$, $p=0.3491$). This is clearly seen in Figure 4.3. There was also a Target x Object interaction: $F_1(1,56)=22.53$, $p<0.0001$; $F_2(1,31)=10.77$, $p=0.0026$; Min $F'(1,60)=7.29$, $p<0.01$.

Given that there was no effect of Instrument, the data were collapsed across the Instrument condition. The new figures are given in Table 4.3 and shown in Figure 4.4. Two-way ANOVAs (Object x Target), by subjects and by materials, were performed on the data. A strong main effect of Object was found: $F_1(1,56)=85.74$, $p<0.0001$; $F_2(1,31)=41.51$, $p<0.0001$; Min $F'(1,60)=27.97$, $p<0.005$. A strong main effect of

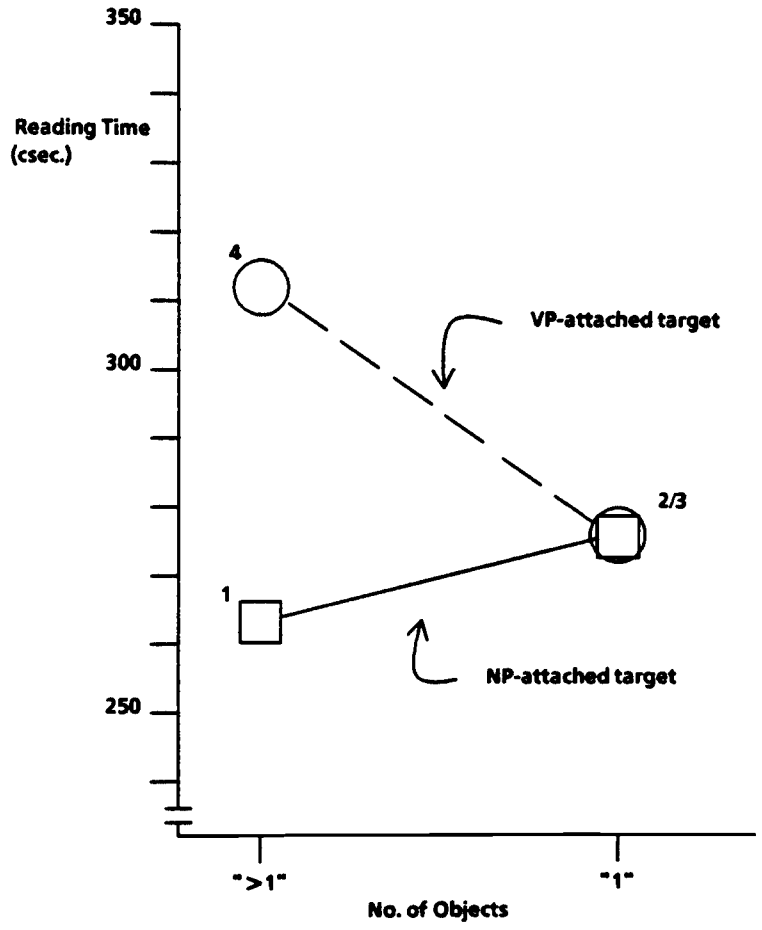


FIGURE 4.4

Mean reading times in centiseconds for Experiment 2, collapsed across instrument, under two conditions of referential context ("No. of Objects"): ">1" versus "1".

TABLE 4.3

READING TIME PER SENTENCE (IN CENTISECONDS)

	Object	
Target		2
VP-attachment	275.3	312.1
NP-attachment	274.8	263.5

Target was also found: $F_1(1,56)=56.05$, $p<0.0001$; $F_2(1,31)=27.00$, $p<0.0001$; Min $F'(1,60)= 18.22$, $p<0.005$. There was also a strong Object x Target interaction: $F_1(1,56)=22.53$, $p<0.0001$; $F_2(1,31)=10.77$, $p<0.0026$; Min $F'(1,60)=7.29$; $p<0.005$.

Multiple Scheffé tests showed the following (where "1" indicates 1 object, and ">1" indicates more than one object). As before, numbers in parentheses refer to the numbered points in Figure 4.4

(1) (4)">1.VP" was significantly greater than all other points ($p<0.005$).

(2) (3)"1.VP" and (2)"1.NP" were not significantly different from one another ($p>0.1$).

(3) (2)"1.NP" was significantly greater than (1)">1.NP" ($p<0.005$).

4.9 Experiment 2: Discussion

The essential difference between the results of Experiment 2 and those of Experiment is that this time effects of referential context were found for both kinds of target sentence. As we had predicted, a strong effect of referential context on the nonrestrictive VP attachment was found once the action denoted by the modified verb was Given. In Experiment 1 the action had not been Given, and so the "VP-inducing" co-text had not in fact been felicitous with the VP attachment.

It is not clear, however, whether the VP-inducing co-text was entirely felicitous with the VP-attached target. It may have been felicitous with respect to referential context, but it may not have been entirely felicitous with respect to other pragmatic factors. For instance, the VP-attached target preceded by the VP-inducing co-text can be interpreted as violating Grice's conversational maxim of quantity (Grice, 1975): make your contribution to a conversation as informative as is required, but no more than necessary. In the co-text, subjects are told that the burglar has some dynamite, that he plans to blow open a safe, and that there is indeed a safe in the bank. They are then told in the target that he blows open the safe with the dynamite. This is perhaps being over informative. The point here is that pragmatic violations of this kind may contribute to increasing reading times to the VP-attached target in the VP-inducing condition. This in turn might account for the observation that the VP-attached targets induced, overall, slower reading times

than the NP-attached targets.[10]

But aside from an effect of referential context on both targets, Experiment 2 also demonstrates that the processing of VP-attached prepositional phrases (as measured via reading times) is not facilitated by setting up an explicit contrast in the co-text between one instrument with which to modify the action denoted by the verb and another. Also, the effect of context which was found in Experiment 1 for the NP-attached targets was not artefactual on the indeterminacy of the co-texts used in that experiment. We would not have expected to find an effect of referential context in Experiment 2 if the reading time differences in Experiment 1 had been determined only by differences in the determinacy of the co-texts. We do see from Figure 4.5 below, however, that there was a smaller effect of referential context ("Object" in Experiment 2) in this experiment (11.3 csec.) than in Experiment 1 (23 csec.).[11] This decrease in magnitude seems to result from a decrease in reading times to the "1.NP" condition, which is just that condition for which the target sentence in Experiment 1 resolved the local indeterminacy of the preceding co-text. This suggests that the indeterminacy of the co-texts used in Experiment 1 did contribute to (but was not solely responsible for) the differences in reading times to the NP-attached

[10] My thanks to Gisela Redeker for pointing this out.

[11] Figure 4.5 should be treated with a certain degree of caution. In Chapter 5 we shall consider some of the dangers of superimposing the results of two separate experiments.

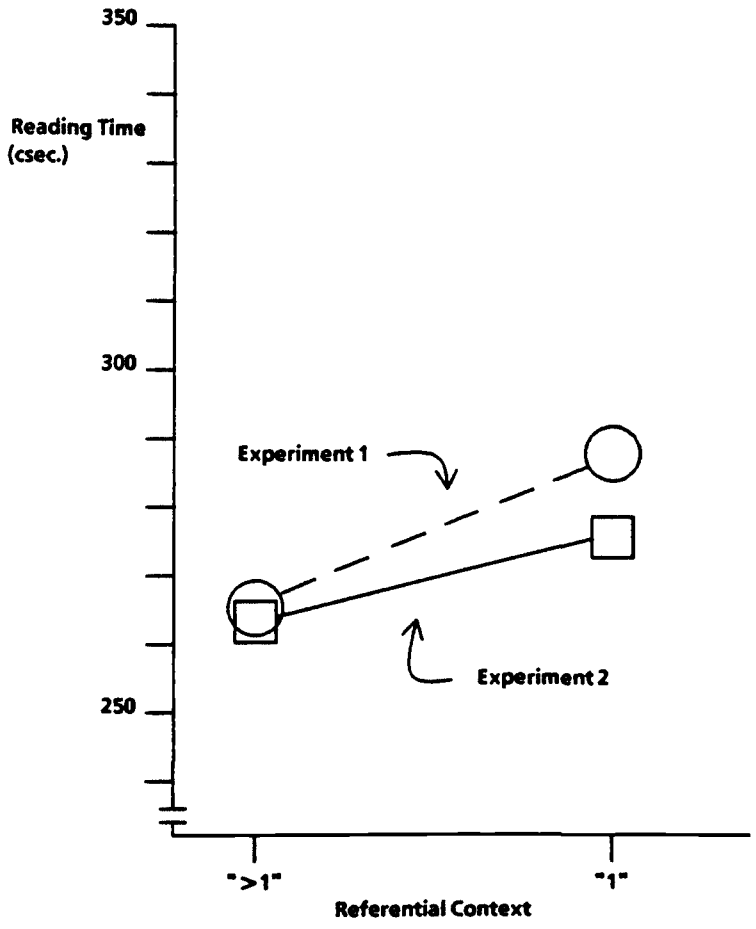


FIGURE 4.5

Mean reading times in centiseconds to the NP-attached targets under two conditions of referential context: ">1" versus "1". The data for Experiments 1 and 2 are superimposed.

targets.[12]

So the results of Experiment 2 support the predictions made during the discussion of Experiment 1. If Minimal Attachment did nonetheless guide the operation of the HSPM, then the effect of referential context found on the VP-attached targets would have to be attributed to context-sensitive processes. We considered in regard to the results of Experiment 1 two possibilities concerning the nature of these processes. First, they might be the processes responsible for the rejection of the initial minimally-attached structure assigned to the sentence in favour of the alternative nonminimally-attached structure; and second, they might be interpretive processes which occur only after the final (nonminimal) structure is chosen.

With regard to the first of these suggestions, we concluded that there could be no motivation for a model in which Minimal Attachment selected the initial hypothesis but in which the alternative

[12] The reading time task used by Stenning (1984), involved the use of texts such as

(4.17) There are just two objects. There is a small black. There is a small square. There is a large black. There is a large circle.

(In this example, the indeterminacy is resolved by sentence 4, from which we can infer that the small black and the small square are not the same object.) The discussion on NP attached prepositional phrases and determinacy suggests that the effects Stenning found have been "replicated", but using a task which involved more "natural" texts, and which, from the subjects' point of view, is less "problem-solving" oriented and hence less likely to have led to the operation of "artefactual" processing strategies.

hypotheses would in any case have to be examined in considering the plausibility of that first choice. It is unclear what the computational saving of such a procedure might be. With regard to the second of these suggestions, we concluded that it would not be possible to distinguish empirically between the predictions of a model which exhibited arbitrary post-syntactic "masking" effects, and any other kind of model.

We can only conclude that the present results provide no evidence for the existence of a Minimal Attachment strategy.

4.10 Summary and Conclusions

Rayner et al. (1983) concluded on the basis of their experimental results that semantic/pragmatic information does not affect the HSPM's initial choice of which analysis to pursue when more than one analysis is permissible. It was suggested at the beginning of this chapter that Crain's (1980) own results, which suggested the opposite, did not constitute sufficient justification for the outright rejection of Rayner et al's claim. In order to test both Crain & Steedman and Rayner et al's claims more fully, a replication of the Rayner et al. study (using reading time as the dependent measure) was attempted in which the target sentences were preceded by minimally-differing co-texts. These co-texts differed to the extent that they introduced either one candidate referent to the postverbal referring expression in the target sentence, or they introduced two such candidates.

Experiment 1 included a "null" context condition for which the results reported by Rayner et al. (1983) were replicated: the minimally-attached targets induced significantly shorter reading times than did the nonminimally-attached targets. Reading times for both targets in this condition, however, were significantly longer than reading times to these targets when preceded by contexts. It was argued that these results were inconsistent with the Minimal Attachment Hypothesis proposed by Rayner et al. (1983). Also inconsistent with this hypothesis was the effect of referential context which was found for the nonminimally-attached target sentences.

As far as the Referential Success Hypothesis was concerned, this first experiment was only partially successful: although an effect of referential context was found for the nonminimally attached targets, no such effect was found for the minimally attached targets. Moreover, it was suggested that even the effect of context on the nonminimally-attached targets might have been artefactual on differences between the texts which were not referential in nature. It was suggested that a second experiment should be run in order to eliminate these additional differences.

We proposed that minimally attached targets showed no context effects because neither co-text was felicitous for the (minimal) attachment of a prepositional phrase to a verb phrase: the action denoted by the verb was not Given.

In a second experiment, Experiment 2, it was ensured that the action denoted by the verb was Given, and this time, strong effects of referential context were found for both the nonminimally-attached targets and the minimally-attached targets. Moreover, the differences between the experimental texts which may have contributed in Experiment 1 to the effect of context on the nonminimally-attached targets were removed. A strong effect of context was still found. Finally, as in the case of Experiment 1, it was found that reading times to the minimally-attached targets were significantly longer than reading times to the nonminimally-attached targets.

We discussed in Sections 4.5 and 4.9 the implications of these results for the Minimal Attachment Hypothesis. According to this hypothesis, the minimally-attached targets should always be processed faster than the nonminimally-attached targets. This was clearly not the case. The hypothesis also seems incompatible with the effects of referential context which were found for both kinds of target. Proponents of the Minimal Attachment Hypothesis would be forced to attribute these effects to comprehension processes which somehow mask the effects which they would expect here. We concluded from these discussions that an explanation based on the existence of such masking effects would either question the computational motivation for a Minimal Attachment strategy, or else would question the empirical value of such a hypothesis, in that the Minimal Attachment Hypothesis would become, empirically speaking, indistinguishable from the alternative Referential Success Hypothesis. It was therefore concluded on the basis of Experiments 1 and 2 that there is no

empirical evidence to support the operation of a Minimal Attachment strategy.

The empirical motivation for the Minimal Attachment Hypothesis seems to be undermined by the results of Experiments 1 and 2 simply because we would need to account for these results by invoking "post-syntactic" masking effects. Such effects could be used to account for any reading time differences that might otherwise be taken as support for the operation of Minimal Attachment. But suppose that we do invoke such effects, would that not also question the empirical motivation for the Referential Success Hypothesis? How do we know that the observed effects of context on reading times to each target sentence stemmed from contextually-induced garden paths? In the absence of any suitable on-line measure it is clear that we do not. What is important, though, is that Referential Success accounts for the data more adequately than Minimal Attachment does. Although experimental data will in general permit a variety of hypotheses, they will also restrict the class of permissible hypothesis. The results of Experiments 1 and 2 seem to exclude Minimal Attachment and other structurally-based strategies (e.g. Wanner, 1980; Ford, Bresnan, & Kaplan, 1983), as plausible hypotheses concerning the resolution of local syntactic ambiguity.

We are still left, however, with the possibility that some of the reading time differences in Experiments 1 and 2 were caused (perhaps only in part) by processes which were unrelated to those responsible for the resolution of local syntactic ambiguity. In the

following chapter, we shall return to the processing of relative clauses, and take into account some early, and often cited, results by Wanner, Kaplan, & Shiner (1974). They essentially collected reading times for globally ambiguous sentences which contained a "that"-clause which could be interpreted either as a complement clause or as a relative clause (cf. the examples used by Crain, 1980). They made two observations: first, that the ambiguous that-clause tends to be interpreted most often as a complement clause; and second, that even when interpreted as a relative clause, reading times are still longer than when interpreted as a complement clause. We shall consider in Chapter 5 whether we can account for this residual difference between the two kinds of clause, and whether we can thereby shed any further light on the processes which must be involved during the comprehension of the kinds of example with which we have so far been concerned. In particular, we shall consider not just why particular syntactic constructions are used, but also what kinds of process (concerned with the accessing and maintenance of the discourse model) are demanded by their use.

CHAPTER 5

Referential Context, Relative Clauses, and Inference

Crain (1980) and Crain & Steedman (1985) were primarily concerned with restrictive relative clauses. Their discussions centred around the presuppositions which are implicated by the use of various kinds of noun phrase. Their approach takes into account just why certain constructions are used during normal discourse. In the preceding chapter we saw that this same approach carries over to the study of other kinds of construction, and in particular, the attachment of prepositional phrases.

In all these discussions we have been concerned primarily with the reasons underlying the use of a particular construction. But associated with this issue is another: what kinds of comprehension process are implicated by the use of a particular construction? In other words, what kinds of process does a speaker elicit in the hearer when he/she uses a particular kind of syntactic construction? We have assumed during previous discussions that sentence comprehension involves integrating the information conveyed by the current sentence into a discourse model. It follows that certain kinds of processing are for accessing and generally "managing" the discourse model. In this chapter we shall consider the kinds of process that would seem to be demanded by the use of relative clauses. In particular, we shall demonstrate that evidence which was previously thought to favour structural accounts of syntactic

ambiguity resolution do not bear on the issue of ambiguity resolution at all.

Many theories of ambiguity resolution use as evidence the observation that relative clauses take longer to process than complement clauses (e.g. Wanner, Kaplan, & Shiner, 1974). In an example like

(5.1) The boy told the girl that he liked the story

the noun phrase "the girl" is preferentially treated as a simple NP, and not as the first NP constituent in a complex NP. Furthermore, when the noun phrase actually is interpreted as complex, these examples take longer to process (via a reading time measure) than when the preferential reading is given. Wanner et al. (1974) and Wanner (1980) model these effects using an ATN, and show that they can be made to arise from the order in which arcs leave certain states. Frazier & Fodor (1978) cite the effects in support of Minimal Attachment, whilst Ford et al. (1982) would predict this effect on the basis of their theory of lexical and syntactic preferences, in which the simple NP expansion must be ordered before the complex NP expansion. We will claim here, however, that the differences found by Wanner et al. (1974) need not have been due solely to garden path effects, and that they therefore do not constitute empirical support for these structural theories.

The chapter is organised as follows: in Sections 5.1 and 5.2 we shall consider a reading time replication of Crain's (1980)

experiment with ambiguous relative/complement clause sentences. We shall see in Experiment 3 that strong effects of referential context are found, but that there is also a residual difference in reading time between relative clauses and complement clauses (cf. Wanner et al.'s result). In Section 5.3 we shall argue that this residual difference is brought about by a difference in the kinds of process that are required when interpreting relative or complement clauses. More specifically, we shall argue that the propositional content of a restrictive relative clause must be matched against the contents of a discourse model, and that it is this matching process which is responsible for the residual differences between relative clauses and complement clauses found in Experiment 3. In Sections 5.4 and 5.5 we shall consider a second experiment on relative/complement clauses (Experiment 4), in which the amount of inferencing required during the matching process is minimised. A further experiment in Sections 5.6 and 5.7 will confirm the results of Experiment 4, namely that the residual differences in reading time between relative clause targets and complement clause targets can be eliminated by minimising the amount of inferencing required during the processing of relative clauses.

5.1 Relative Clauses and Processing Complexity

In Section 4.5.1 we saw that Rayner et al. (1983) commit themselves to a notion of sentence complexity which is merely dependent on whether or not the HSPM is forced to recompute a syntactic analysis. Wanner et al. (1974) took this same approach

and interpreted their results as supporting a structural account of syntactic ambiguity resolution. However, there are many factors other than syntactic reanalysis which, though not necessarily "structural" in origin, may affect the complexity of a sentence. It follows that if reading time is a measure of a sentence's complexity, then it too will be affected by such factors. Just what these factors might be must be considered when using sentence complexity (via, for instance, reading times) to investigate syntactic processing. In Wanner et al.'s (1974) study, it is unclear just how much of the observed differences were due to garden-path effects, and how much might have been due to differences in processing load which were unrelated to these structural effects.

Crain (1980) showed that syntactic preferences can be overcome (or induced) with the provision of adequate contexts by using an incremental grammaticality judgement task (see Section 3.3). The incremental grammaticality judgement task, however, is not necessarily sensitive to sentence complexity: although Crain might have shown that the relative clause analysis can be chosen as frequently as the complement clause analysis once context is controlled for, he cannot say whether the relative clause analysis in nonetheless more complex, in terms of processing load, than the complement analysis. Of course, once context is controlled for, there might not exist any such differences in complexity. Whether or not they would exist is, at present, unknown. Clearly, there is a need to replicate Crain's basic experiment, but using an experimental task which is sensitive to sentence complexity. Global reading time

is such a task.

In the following section, we shall describe such an experiment.

5.2 Experiment 3

The aim of Experiment 3 is to attempt a replication of Crain's (1980) experiment with ambiguous relative/complement sentences. We should find, as in the case of Experiments 1 and 2, that referential context will have an effect on the reading times to each target sentence. On the basis of Wanner et al.'s (1974) results, we should also expect to find an absolute difference in reading time between the relative clause targets and the complement clause targets: the former should induce longer reading times than the latter. If Wanner et al.'s results are merely artefactual on the fact that all the materials in their experiment were presented in the null context, then we might expect to find no effect of target construction in this experiment.

5.2.1 Materials

Twenty-four examples were constructed which all conformed to the following format:[1]

[1] Experiment 3 is replicated by Experiment 5 (see Sections 5.7 and 5.8). Consequently Appendix C lists only the materials used in that experiment.

Complement-inducing co-text (henceforth "1").

A psychologist was counseling a man and a wife.
He was concerned about one of them
but not about the other.

Relative-inducing co-text (henceforth ">1").

A psychologist was counseling two wives.
He was concerned about one of them
but not about the other.

Complement target sentence ("comp").

The psychologist told the woman that he was worried about
her marital problems.

Relative target sentence ("rel").

The psychologist told the woman that he was worried about
to visit him again.

Note that with these materials, the change from "a man and a wife" to "two wives" is the only change which is made in transforming the co-texts from one condition to the other. Thus any differences in reading time for the same target sentence can only be due to this one modification, the effect of which, in theory, is to change the cardinality of the set of "women" introduced into the story from "1" to ">1"

The two versions of each target sentence were matched as closely as possible for length (i.e. number of characters). There were four versions of each example (2 targets x 2 co-texts). The design was a factorial repeated measures design incorporating a Latin square. The

twenty-four test items were embedded in thirty-six fillers which bore varying degrees of resemblance to the test items.

5.2.2 Apparatus and Procedure

The same apparatus and procedure was used as in Experiment 1

5.2.3 Subjects

Thirty-two members of the University of Edinburgh were paid to participate in the experiment. All were native speakers of English and naive as to the purpose of the experiment.

5.2.4 Results

Reaction times which fell more than 1.96 standard deviations away from the mean were rejected and replaced by the critical cut-off value. 5.7% of the data points were affected by this procedure. The results are summarised in Table 5.1 and Figures 5.1 and 5.2.

TABLE 5.1

READING TIME PER SENTENCE (IN CENTISECONDS)

	Context	
Target	1	>1
complement	282.5	310.5
relative	364.3	324.0

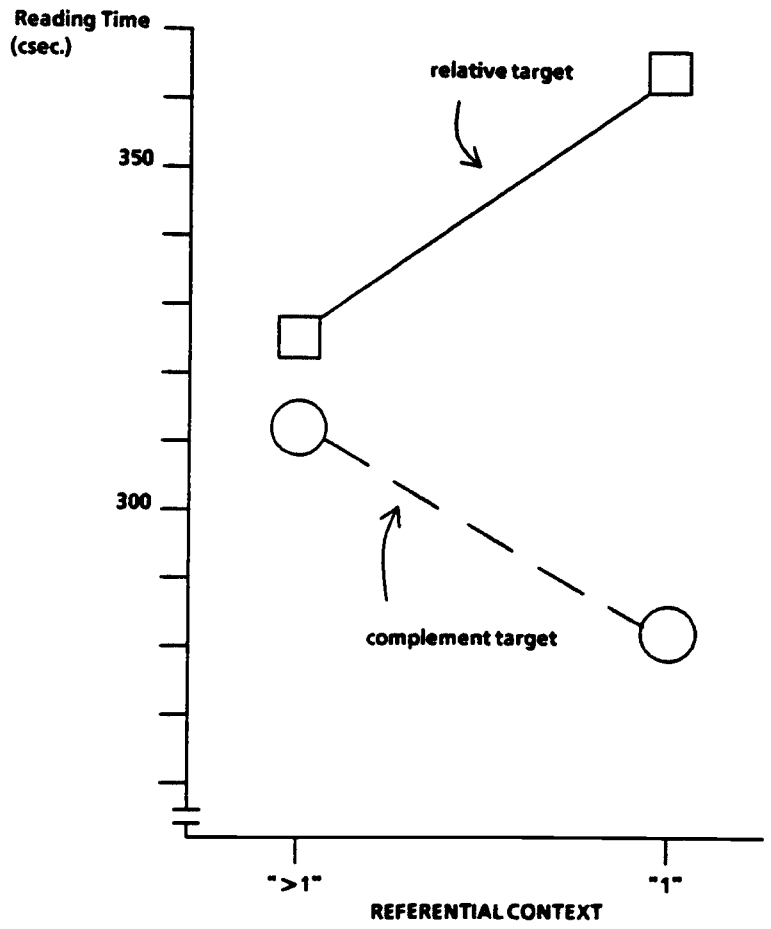


FIGURE 5.1

Mean reading times in centiseconds for Experiment 3 under two conditions of referential context: ">1" versus "1".

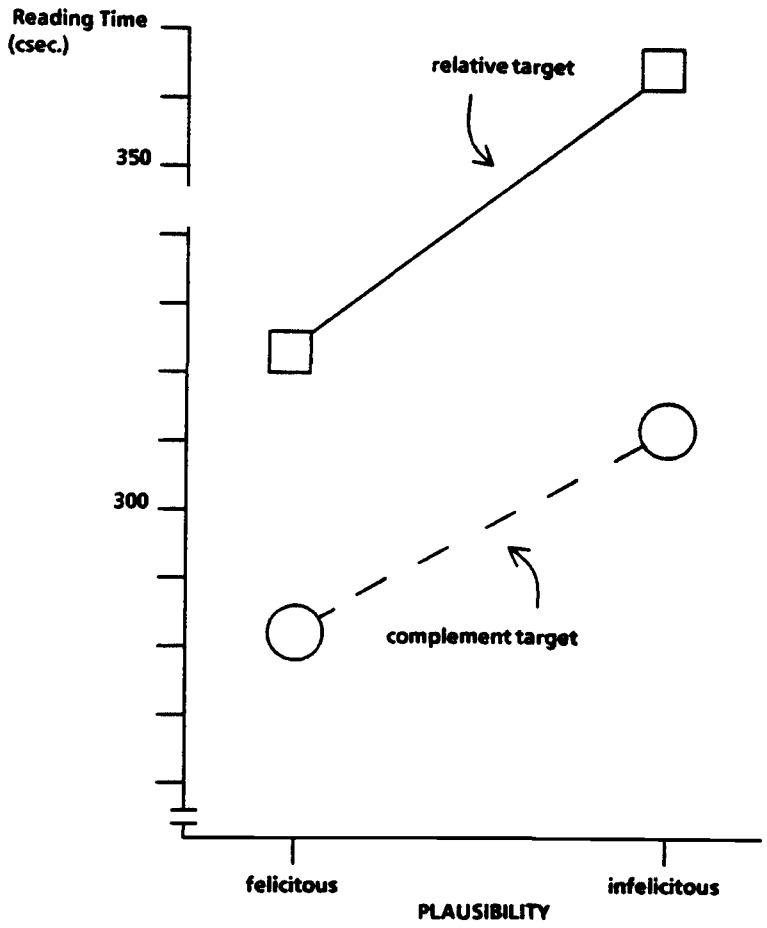


FIGURE 5.2

Mean reading times in centiseconds for Experiment 3 under two conditions of plausibility: ">1" versus "1".

As in the case of Experiment 1, the context dimension was transformed into a Plausibility dimension (defined in terms of felicity - cf. Figure 5.2): for the relative clause targets, the relative-inducing co-text is felicitous and the complement-inducing co-text infelicitous; for the complement clause targets, the complement-inducing co-text is felicitous and the relative-inducing co-text infelicitous. A main effect of Plausibility thus corresponds to what would have been an interaction between Context (as defined by the number of candidate referents introduced) and Target.

Two-way ANOVAs (Plausibility x Target), by subjects and by materials, were performed on the data. A strong main effect was found for Plausibility: $F_1(1,28)=21.09$, $p=0.0001$; $F_2(1,23)=6.11$, $p=0.0212$; Min $F'(1,36)=4.74$, $p<0.05$. A strong main effect of Target was also found: $F_1(1,28)=29.77$, $p<0.0001$; $F_2(1,23)=12.39$, $p=0.0018$; Min $F'(1,40)=8.75$, $p<0.01$. There was no Plausibility x Target interaction ($F_1(1,28)=1.22$, $p=0.2781$; $F_2(1,23)=0.24$, $p=0.6306$). A multiple comparison procedure using Scheffé's test was performed on the four data points: all six possible pair-wise comparisons were found to be significant ($p<0.005$).

5.3 Experiment 3: Discussion

Experiment 3 shows, as might be predicted from the findings of Crain (1980), that the provision of a co-text which is felicitous with the use of a restrictive relative clause does in fact facilitate such use as compared with a non-felicitous co-text. Furthermore, there is associated with this facilitatory effect on relatives, a

corresponding inhibitory effect on the use of complement constructions (see Figure 3.1). However, Experiment 3 also shows that there is a strong effect of target construction: even with co-text (and length) controlled for, relative clauses were found to induce longer reading times than complement clauses.

Although an explanation based on Referential Success can account for the observed effects of referential context, it does nothing to account for the observed effects of Target.

Could these effects be due to the operation of Minimal Attachment? Minimal Attachment would predict that the relative clause targets will always take longer to process than the complement clause targets. This much was found. But if, as Rayner et al. (1983) suggest, sentence complexity is determined only by whether or not the processor has to recompute its initial syntactic analysis, then there should be <u>no</u> effect of referential context on either the relative clause targets or the complement clause targets. Referential context should not induce or prevent garden pathing. We would have to account for the effects of referential context on each target by assuming that they do not reflect differences in the frequency of garden pathing. We would have to assume, if we believed the Minimal Attachment Hypothesis to be correct, that these reading time differences arose during some "interpretive" stage which occurs only after the assignment of an initial structure, the choice of which is not dependent on contextual considerations.

In the preceding chapter the implications of such an assumption for the Minimal Attachment Hypothesis were discussed at length. We concluded that Minimal Attachment and Thematic Selection could not account for the observed effects of referential context. To summarise the argument very briefly: if we assumed that the reading time differences were brought about by non-syntactic factors, then we would have to either question the empirical motivation for the Minimal Attachment Hypothesis (in terms of whether there are any reading time experiments which support the hypothesis), or else we would have to question the computational motivation for the hypothesis (in terms of the computational saving that Minimal Attachment and Thematic Selection would lead to). For the full discussion, refer to Sections 4.5 and 4.9.

The most consistent interpretation of the results is therefore to assume that the observed effect of context is due to referential factors as predicted by the Referential Success Hypothesis. We must also assume, then, that the observed effect of target construction on processing time is somehow dependent on an increased complexity of relative clauses as compared to complement clauses. In the following section, we shall consider the reasons for this increased complexity, and in Sections 5.5 and 5.6 we shall describe an experiment in which these effects of complexity are successfully eliminated.

5.4 Relative Clauses and Inference

Examples (5.2) and (1.6) below differ with regard to their syntactic structure. As described before, each is locally ambiguous, between a relative clause interpretation of the that-clause, and a complement clause interpretation of the that-clause.

(5.2) The psychologist told the woman that he was worried about her marital problems.

(1.6) The psychologist told the woman that he was worried about to visit him again.

More importantly, though, these examples differ in terms of the presuppositions (or "felicity conditions") which are implicated by the use of the two kinds of clause. Furthermore, they differ with respect to the kinds of processing which these presuppositions demand of them.

The "function" of a complement clause, as used in (5.2), is to represent a "message". The content of the message does not presuppose anything with respect to the discourse which has preceded this sentence. In terms of the "Given-New" contract (e.g. Clark & Haviland, 1977) it represents "New" information. The "function" of a restrictive relative clause is to make the noun phrase it modifies specific enough to refer to a single identifiable referent. But to do this, the relative must convey some proposition which is already represented in the model of the preceding discourse. In example (1.6), there must already exist in the model of the discourse

constructed thus far the information which is now being used to distinguish between which of a number of "women" is being referred to. That is, we must already know that the psychologist was worried about a particular woman (although we don't know which woman, we do know that his being worried about one of them is a distinguishing characteristic). The relative clause, if restrictive, contains Given information.[2] It follows that some attempt must be made to "match" the propositional content of the relative with information already available in the discourse model. Fundamental to this (albeit vague) notion of matching is the notion of "Inference".

It is commonly asserted that the comprehension of an utterance relies on a tacit agreement between speaker and hearer concerning the Given-New structure of the language (Halliday, 1967; Clark, 1975; Haviland & Clark, 1977). Given information must be assumed by the speaker to be uniquely identifiable by the hearer, while New information must be assumed by the speaker to be unknown to the hearer. Sanford & Garrod (1981) comment:

> " ... in interpreting any sentence, the listener first identifies the Given and the New, realises that he is expected to know about the Given already, and so searches back in his memory for something to match it. When he finds the matching information, this is then set up as antecedent to the particular piece of Given information in the current sentence." (Sanford & Garrod, 1981. p.94)[3]

[2] Non-restrictive relative clauses, which tend to contain parenthetical information, do not necessarily provide Given information. For the purposes of the present study, we shall limit discussion to restrictive relative clauses.

[3] For further discussion, see Langford & Holmes (1979) and Kopple

Their discussion is framed within the context of anaphoric referring expressions (in particular, noun phrases), and arose in connection with the following pair of sentences:

(5.3) a. Mary unpacked the picnic things.
b. The beer was warm.

(After Haviland & Clark, 1974)

In this case, "the beer" is interpreted as Given information by the hearer, who then attempts to "match" it against a discourse model. Because "the beer" has not been explicitly mentioned before, this matching process (an act of reference) has to rely on a "bridging" inference in order to be successful: beer is an example of a picnic thing, therefore it is reasonable to infer that the antecedent to "the beer" is one of these "things".

It should be clear that there is a sense in which the relative clause in (1.6) above is itself a referring expression: it refers to some proposition represented (perhaps only implicitly) in the discourse model which somehow enables a distinction to be made between one "object" in the model and another. It follows that on encountering the relative, the hearer has to attempt to match it to the discourse model. In so doing, it may be necessary, as in the cases discussed by Haviland & Clark (1974) and Sanford & Garrod (1981), to infer the relationship between the proposition expressed by the relative, and the corresponding proposition represented in the

(1982), who describe a number of experiments which aimed to test this Given-New strategy.

discourse model.

One might then expect a difference in processing time between relative clauses and their complement counterparts simply because of this process of inference which must link the content of the relative to some previously established proposition. No such inferencing is necessary in the case of the complement versions. Thus, although for any one target (complement vs. relative) we might expect a difference in processing time to be induced by using different co-texts (cf. Crain), there might nonetheless exist an absolute difference between the two.

Is there any way in which such a theory could be tested? Consider again one of the "relative-inducing" co-texts (">1") and one of the relative target sentences which were used in Experiment 3.

(5.4) Relative-inducing co-text (>1).

> A policeman was talking to two women claiming to be CIA agents. He was very suspicious of one of them but not of the other.

Relative target sentence.

> The policeman told the woman that he had his doubts about to tell the truth.

In this example (as in all others) the wording of the relative clause does not match the wording of the second sentence in the co-text: "he had his doubts about .. vs. "he was very suspicious of. This disparity between co-text and target was intentional. In the example above, when subjects first encounter the fragment "he was very suspicious of it is followed by an object ("one of them"). If

they were to encounter it a second time, they might, on the basis of their previous experience of the fragment, "expect" an object. This would predispose subjects towards a complement analysis of the ambiguous portion of the target (in which the fragment is followed by an object) and away from the relative analysis (in which the fragment is not followed by an object). So we might expect a "syntactic carry-over effect" if the ambiguous target fragment contains the same wording as was used in the preceding co-text.

The original question being addressed here was whether we could test the hypothesis that differences between relatives and complements are due to the inferencing which necessarily accompanies the use of a relative. If we were to disregard the possibility of syntactic carry-over effects, then we could test the hypothesis by using the same sequence of words in the target as were used in the preceding co-texts. This should have the effect of minimising the inferencing required in interpreting the relative clause. The example given above would thus become:

(5.5) Relative-inducing co-text (>1).

> A policeman was talking to two women claiming to be CIA agents. He had his doubts about one of them but not about the other.

Relative target sentence.

> The policeman told the woman that he had his doubts about to tell the truth.

If such a modification does minimise the required inferencing, and if it is this inferential process which is responsible for the

difference between relatives and complements as found in Experiment 3, then we would expect to find a smaller effect of target construction on reading time with these examples.

5.5 Experiment 4

The aim of Experiment 4 is to test the hypothesis that the effect of target construction found in Experiment 3 can be reduced by minimising the inferencing which is associated with the analysis of the relative clause targets.

5.5.1 Materials

The twenty-four examples from Experiment 3 were all modified as discussed above.[4] As with Experiment 1, target sentence-pairs were matched as closely as possible for length. There were four versions of each example (2 co-texts x 2 targets). The design of the Experiment was a factorial repeated measures design incorporating a Latin Square (see Experiment 1). The twenty- four test items were embedded in thirty-six fillers.

5.5.2 Apparatus and Procedure

The same Apparatus and Procedure were used as in Experiment 1.

[4] Experiment 4 is replicated by Experiment 5 (see Sections 5.7 and 5.8). Consequently Appendix C lists only the materials used in that experiment.

5.5.3 Subjects

Thirty-two members of the University of Edinburgh were paid to participate in the experiment. All were native speakers of English and naive as to the purpose of the experiment.

5.5.4 Results

Reaction times falling more than 1.96 standard deviations from the mean were replaced by the critical cut-off value. Less than 5% of the data points were affected by this procedure. The results are summarised in Table 5.2 and Figures 5.3 and 5.4.

As in the case of previous experiments, the Context dimension was transformed into a Plausibility dimension (cf. Figure 5.4). Two 2-way ANOVAs (Plausibility x Target), by subjects and by materials, were performed on the data. A strong main effect of Plausibility was found: $F_1(1,28)=54.85$, $p<0.0001$; $F_2(1,23)=20.35$, $p=0.0002$; Min $F'(1,39)=14.84$, $p<0.005$. There was also a relatively weaker effect

TABLE 5.2

READING TIME PER SENTENCE (IN CENTISECONDS)

	Context	
Target	1	>1
complement	301.4	339.3
relative	393.9	310.9

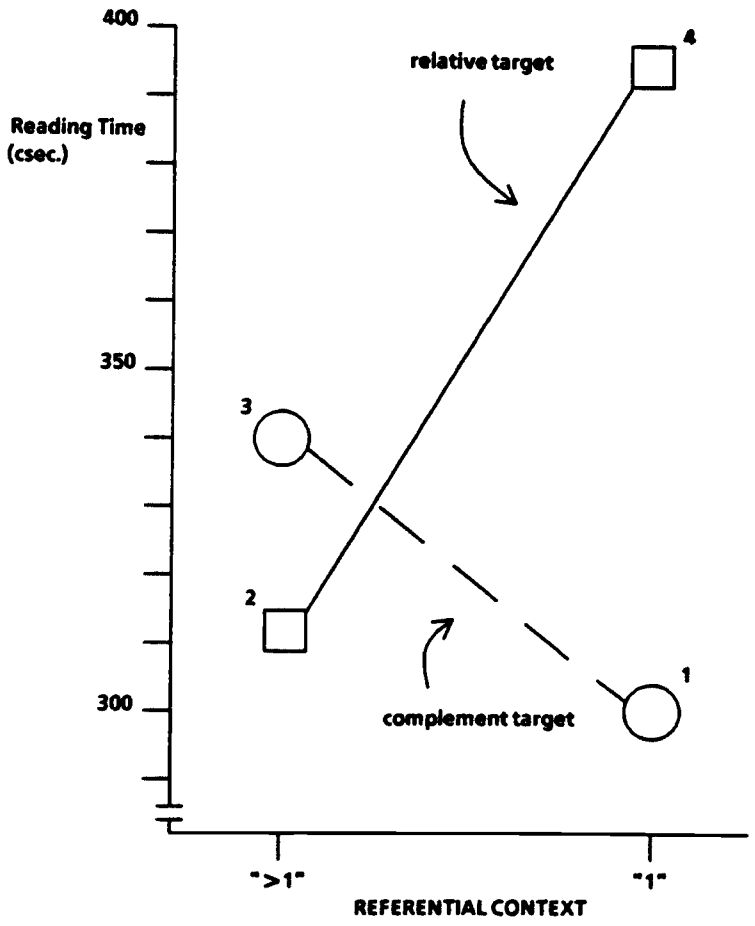


FIGURE 5.3

Mean reading times in centiseconds for Experiment 4 under two conditions of referential context: ">1" versus "1".

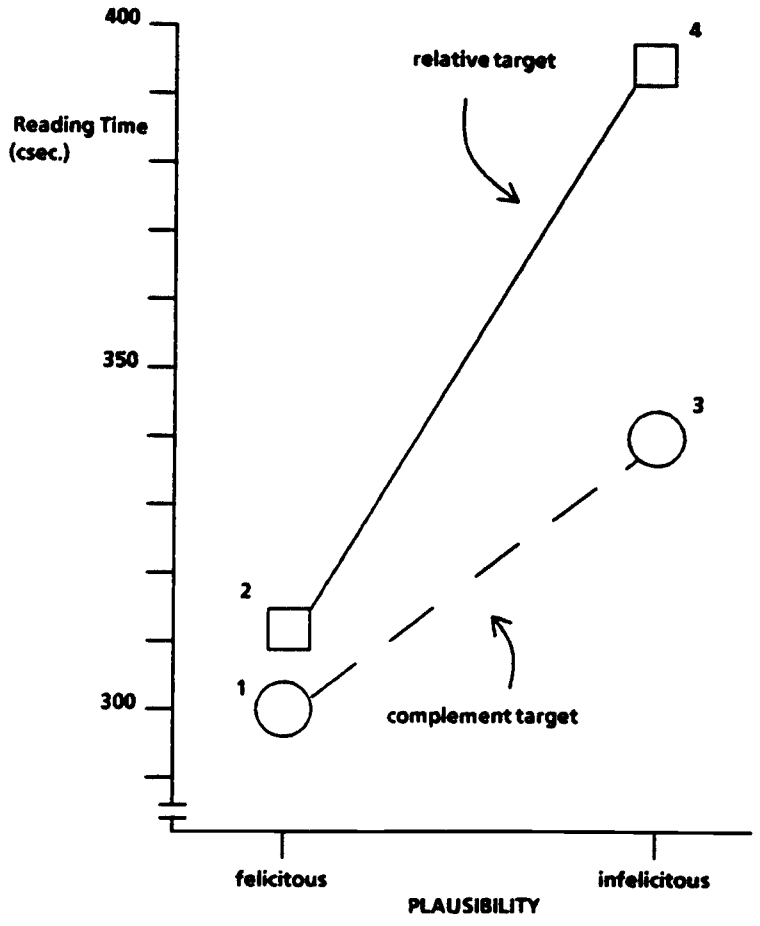


FIGURE 5.4

Mean reading times in centiseconds for Experiment 4 under two conditions of plausibility: "felicitous" versus "infelicitous".

of Target, but this was only significant on the "by subjects" analysis: $F_1(1,28)=11.03$, $p=0.0025$; $F_2(1,23)=2.54$, $p=0.1250$. Similarly, there was a weak Plausibility x Target interaction: $F_1(1,28)=5.24$, $p=0.0298$; $F_2(1,23)=2.47$, $p=0.1298$.

Scheffé's test on the data showed the following significant differences. (where the numbers in brackets refer to the numbered points on the graphs):

(1) (4)"1.rel" was significantly greater than all other points ($p<0.005$).

(2) (3)">1.comp" was greater than (2)">1.rel" and (1)"1.comp" ($P<0.005$).

(3) There was no significant difference between (2)">1.rel" and (1)"1.comp" ($p>0.1$).

5.6 Experiment 4: Discussion

In Experiment 3, relative target sentences took longer to process than their complement counterparts. Experiment 4 was identical to Experiment 3 except that the materials were modified to ensure that the wording used in the relative targets was the same as that used in the co-texts which would precede them. It was suggested that such a modification would "minimise" the inferencing required with which to match the relative against a representation, or model, of the preceding co-text. As predicted, a much smaller effect of target construction was found in Experiment 4 than was found in

Experiment 3. Furthermore, there was no observed difference between relatives and complements when preceded by their respectively facilitating co-texts.

There is, however, one problem with this interpretation. We have assumed that the modification of the materials used in Experiment 3 to produce the materials used in Experiment 4 led to a decrease in the processing time required for the relative targets (a difference of 13.1 csec.), and hence the difference between the two experiments. But it might equally well be the case that in fact this modification led to an increase in the processing time required for the complement targets (an increase of 28.8 csec. was found). Thus, it may not have been the relative targets which were affected but rather the complement targets. Experiment 4 alone does not provide the necessary information with which to resolve this problem.

On the other hand, the two experiments were run at different times and with different subjects. This means that whereas the trends within each experiment should generalise, it is not necessary that the absolute values of each data point should remain constant across the two experiments. To resolve the issue of whether it was the relative targets which were affected or the complement targets which were affected one must include all the conditions (the two levels of Context, the two levels of Target, and the two levels of Inference) within just one experiment.

If the two levels of Inferencing are labelled "Bridging" and "Minimal-Bridging", then we could expect an effect of Target in the

Bridging condition, but not in the Minimal-Bridging condition. Furthermore, we would also expect there to be an effect of Inferencing on the relative clause targets but not on the complement clause targets.

A third experiment (Experiment 5) was therefore run in which all 8 conditions described above were represented.[5]

5.7 Experiment 5

The aim of Experiment 5 was to test the inferencing hypothesis more fully. Experiments 3 and 4 contained just one level of inferencing each. It was unclear from Experiment 4 whether only the relative clause targets were affected when the amount of inferencing associated with their processing was reduced, or whether the complement clause targets were also affected. Experiment 5 contains two levels of inferencing. If the inferencing hypothesis is correct, then we should expect to find an effect of inferencing only on the relative clause targets, and not on the complement clause targets.

[5] This design ensures that in comparing across the levels of Inference (which corresponds to comparing Experiment 3 with Experiment 4) the same target sentences are used. In designing Experiment 4, the materials of Experiment 3 were modified by either changing the co-texts or by changing the targets. Hence the targets used in Experiment 4 were not necessarily the same as those used in Experiment 3 - another reason for not comparing the two experiments directly.

5.7.1 Materials

The materials used in Experiment 5 all conformed to the format illustrated by the following examples. The underlining is for expository purposes only and was not included in the experimental presentations. The full set of materials is listed in Appendix C.

Complement-inducing co-text: "Bridging" (henceforth "B.1")

A psychologist was counselling a man and his wife.
He was particularly concerned with the problems of
one of the pair but wasn't concerned about the other.

Relative-inducing co-text: "Bridging" (B.>1)

A psychologist was counselling two wives.
He was particularly concerned with the problems of
one of the pair but wasn't concerned about the other.

Complement-inducing co-text: "Minimal Bridging" (~B.1)

A psychologist was counselling a man and his wife.
He was worried about one of the pair
but wasn't concerned about the other.

Relative-inducing co-text: "Minimal Bridging" (~B.>1)

A psychologist was counselling two wives.
He was worried about one of the pair
but wasn't concerned about the other.

Complement target sentence

The psychologist told the woman that he was worried about
her marital problems.

Relative target sentence

The psychologist told the woman that he was worried about to visit him again.

Each target sentence was presented on a single line, and not on two lines as shown here.

Thirty-two examples were constructed which were similar to these. Target sentence-pairs were matched as closely as possible for length. There were eight versions of each example (two Inference x two co-texts x two targets). The design was a factorial repeated measures design incorporating a Latin Square. The thirty-two test items were embedded in forty-eight fillers.

5.7.2 Apparatus and Procedure

The same apparatus and procedure were used as described in Experiment 1.

5.7.3 Subjects

Fifty-six members of the University of Edinburgh were paid to participate in the experiment. All were native speakers of English and naive as to the purpose of the experiment.

5.7.4 Results

Reaction times falling more than 1.96 standard deviations away from the mean were replaced with the critical cut-off value. Less than 4.7% of the data were affected by this procedure. The results

are summarised in Table 5.3 and Figures 5.5 and 5.6.

As before, Context was transformed into a Plausibility dimension.

Overall, a strong main effect of Plausibility ("felicity") was found: $F_1(1,48)=64.54$, $p<0.0001$; $F_2(1,31)=14.99$, $p=0.0005$; Min $F'(1,45)=12.16$, $p<0.005$. A weak effect of Target was found: $F_1(1,48)=14.20$, $p=0.0004$; $F_2(1,31)=2.19$, $p=0.1486$; but there was no main effect of Inference ($F_1(1,48)=1.65$, $p=0.2055$; $F_2(1,31)=0.79$, $p=0.3824$.). The weak effect of Target is due to an interaction between Inference and Target: $F_1(1,48)=17.63$, $p=0.0001$; $F_2(1,31)=8.4$, $p=0.0068$; Min $F'(1,59)=5.69$, $p<0.025$. There were no other

TABLE 5.3

READING TIME PER SENTENCE (IN CENTISECONDS)

Inference - "Bridging"

	Context	
Target	1	>1
complement	311.6	348.1
relative	405.1	350.1

Inference - "Minimal-Bridging"

	Context	
Target	1	>1
complement	319.0	365.7
relative	381.2	322.1

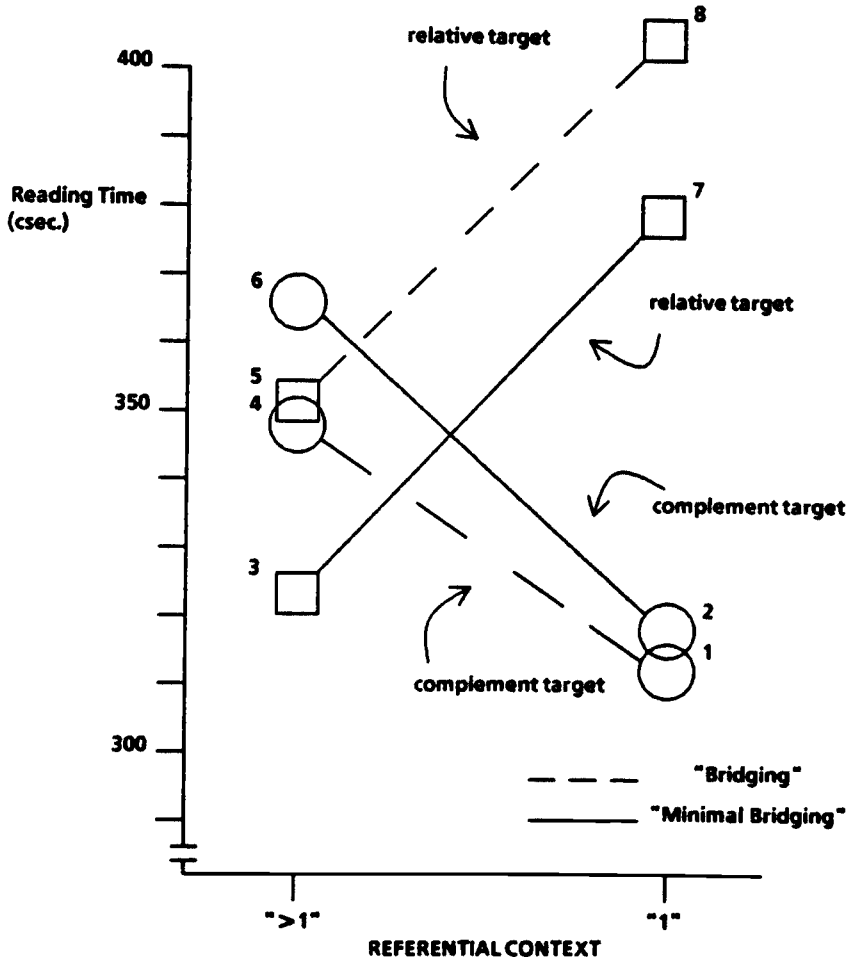


FIGURE 5.5

Mean reading times in centiseconds for Experiment 5 under two conditions of referential context (">1" versus "1") and two conditions of inferential context ("Bridging" versus "Minimal Bridging").

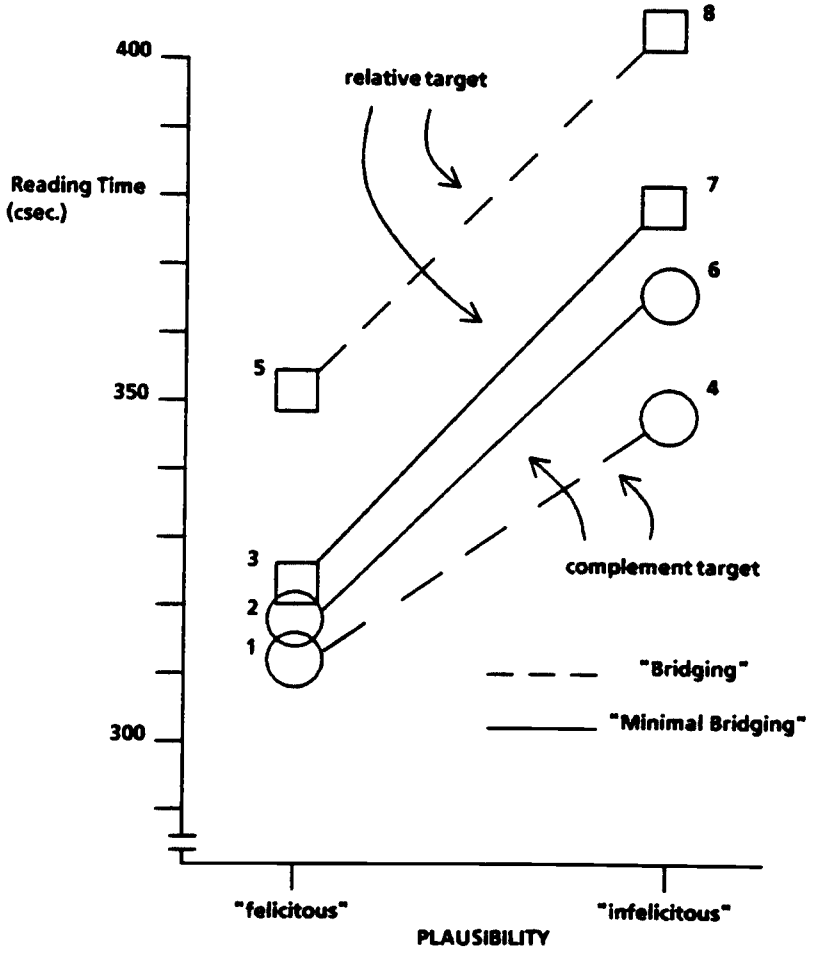


FIGURE 5.6

Mean reading times in centiseconds for Experiment 5 under two conditions of plausibility ("felicitous" versus "infelicitous") and two conditions of inferential context ("Bridging" versus "Minimal Bridging").

interactions.

Multiple Scheffé's tests gave the following additional information (the numbers in brackets refer to the numbered points on the graphs. Due to the complexity of the design, the reader is advised to refer to Figures 5.5 and 5.6 during the following description):

1. (6)"Minimal-Bridging:>1.comp" and (4)"Bridging:>1.comp" were <u>not</u> significantly different from one another. Neither were (2)"Minimal-Bridging:1.comp" and (1)"Bridging:1.comp". In other words, there was no effect of Inference on the complement targets.

2. (3)"Minimal-Bridging:>1.rel" was significantly lower than (5)"Bridging:>1.rel": $p<0.05$, though (7)"Minimal-Bridging:1.rel" was not significantly lower than (8)"Bridging:1.rel" ($P>0.1$). In other words, there was a strong effect of Inference on relative clause targets preceded by felicitous co-texts. The effect failed to reach significance when the relative clause targets were preceded by infelicitous co-texts (any effect may have been masked by the increase in latencies due to the garden path effect). The results do indicate, however, that in felicitous conditions (cf. Figure 5.6), there was an effect of Inference only on relative targets, and not on complement targets.[6]

[6] The data were separated by Target and separate ANOVAs performed. For complement targets, no main effect of Inference was found ($F_1(1,48)=3.59$, $p=0.0643$; $F_2(1,31)=3.43$, $p=0.0736$), though again there was a strong effect of Plausibility: $F_1(1,48)=26.60$, $p<0.0001$; $F_2(1,31)=8.56$, $p=0.0064$; Min $F'(1,51)=6.48$, $p<0.025$. For relative

3. (3)"Minimal-Bridging:>1.rel" was not significantly different to (2)"Minimal-Bridging:1.comp" or (1)"Bridging:1.comp", but it was significantly lower than (6)"Minimal-Bridging:>1.comp": $p<0.005$.

Finally, all pairs of points which represent conditions differing only along the Context/Plausibility dimension (i.e. all pairs of points the members of which are "joined" by lines on Figures 5.5 and 5.6) were significantly different from one another ($p<0.005$).

5.8 Experiment 5: Discussion

Experiment 5 was designed both to replicate the effects found in Experiments 3 and 4, and to show any differences between the two. The dotted lines in Figures 5.5 and 5.6 represent the conditions in Experiment 5 corresponding to those of Experiment 3. The solid lines in Figures 5.5 and 5.6 represent those conditions in Experiment 5 which correspond to those of Experiment 4. Clearly the effects found in the two earlier experiments have been replicated. More interesting, though, is the relationship between the two. It was suggested earlier that there were two possible "explanations" as to the observed differences: they might have arisen either because of a decrease in reading times to relative targets (the predicted shift), or because of an increase in reading times to complement targets. However, the data as described in Figure 5.7 show that there was no

targets, a main effect of Inference was found: $F_1(1,48)=12.56$, $p=0.0009$; $F_2(1,31)=4.31$, $p=0.0463$; Min $F'(1,52)$ $(=3.21)$[1] just failed to reach[2] significance. Again, there was a strong effect of Plausibility: $F_1(1,48)=50.05$, $p<0.0001$; $F_2(1,31)=15.06$, $p=0.0005$; Min $F'(1,50)=11.58$,[1] $p<0.005$.

3. (3)"Minimal-Bridging:>1.rel" was not significantly different to (2)"Minimal-Bridging:1.comp" or (1)"Bridging:1.comp", but it was significantly lower than (6)"Minimal-Bridging:>1.comp": $p<0.005$.

Finally, all pairs of points which represent conditions differing only along the Context/Plausibility dimension (i.e. all pairs of points the members of which are "joined" by lines on Figures 5.5 and 5.6) were significantly different from one another ($p<0.005$).

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effect of Inferencing on the complement targets whilst there was an effect of Inferencing on the relative targets.

The results of Experiment 5 have shown, then, that the original differences observed between relatives and complements can be eliminated by minimising the inferencing needed to match the propositional content of a relative to some proposition already represented in a discourse model.

By taking into account the functional differences between relative clauses and complement clauses, one can create a situation in which there is no effect of target construction on reading time. More specifically, the present approach has involved taking into account the different processes which are implicated by the use of each construction: relative clauses "demand" a different kind of process than do complement clauses. The difference arises because of the nature of the information each clause conveys: as described earlier, restrictive relative clauses provide Given information, whilst complement clauses provide New information. We shall see in Chapter 7 that this distinction, between Given and New, is fundamental in terms of providing an explanation for syntactic parsing preferences.

5.9 Summary and Conclusions

The aim of this chapter was to consider some of the processing demands which are made of the hearer/reader when he or she encounters a particular syntactic construction. In particular, we were

concerned with the process by which information conveyed by a sentence is matched against information which already exists in the hearer/reader's mental model of the discourse. This matching process is essential during the processing of restrictive relative clauses (or their functional equivalents) as well as during the processing of any definite referring expression for which a unique referent must be identified in the discourse model.

Specifically, we considered the reasons why relative clauses might take longer to process than complement clauses. It was suggested that this difference stemmed from differences in referential function between the two constructions. Fundamental to this difference was the notion of inferencing. A method was suggested whereby the amount of inferencing which was required in order to process a relative clause could be reduced. Two experiments were described which attempted to do this. The second of these (Experiment 5) showed that differences in reading time between the two target constructions could be eliminated by reducing the amount of inferencing required during relative clause processing.

The reason for considering the matching process to be so important is that it must operate during the processing of any referring expression for which the speaker/writer presupposes that the hearer/reader can identify a unique referent. Most of the examples in the literature to which the Minimal Attachment Hypothesis has been applied contain local ambiguities which stem from the interpretation of a noun phrase as either simple or complex. It

follows that any study which uses such examples, and which uses an experimental measure which is sensitive to sentence complexity, must take into account the possibility that this measure will be sensitive not just to whether or not the processor has to recompute its syntactic analysis of a sentence, but also to the operation of (and the increased processing load caused by) this matching process.

These considerations are clearly important when we consider the processing of relative clauses within the context of a discourse. But what are the implications of this discussion for the original results of Wanner et al. (1974)? Most of the theories relating to the processing of restrictive relative clauses (or their equivalents), such as those of Wanner et al. (1974), Frazier (1979), Rayner et al. (1983), Wanner & Maratsos (1978), and Ford (1983), have studied sentence processing in isolation. But to study the processing of relative clauses in the absence of any explicit context is to study the processing of a referring expression without allowing it to refer. It is possible, therefore, that some of the effects observed in connection with the processing of relative clauses were partly artefactual on the fact that the relative clauses in these studies were not preceded by adequate co-texts. As such, the processing of these relatives must be considered to have been "abnormal

But even if we assume their processing to have been abnormal, how do we account for the increased reading times to relative clauses in the null context? Presumably, we cannot ascribe this increase to

differences in the amount of inferencing required during the processing of relative and complement clauses. It is impossible to match the propositional content of the relative clause to the contents of a discourse model if the model contains no information at all. There is an alternative explanation, however, which is derived from the original suggestions concerning the null context which were made by Crain & Steedman (1985), and which was discussed in Section 4.5.2 in connection with the increased reading times to target sentences in the null context.

In Section 3.3 we discussed the reasons, given by Crain & Steedman, for preferring the complement clause analysis of an ambiguous that-clause over the relative clause analysis. To summarise the argument briefly: the relative clause analysis would require more extensions to the discourse model than would the complement clause analysis. Crain & Steedman's Principle of Parsimony ensures that an analysis is chosen which requires fewest extensions to the model. In Section 4.5.2 we suggested that processing time increases as more extensions are required of the discourse model. Relative clause analyses therefore require more processing time than complement clause analyses simply because the former require more extensions to the model than do the latter.[7]

We began Part II in Chapter 4 by considering whether Crain's experimental results justified the rejection of experimental evidence

[7] We shall discuss this further in Chapter 7.

collected in those studies of ambiguity resolution which did not take into account referential function. On the basis of results obtained in further experiments (Experiments 1 - 5), we must now conclude that there does exist sufficient evidence to question these studies. The current experiments, which do take referential function into account, all cast serious doubt on the Minimal Attachment Hypothesis.

The experimental data we have gathered in the course of this and the preceding chapter also call into question the motivation for one of the assumptions underlying an implementation of Minimal Attachment which we discussed in Section 2.3.2. Wanner (1985) proposes an ATN model of syntactic ambiguity resolution which exhibits limited parallelism: all the arcs leaving the current state in the network are attempted simultaneously, and as soon as an analysis path successfully attaches the current word to the parse tree then all other analyses are abandoned, and control returns to the state at the end of the successful analysis path. Wanner points out that Minimal Attachment (and Right Association) falls out of such a model as an automatic consequence of the limited parallelism. The assumption underlying this model, however, is that processing time is merely dependent on the number of arcs traversed, and <u>not</u> on the kinds of action that are associated with each arc, and which must therefore be performed in order to traverse successfully that arc. But the matching and inferencing processes which we have studied during the course of this chapter are examples of the kinds of process which would be associated with an arc in an ATN, and the experimental results clearly indicate that processing time <u>is</u> dependent on the

operation of these processes.

In conclusion, the experimental data support a "functional" approach to syntactic processing: it is not enough to consider syntactic structure alone. We must also consider the function of each syntactic construction, and the processes which must be associated with each such function.

In Part III we shall consider a number of theoretical issues concerning the hypothesis with which we have been working, namely that the resolution of local syntactic ambiguity is governed, where appropriate, by Crain & Steedman's principles of Referential Success and Parsimony. In Chapter 6 we shall reconsider the adequacy of their account. In particular we shall consider whether it adequately describes <u>when</u> referential information is brought to bear during the resolution of local syntactic ambiguity. We shall conclude that the Principle of Referential Success is inadequate in this respect. An alternative principle will be offered in its place. In Chapter 7, we shall return to Crain & Steedman's Principle of Parsimony, and revisit their account of the original null context data. We shall consider whether or not Parsimony might also account for the second structural strategy which we encountered in Part I: Right Association/Late Closure.

Part III: The Theory

"It is more important that my equations be beautiful than that they fit experiments."

P.A.M. Dirac.

CHAPTER 6

Interactions, Autonomy, and Referential Failure

The aim of Parts I and II of this thesis has been to characterise the information that is used by the HSPM in order to resolve local syntactic ambiguities. We concluded in Part I that there are two kinds of approach to this problem: the structural approaches on the one hand, and the interactive, contextually-based approaches on the other.[1] In Part II we considered the empirical adequacy of both the structurally-based Minimal Attachment Hypothesis (Frazier, 1979; Rayner et al., 1983) and the contextually-based Referential Success Hypothesis (Crain & Steedman, 1985). We concluded as a result of a number of experiments that the data were not consistent with the former hypothesis, but were consistent with the latter. In other words, we concluded that the HSPM resolves local syntactic ambiguities not according to a structural bias, but on the basis of semantic/pragmatic information.

In Chapter 1 we suggested, however, that we should do more than simply characterise the information that is used by the HSPM in order to resolve local syntactic ambiguities. In addition, we should

[1] There has been a certain amount of discussion concerning the term "interaction" (Forster, 1979; Cowart, 1982; Norris, 1982; Crain & Steedman, 1985). For present purposes the term will be used as it applies to the resolution of local ambiguity: an interaction will be said to occur if an ambiguity at one level of description is resolved on the basis of information at another.

characterise just when this information is used during the "processing sequence". Theories which advocate contextual interactions during syntactic processing have been criticised in the past for underspecifying the nature of the contextual information being used (cf. Forster's (1979) criticism of Tyler & Marslen-Wilson, 1977). Crain & Steedman (1985) propose a theory of contextual interaction which is not subject to such criticism. But theories of contextual interaction can also be criticised to the extent that they underspecify the precise nature of the interaction, and fail to describe the time-course of the events surrounding it. In other words, they fail to specify exactly <u>when</u> a particular decision is made. It is to this issue that the present chapter is addressed.

As we shall see during the course of this chapter, the timing of the HSPM's decisions is sensitive to a number of factors concerning the organization of the language processor. In Section 6.1 we shall consider the different kinds of interaction that might be possible between syntax and semantics. Within the taxonomy of such interactions provided by Crain & Steedman, we shall review the Referential Success Hypothesis in detail. We shall conclude in Section 6.2 that the Principle of Referential Success makes certain predictions concerning the timing of the HSPM's decisions which are implausible and inadequate. An alternative principle, of Referential Failure, will be offered in its place.

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In Section 6.3, we shall discuss the implications of this extended theory for the relationship between syntax and semantics during sentence comprehension. The emphasis will switch away from the different kinds of interaction that are possible and towards the different degrees of autonomy that could be exhibited by the language processor. This is another factor on which the timing of the HSPM's decisions depends. In particular, we shall consider the degree of autonomy that is implied by the Principle of Referential Failure. We shall contrast these implications with those of Rayner et al.'s (1983) Minimal Attachment Hypothesis and Ford, Bresnan, & Kaplan's (1982) theory of Lexical and Syntactic Preferences.

6.1 Selection versus Instruction

Crain & Steedman (1985) distinguish between "weak" (or "selective") interactions on the one hand, and "strong" (or "instructional") interactions on the other. A weak interaction involves the filtering out by one component of alternative analyses proposed (either serially or in parallel) by another. A strong interaction allows one component to direct or guide another in an active fashion: it thus allows one component to determine which is the first or only analysis to be pursued by another. The nature of the interaction will have crucial consequences for the temporal sequencing of the different operations of the HSPM. Of the two positions Crain & Steedman say:

> ".. under the weak hypothesis, syntactic processing independently 'proposes' alternatives, either serially or in parallel, while semantics 'disposes' amongst these alternatives.

> According to the strong form of the hypothesis, on the other hand, semantics and context actually influence which syntactic entities get proposed in the first place, perhaps by adjusting the order in which alternative syntactic rules are applied, or by temporarily making certain rules entirely unavailable." (Crain & Steedman, 1985, p.325)

Strong interactions commit the processor to "resolving" local ambiguities irrespective of whether or not there exists sufficient information with which to do so correctly. Weak interactions in principle allow the processor to "wait and see" before making any committing decisions one way or another (cf. the computational models of Marcus, 1980; Mellish, 1981; Milne, 1984; Schieber, 1983; and others). In other words, the processor in models exhibiting weak interactions can wait until there is sufficient information with which to resolve the ambiguity (pursuing the alternative analyses either serially or in parallel, see below). Fodor (1983) contrasts weak interactions with strong interactions as follows:

> "... semantic information is never used to predict syntactic structure, but a line of analysis on which the parser is engaged can be aborted whenever it produces structures that resist contextual integration. Boxologically, this means that the parser feeds information freely to the context analyzer, but all that the context analyzer is allowed to say to the parser is either 'yes' (continue with the present line of analysis) or 'no' (try something else, I can't fit what you're giving me to the context). What the context analyzer is prohibited from doing is telling the parser which line of analysis it ought to try next - i.e., semantic information can't be used predictively to guide the parse." (Fodor, 1983, p.78, note 26.)

Crain & Steedman adopt a model in which only weak interactions govern the resolution of local syntactic ambiguity. They thus have to specify both the basis on which semantics "disposes" of the

alternative syntactic analyses, and the manner in which these alternative analyses are first "proposed" by syntax.

With regard to the former, they propose the following principle:

> "The Principle of A Priori Plausibility. If a reading is more plausible in terms either of general knowledge about the world, or of specific knowledge about the universe of discourse, then, other things being equal, it will be favoured over one that is not." (Crain & Steedman, 1985. p.330)

Plausibility can be thought of as a criterion by which analyses are either selected or rejected. It can be treated either as an "absolute measure" or as a "relative measure". If plausibility is an absolute measure, then the evaluation of plausibility must be an "all-or-none" judgement. If there existed a scale of plausibility, then there would have to exist some absolute value (threshold) on the scale above which a reading is deemed to be plausible, and below which it is deemed to be implausible. If, on the other hand, plausibility is a relative measure, then there is no such absolute threshold. Whichever analysis scores highest on the scale will be deemed to be the most plausible analysis. Crain & Steedman argue that plausibility must be considered to be a relative measure, and that its evaluation involves comparisons over two or more alternative analyses (see also Rayner et al., 1983, and Kurtzman, 1984, for further, albeit brief, discussion on the subject). Given this assumption (implicit in the Principle of A Priori Plausibility), Crain & Steedman are committed to a model in which alternative

analyses are proposed not serially, but in parallel.[2] But given this framework, what claims do Crain & Steedman make with regard to just when it is that one analysis is chosen and the other alternative analyses are rejected?

6.2 Referential Success versus Referential Failure

Crain & Steedman expand on the notion of "specific knowledge about the universe of discourse", as introduced in the Principle of A Priori Plausibility, and provide a particular instance of this principle which we have already encountered:

> "The Principle of Referential Success: If there is a reading which succeeds in referring to an entity already established in the hearer's mental model of the domain of discourse, then it is favoured over one that does not." (Crain & Steedman, 1985, p.331)

So when is the decision to favour one analysis over another made? If Referential Success is taken literally, then only after encountering the item "about" in the last sentence of the example below is it possible to determine that a reading has succeeded "in referring to an entity already established in the hearer's model of the domain of discourse"

[2] There does exist a serial interpretation of the weak interaction, in which alternative analyses are considered not in parallel, but serially. But as Crain & Steedman point out, there is little distinction to be drawn between a model in which all the possible analyses are pursued and considered simultaneously, and a model in which all the possible analyses are pursued, but considered one after the other. There is a sense in which the two models are equivalent.

(6.1) A psychologist was counseling two women. He was worried about one of the pair but not about the other. The psychologist told the woman that he was worried about...

Moreover, until this last item is reached, both possible analyses (relative clause and complement clause) must be pursued.

If, on the other hand, the HSPM can select an analysis on the basis of the inevitable referential failure of that analysis' competitors, then a decision as to which analysis to choose could be made before completing the relative clause. In example (6.1), the decision could be made as soon as the ambiguous "that" was encountered, because neither the critical referring expression "the woman" nor this NP followed by a that-complement clause will be referentially successful. The use of a relative clause presupposes not only that the head nominal on its own is not referentially successful, but also that the modification of the head nominal with the relative may lead to a state of referential success.[3] It should thus be possible to make a choice of analysis as soon as the "that" is encountered: of the two readings only the relative clause "promises" to be referentially successful.

Clearly, then, it is not so much referential success which determines the choice of analysis, but rather referential failure:

> The Principle of Referential Failure: If a definite referring expression fails to restrict the set of candidate referents

[3] In Chapter 7 we shall return to this point, and the notion of "presupposition" as it applies here.

> in the hearer's mental model of the domain of discourse to a single member, then an analysis which treats subsequent material as a modifier for that referring expression (i.e. as providing information which may lead to further restriction), will be favoured over one that does not.

Unlike the Principle of Referential Success, Referential Failure implies that a decision to interpret the ambiguous item "that" one way or the other is made before the complete that-clause is encountered. A restrictive analysis is chosen not because it "succeeds in referring to an entity.." but because of the "presuppositional promise" that it will succeed, and because of the promise that the alternative analyses will not succeed.

Clearly, Referential Failure applies only in the case of definite referring expressions. Indefinite referring expressions, if used to introduce new entities into the model, will fail to refer to any entity already established in that model. It is an empirical question whether, or in what circumstances, parsing preferences would be exhibited in the case of indefinite referring expressions embedded in the kinds of example we have so far considered.[4]

A number of predictions follow from the Principle of Referential Failure, one of which is exemplified by the following:

(6.2) The psychologist told the woman that he thought she should take a holiday and visit him afterwards.

[4] See Chapter 7 for further discussion.

In this example, the presence of the pronoun "she" after the phrase "he thought" indicates that the correct interpretation of the that-clause is as a complement clause. If this example were to be preceded by a co-text which introduced more than one "woman", then subjects should, if Referential Failure is correct, garden path on encountering the disambiguating "she". That is, they should choose the restrictive relative reading on encountering "that".

The Principle of Referential Failure, like Referential Success, is just a special case of Crain & Steedman's Principle of Parsimony (see Section 3.3 and below). Unlike Referential Success, Referential Failure is different with respect to just <u>when</u> the Principle of Parsimony is invoked.

> <u>"The Principle of Parsimony:</u> If there is a reading which carries fewer unsatisfied but consistent presuppositions or entailments than another, then, other criteria of plausibility being equal, that reading will be adopted as most plausible by the hearer, and the entailments in question will be incorporated in his or her model." (Crain & Steedman, 1985. p.333)

The Principle of Parsimony predicts that if a referring expression <u>has succeeded</u> in referring to an entity in a model, then an analysis which <u>does not</u> treat subsequent material as a modifier for that referring expression will be favoured over one that does. In (6.3) below, the absence of a pronoun after "he thought" indicates that the correct interpretation for the string "that he thought.." must be a relative clause.

(6.3) The psychologist told the woman that he thought should take a holiday to visit him again afterwards.

If this example were to be preceded by a complement-inducing context, then according to the Principle of Parsimony, subjects should garden path on the disambiguating item "should".

Kurtzman (1984) describes an experiment which appears to counter this prediction. Examples like the following were used:

(6.4) Relative Sentence:

The psychologist told the wife that was yelling to shut up.

(6.5) Complement Sentence:

The psychologist told the wife that to yell was not constructive.

In these examples the local ambiguity is disambiguated immediately after the pronoun "that". In a variant of the incremental grammaticality judgement task, words were presented at 240 msec intervals in the centre of a screen (with 60 msec of blank screen between each presentation), and subjects were instructed that when they heard a tone (which would stop any further presentation of words), they should indicate whether or not the sentence had been "grammatical" up to that point. In the example above, the tone would be heard immediately following the disambiguating items "to" and "was". Kurtzman preceded each target sentence by either a relative-inducing, or complement-inducing co-text. Contrary to the predictions of Referential Failure and Parsimony, Kurtzman claimed to find no effect of context on subjects' responses, and to find that

subjects responded correctly more often to the complement target (6.5), for which there were "practically no errors", than to the relative target (6.4).

At present few details are available of Kurtzman's experiment, and it is unclear whether or not his findings do pose a problem for the theory which is being advocated here. It is possible that no effect was found because of the relative ease with which subjects can recover from a garden path in cases where disambiguation occurs so soon after the onset of the ambiguity. Unfortunately, Kurtzman provides no details of his statistical analyses or of the materials he used (other than the example above). It is certainly surprising that there were "practically no errors" in subjects' responses to the complement target. It remains to be seen whether or not the predictions made of the earlier examples (6.2 and 6.3) are borne out by the experimental data.[5]

[5] The problem with studies like these is to find a task which can provide an on-line measure of garden-pathing. The naming latency task developed by Tyler & Marslen-Wilson (1977) is one candidate, except that the aural presentation of the relative/complement examples may provide intonational cues as to which analysis should be pursued (see Lehiste, 1973; Wales & Toner, 1979; Berkovits, 1981; and Berkovits, 1982, for evidence concerning the role of intonation in disambiguating surface structure ambiguity). In an inconclusive series of experiments not reported here, a visual analog of the naming-latency task was attempted in which the target disambiguating items appeared on the screen in upper case. Ungrammatical controls (e.g. "The psychologist wanted the woman that he was having trouble with THE husband" and "The psychologist told her that he was having trouble with TO leave") were included but the measure was found to be insensitive even to these. Just why is a matter for conjecture. In the original version of the task subjects just had to say out aloud whatever appeared on the screen. In the visual analogue subjects had to be selective (selecting between upper and lower case). Any effects of grammaticality may have been masked by delays introduced

In conclusion, the Principle of Referential Failure claims that the resolution of local syntactic ambiguity can occur without the need to pursue to completion any of the alternative analyses. Results reported by Kurtzman (1984) do appear to question such a position but in the absence of more detailed information about his experiments it is impossible to assess the severity of the problem they pose to the present theory.

The Principle of Referential Failure differs from Crain & Steedman's Principle of Referential Success in a number of important respects. In the following section we shall consider the implications of a model based on Referential Failure for the relationship between syntax and semantics.

6.3 Autonomy and Ambiguity Resolution

We have so far assumed that a distinction can be made within the language processor between a "syntactic" component and a "semantic" component. The extent to which these components can freely communicate with one another, if indeed they are separable components, will influence the degree to which different kinds of information can be brought together during the processing of a sentence. This is clearly important when considering the time course of the HSPM's decisions during the resolution of local syntactic ambiguity. In this section we shall consider a number of possibilities with regard to the relationship between syntax and

by the attention-shift necessitated by such selection.

semantics, and the different demands that the principles of Referential Failure and Referential Success make with respect to this issue.

Crain & Steedman suggest that there are two ways in which the distinction between syntax and semantics can be realized: as one of "Formal Autonomy" or as one of "Representational Autonomy". If the syntactic and semantic components are representationally autonomous, then there is an intermediate level of purely syntactic representation which is subsequently reinterpreted into a semantic representation. If the syntactic and semantic components are only formally autonomous, then there is no such intermediate representation; syntax is merely part of a description of how the semantic interpretation is arrived at.

If there does exist an intermediate level of syntactic representation, there then exists a further issue concerning the "extent" of a syntactic representation: how large is the "unit" of autonomy? If a purely syntactic representation of an entire sentence were built before semantics could begin to interpret it, then the autonomous unit would be the sentence. If semantics had to wait for syntactic units corresponding to the clause, then the autonomous unit would be the clause (cf. theories of clausal processing, e.g. Fodor, Bever, & Garrett, 1974). One can characterise the notion of autonomy in terms of a dimension. At one extreme of the dimension is "total" autonomy, in which syntactic representations of whole sentences are constructed before being passed to the semantic processor. At the

other extreme, is a situation in which the autonomous unit corresponds to the word. One can think of this extreme as representing total (representational) non-autonomy. Such considerations play an important part when considering the timing of the HSPM's decisions. However, the degree to which the language processor is representationally autonomous is not the only factor which will influence the time-course of the HSPM's decisions. Much depends on the size of the unit at which the semantic component chooses to interact with the syntactic component.

The extent to which a model is representationally autonomous is independent of whether or not the model is interactive: Crain & Steedman characterise weakly interactive models according to the "intimacy with which semantics is allowed to intervene". This notion of "intimacy" depends on the size of the syntactic representations which, when completed, are passed up to semantics for <u>disposal</u> (as opposed to being passed up to semantics for <u>interpretation</u>; the two are not necessarily the same). Weak interactions are implementable at any position along the autonomy dimension: at one extreme, semantics and syntax might only interact on a sentence-by-sentence basis; at the other, they might interact on a word-by-word basis; and in between, they might interact on a phrase-by-phrase or clause-by-clause basis. This is not to say, however, that the unit of autonomy necessarily defines the unit at which syntax and semantics interact. For instance, the unit of autonomy might be the word, but the unit at which semantics instructs syntax to abandon analyses might be the phrase, or clause, or even sentence. There is a sense in which the

unit of autonomy defines the smallest unit at which syntax and semantics could in principle interact (though see Crain & Steedman's footnote 5, which is discussed below).

So the notion of weak interaction is fully compatible with any position along the autonomy dimension. However, a model based on Referential Failure makes certain assumptions concerning autonomy and interactions. It assumes that the relative clause interpretation of the that-clause, or the NP-attachment of the prepositional phrase in the case of the PP-examples, can be chosen as soon as the referring expression which these constructions could modify is found to be referentially unsuccessful. The implication, during the discussion of this model, was that referential success or referential failure could be determined as soon as the target noun phrase was encountered. For this to happen the simple noun phrase must be immediately available to "semantics" for reference evaluation (a broad term which is intended to cover that aspect of semantics concerned with the process of establishing reference).

Such a proposal is consistent with the approach to reference evaluation developed by Mellish (1981). Mellish views the task of discourse processing as narrowing down the set of things that might be under discussion to the set of things that are. He proposes that the evaluation of reference is a by-product of a process of constraint satisfaction: as the reader progresses through a text, he or she accumulates an ever increasing number of constraints which the referents of the referring expressions must satisfy. This process of

"incremental evaluation" involves having available representations of "partially evaluated" referents. These are simply the members of the set of referents which all satisfy the available constraints. This set gradually becomes more and more refined as the analysis proceeds, until just one candidate referent remains. Within this model, Referential Failure is interpreted as follows; as long as the cardinality of the candidate referent set is greater than one, then an analysis which provides further constraints to be satisfied by these candidates will be favoured over one that does not.

In a non-autonomous incremental model like that of Mellish, there would be no "syntactic" distinction (to use the term "syntactic" in the Formal sense) between a simple NP and the head NP of a complex NP. In other words, there would be no difference in the manner in which the semantic interpretation proceeds. To use the terminology of Mellish: for both the "simple" NP and the head NP of the "complex" NP, the processor would attempt to derive constraints which must be satisfied by the candidate referent(s). In the case of the former, constraint satisfaction would identify a single candidate, whilst in the case of the latter, it would not. It is only <u>after</u> these two NPs have been encountered, and evaluated, that any "syntactic" differences (e.g. in the resolution of a local ambiguity) might appear.

In a model which exhibits some degree of autonomy of syntax, the processes responsible for incremental reference evaluation must wait until they receive from syntax some output over which to start

operating. To put it more generally, semantics can only operate on the "chunks" which are given it by syntax, where each chunk is an autonomous unit. Semantics cannot operate on only partially built chunks.[6] If the size of the autonomous unit which contains the critical NP is such as to extend beyond the ambiguous item "that" (or "with"), then referentially-oriented information will not be available as soon as the target NP has been encountered.

Clearly, a model based on Referential Failure and the immediacy of semantic processing is incompatible with a model in which the level of constituent up to which syntax is autonomous is the sentence, or even the clause.[7] So, whilst weak interactions in general are compatible with any position concerning the autonomy of syntax and the immediacy of semantic processing, a model based on Referential Failure is not. Instead it tends towards the non-autonomous extreme of the autonomy dimension, which permits the early

[6] If semantics could operate on partial syntactic "chunks", then the ensuing model would be equivalent to one in which the unit of autonomy corresponded to the "partial chunk". In terms of grammatical "constituents", a model which allows the semantic interpretation of partial constituents is equivalent to a model in which each of these partial constituents is itself a constituent. (cf. the approach to grammatical structure taken by Ades & Steedman, 1982)

[7] Crain & Steedman (1985; footnote 5) permit the possibility of a model in which there exist non-autonomous representations at some "lower" level, such as the noun phrase, but autonomous representations at some "higher" level, such as the clause. Presumably this means that syntax and semantics can interact either at the level of the noun phrase, or at the level of the clause, but not in between (at, for instance, the level of the verb phrase). Such a situation would be compatible with the assumption concerning the "immediacy" of referential evaluation.

integration of information across the different levels of linguistic structure.[8]

There are a number of studies which provide evidence for the early interpretation of noun phrases: Swinney (1979) shows that contextual information can be used to disambiguate lexically ambiguous nouns (such as "bug" - an insect or a radio device) within 250 msec (or approximately 3 syllables) of their occurrence. Shillcock (1982) shows that anaphoric pronouns are resolved within an equally short space of time. Garrod & Sanford (forthcoming) show that the isolation of the appropriate antecedents to full noun phrases seems to occur as soon as they are encountered (though this does not seem to be the case for certain pronouns used in their study). And finally, Marslen-Wilson and colleagues, in a number of experiments, produce evidence in support of the early integration of information across the lexical, syntactic, and semantic levels of linguistic structure (e.g. Marslen-Wilson, 1975; Tyler & Marslen-Wilson, 1977; Marslen-Wilson & Tyler, 1980).

Minimal Attachment and autonomy

Rayner et al. (1983) discuss the advantages of such early integration and concede that it could potentially lead to considerable computational savings:

[8] In a totally non-autonomous model, the existence of different levels of linguistic structure would be a Formal property of the model, but not a Representational property (cf. the earlier discussion on this distinction as drawn by Crain & Steedman, 1985).

> "If assignment of structure within each level occurred as the words of a sentence were encountered but without integration between the different levels of structure, much needless processing effort would be expended. Integration across different levels of linguistic structure might eliminate or drastically reduce the need for multiple representations and/or reduce the extent to which the garden-path phenomenon occurs." (Rayner et al., 1983, p.359)

A model which did not permit such early integration would require of the ambiguity resolution process either that decisions be delayed until integration were possible (the implication being that during the course of this delay, multiple analyses may have to be pursued), or else that a decision on which analysis to pursue be made on syntactic grounds alone (cf. the Minimal Attachment/Thematic Selection model). But if semantics could determine the referential success or failure of a noun phrase as soon as it encountered that noun phrase, then a model like Minimal Attachment would cease to be plausible: it would require that readily available referential information, which could lead to successful resolution, be temporarily ignored in favour of non-referential information which would not necessarily lead to a correct choice of analysis. Presumably, then, non-interactive models of language processing such as Rayner et al.'s are plausible only if they also tend away from representational non-autonomy and incremental evaluation.[9]

[9] The argument that non-interactive models provide a plausible account of sentence processing only if we assume a degree of representational autonomy is dependent on the notion that the HSPM is maximally efficient. If the HSPM were not maximally efficient, then it might well ignore information which would lead to the successful resolution of ambiguity.

The interactive model of ambiguity resolution, based on Referential Failure, has been shown to make a number of assumptions concerning the early availability of referential information. These assumptions differ from those which have been associated with non-interactive models such as Minimal Attachment (which we have already seen, in previous chapters, to be lacking in empirical adequacy). They also differ from those which must be associated with "strongly" interactive models. In fact, it is not even clear that there exists a coherent set of such assumptions for strong interactions.

Syntactic Preferences and autonomy

Ford et al. (1982) propose that contextual information may dynamically change the strengths of the lexical forms of a verb.[10] In the cases of (5.2) and (1.1) below, Ford et al. would presumably postulate that context can also change the strengths of syntactic preferences: in a relative-inducing context, the complex NP expansion would have to be re-ordered before the simple NP expansion.

(5.2) The psychologist told the woman that he was worried about to visit him again.

(1.1) The oil tycoon sold the off-shore oil tracts for a lot of money wanted to kill J.R.

Ford et al.'s model is "strongly" interactive: semantics can adjust

[10] For a detailed review of their proposals (excluding their treatment of context), refer to Chapter 2, Section 2.3.4.

the order in which the alternative syntactic rules are first attempted. It requires that a decision is made, for instance, to treat all noun phrases occurring in a particular position as being either simple NPs or complex NPs, depending on the context.[11] In the above example, the processor must decide to reorder the noun phrase expansion rules before it has identified that the critical noun phrase refers to a "woman". If the processor did not reorder the NP expansion rules until after identifying the critical noun phrase, and if it reordered them on the basis of referential considerations, then we might ask what motivation there is for ordering the rules in the first place. But if the NP expansion rules are ordered before the interpretation of the NP, on what basis would they be reordered in the following situation?

(6.6) A psychologist was counselling a man and two women.
.....
He told the ...

In a strongly interactive model, there could be no a priori reason on which to base a re-ordering of the syntactic preferences for noun phrase expansion. If some order were chosen, then there would be no guarantee that it would result in the correct choice of analysis.

[11] In Chapter 2 we considered in detail Ford et al.'s model of Lexical Preference. We saw that all options for rule expansion must be stored on an "option list", and that their principle of Invoked Attachment seems to be equivalent to Minimal Attachment. It follows that the suggestion Ford et al. make with regard to the effects of context is inadequate: even if lexical or syntactic preferences were changed, minimal attachments would still be preferred: the different expansions would be re-ordered on the option list, but Invoked Attachment would still choose the Minimal Attachment.

Ford et al.'s model (and strongly interactive models in general) seems to be subject to the same criticism that was levelled at Minimal Attachment: if the incremental evaluation of noun phrases were possible, then information which would otherwise lead to successful resolution would be ignored. However, although Minimal Attachment might be a strategy which would be well-suited to a model in which such early referential information is not available, the *a priori* re-ordering of syntactic rules would not: there can be no *a priori* basis for re-ordering rules.

6.4 Summary and Conclusions

The present chapter has considered some theoretical issues concerning the alternative models of contextual effects on ambiguity resolution. In Section 6.1 we discussed the proposals advanced by Crain & Steedman (1985) concerning the different kinds of interaction that are possible between syntax and semantics. An extended version of their theory was presented in Section 6.2, which relied on the replacement of their Principle of Referential Success with the "Principle of Referential Failure"

In Section 6.3, Referential Failure, Referential Success, Minimal Attachment, and Lexical Preference were discussed in terms of the assumptions they must make concerning the organisation of the language processor, and specifically, the autonomy of syntax. It was shown that whereas weak interactions in general are compatible with any position along a "dimension" of autonomy, a model incorporating Referential Failure would tend towards the non-autonomy extreme of

this dimension. This is because the Principle of Referential Failure requires that noun phrases be immediately available to semantics for reference evaluation. This would be impossible if the unit of autonomy extended as far as the clause or sentence.

Lexical Preference and Minimal Attachment, on the other hand, tend towards the total autonomy extreme of the autonomy dimension: Minimal Attachment would be a plausible theory if no information but structural information were available with which to resolve these ambiguities. This in itself presumes a particular view of natural language processing, and a particular relationship between syntax and semantics. Consequently, the Minimal Attachment Hypothesis lacks any motivation if viewed within a representationally non-autonomous model. Strongly interactive models (which include Ford et al.'s suggestions for Lexical Preference) lack motivation irrespective of the relationship between syntax and semantics simply because they rely on the existence of some dependable *a priori* basis on which to reorder syntactic rules, when no such basis exists.

There is clearly a certain amount of further work which remains to be done. A number of examples were presented (6.2 and 6.3) which illustrate the different predictions of Referential Failure and Referential Success, and there is much scope for further experimentation. However, it was suggested that there are empirical grounds for favouring a model which assumes "immediacy of referential evaluation" (or "noun phrase interpretation") over one which does not (or indeed, is incompatible with such immediacy). We must conclude,

as a result of these discussions, that the most plausible model of local syntactic ambiguity resolution seems to be the weakly interactive model incorporating Referential Failure and the Principle of Parsimony.

The Principle of Referential Failure has been shown to make strong claims (and predictions) about the resolution of local ambiguity. However, it accounts for parsing preferences only when these examples are preceded by suitable contexts. In Chapter 7 we shall reconsider the account that Crain & Steedman offer with respect to the original null context data with which we started the thesis in Chapter 1. More specifically, we shall re-examine the relationship between the Principle of Parsimony and the null context data. An alternative to Crain & Steedman's account will be offered. We shall also examine the relationship between Parsimony and the Right Association/Late Closure data. In particular, we shall indicate the kinds of question which must be answered before a satisfactory account of the data can be developed.

CHAPTER 7

Implications

The emphasis throughout this study has been on the role of contextual information during the resolution of local syntactic ambiguity. The Principle of Referential Failure, introduced in Chapter 6, is intended to account for parsing preferences in the presence of particular kinds of contextual information. In Section 7.1 we shall revisit Crain & Steedman's account of the null context, and explore an alternative which is perhaps more closely related to the approach to ambiguity resolution which we advocate here.

The approach to ambiguity resolution just alluded to is based primarily on the consideration of the data which have elsewhere been explained by Minimal Attachment (Frazier, 1979; Rayner et al., 1983). In Chapter 2, however, when we considered the range of structural accounts of the resolution of local syntactic ambiguity, we concluded that there are two strategies which these accounts have in common: Minimal Attachment and Right Association/Late Closure. In Section 7.2 we shall consider the applicability of Crain & Steedman's Principle of Parsimony to the phenomena elsewhere explained by Right Association/Late Closure. Discussion will be intended only to illustrate the way in which one might go about approaching, and developing an explanation for, the empirical data. We shall conclude that the data need not necessarily be accounted for in structural terms alone. In Section 7.3 we shall consider some further

implications of the work for research into discourse processing. Section 7.4 is a summary of the thesis, and the study is concluded in Section 7.5.

7.1 The Null Context

The Principle of Referential Failure, reproduced below, relies for its operation on the existence of a set of candidate referents in the hearer's mental model of the discourse. But what if there exists no such set of candidates?

> The Principle of Referential Failure: If a definite referring expression fails to restrict the set of candidate referents in the hearer's mental model of the domain of discourse to a single member, then an analysis which treats subsequent material as a modifier for that referring expression (i.e. as providing information which may lead to further restriction), will be favoured over one that does not.

How do we account for parsing preferences in the null context, when a referent cannot be established, not because there are too many candidate referents (i.e. more than one), but because there are too few candidates (i.e. none)?

Crain & Steedman (1985) propose, of the null context, that an analysis will be adopted which requires fewest "alterations" or "extensions" to the hearer's model (see Section 3.3). In (3.21) below, the reduced relative clause analysis requires more extensions to the discourse model than does the main clause analysis.

(3.21) The horse raced past the barn ... (fell.)

Crain & Steedman's Principle of Parsimony ensures that the main

clause analysis is then preferred simply because it leaves unsatisfied fewer presuppositions.

> "The Principle of Parsimony: If there is a reading which carries fewer unsatisfied but consistent presuppositions or entailments than another, then, other criteria of plausibility being equal, that reading will be adopted as most plausible by the hearer, and the entailments in question will be incorporated in his or her model." (Crain & Steedman, 1982. p.333)

Crain & Steedman's account relies heavily on notions such as "presupposition", "entailment", and "extensions to the model" Accordingly, much hinges on the precise definitions of these notions. If the term "presupposition" is used rather more widely than usual, an alternative account of null context preferences, still based on Parsimony, but not based on notions such as "alteration" and "extension", can be derived.

An informal definition of presupposition might be that a presupposition is anything the speaker assumes to be true of the hearer's mental model of the universe of discourse. If a speaker utters the fragment in (7.1) below, then he or she presupposes that the hearer has available the information that the speaker "met a woman yesterday".

(7.1) The woman whom I met yesterday

We shall say that the relative clause "carries" this presupposition (i.e. the hearer detects the presupposition from the relative clause). In this case, the term "presupposition" applies to a

specific proposition. But the fragment in (7.2) carries a different (more general) presupposition (that is, the hearer can detect, on the basis of the fragment, another presupposition):

(7.2) The woman whom ...

In this case, the hearer detects the following: the speaker presupposes, amongst other things, (i) that there exists in the hearer's model an individual "woman", and (ii) that there exists in this model at least one proposition which can apply to this individual (although just which proposition is as yet unspecified). For the sake of a label, we shall say that such a fragment "promises Given information": the speaker presupposes that what follows in the utterance will be information which can be successfully interpreted as Given information by the hearer. We shall now apply this extended use of "presupposition" to the resolution of local syntactic ambiguity in the null context.

The terms Given and New, as used here, are dependent for their definition on the existence of a model of the discourse: Given information is information which is already established in the model, and New information is information which is to be added to it. In the null context, however, no prior model exists. Nevertheless, the absence of such a model does <u>not</u> mean that the distinction between Given and New must be abandoned. It merely means that <u>nothing is Given</u>.[1] The HSPM must therefore treat <u>all</u> incoming material as if it

[1] With the exception of "general world knowledge" this is true.

provides New information. If there is a choice between an analysis which presupposes that certain information is Given and an analysis which does not, then the processor must, as determined by the Principle of Parsimony, choose the latter.

This model makes straightforward predictions about the structures we have been discussing. If (as in 7.3 below) there is a choice between a relative clause analysis (which "promises" Given information) and a complement clause analysis (which need not be interpreted as Given), then the processor must choose the complement clause analysis.

(7.3) The psychologist told the woman that .

If there is a choice (cf. 7.4) between the NP-attachment of a prepositional phrase (Given information) or the VP-attachment of a prepositional phrase, then the processor must in the null context choose the VP-attachment.

(7.4) The burglar blew open the safe with .

If there is a choice (cf. 7.5) between a subordinate verb, which signals a reduced passive/relative (Given information) and a main clause verb, then the processor must in the null context choose the main clause analysis.

Discourses which begin "The Prime Minister reported.." are more acceptable than those beginning "The minister reported..". In other words, in the null context, nothing can be Given with respect to a <u>discourse</u> model, although it could be Given with respect to a general world model.

(7.5) The oil tycoon sold ...

And in general, if there is a choice between a complex NP, which implicates additional Given information by which the candidate referents are distinguishable in the model, and a simple NP, then in the null context the simple NP analysis must be chosen.

This account of the null context data applies only to materials containing definite referring expressions. We might predict that the HSPM's preferences in favour of simple NPs would be reduced when it encounters indefinite referring expressions, which do not implicate additional Given information. In an experiment to investigate the effects of indefiniteness on ambiguity resolution, Crain (1980) collected grammaticality judgements for examples like the following.

(7.6) The teachers taught by the Berlitz method passed the test.

(7.7) Teachers taught by the Berlitz method passed the test.

Examples like (7.6), containing the definite referring expression "the teachers", were judged ungrammatical significantly more often than examples like (7.7), containing the indefinite referring expression "teachers".

It would seem, then, that we can explain the null context data without necessary recourse to a structural account. The increased latencies which were observed in Experiment 1 for the null context conditions are explained in a manner similar to that introduced in Chapter 4 (Section 4.5.2). In Experiment 2, we found that if a

presupposition concerning Given and New was violated, then processing load (as measured via reading times) increased (presumably in an attempt to "recover" from the violation). In the null context, many such presuppositions are violated, and hence the increased processing times required. In general, we might assume that processing time increases as more presuppositions are violated.

The new account of the null context differs significantly from that offered by Crain & Steedman, even though their Principle of Parsimony is still considered fundamental. Where the two accounts differ is in the expanded notion of "presupposition" which the current explanation uses. Crain & Steedman's account of the null context is correct insofar as the analysis which requires fewest extensions is the preferred analysis. The claim here, however, is that it is not the number of extensions per se which determines the chosen analysis. If analyses were rejected according to the number of presuppositions that are violated, then the HSPM would have to consider the entire sentence before totaling up the number of unsatisfied presuppositions of each alternative analysis. The present account does not require that the entire sentence be considered before making a decision (cf. the replacement of Crain & Steedman's Principle of Referential Success by the Principle of Referential Failure). In the case of the Minimal Attachment data, a decision is made as soon as a noun phrase is encountered: a complex NP analysis presupposes additional Given information, and is therefore rejected in favour of the simple NP analysis.

The account of the null context which we have just developed stresses the importance of the HSPM's attempts to minimise the number of presuppositions which are violated when a sentence is processed within an infelicitous context. It is therefore simply an extension of Crain & Steedman's Principle of Parsimony, just as the Principle of Referential Failure is itself also a special case of this principle. In the following section we shall speculate about whether the Principle of Parsimony might be extendable to account not just for the Minimal Attachment data, but also for the Right Association/Late Closure data.

7.2 Right Association/Late Closure

In (2.2), which we first discussed in Chapter 2, Right Association correctly predicts the preference for associating the adverbial "yesterday" to the verb "died" rather than to the verb "thought".

(2.2) Blake thought Stephen died yesterday.

Right Association, and the equivalent Late Closure, are reproduced below.

> Right Association: "Terminal symbols optimally associate to the lowest nonterminal node." (Kimball, 1973, p.24)

> "Late Closure: When possible, attach incoming material into the clause or phrase currently being parsed." (Frazier, 1979. p.76)

Right Association applies in cases where a constituent (such as an

adverbial) can optionally attach to more than one site in the phrase marker.[2] Like Minimal Attachment, neither Right Association nor Late Closure permit any role for contextual considerations. If, as Crain & Steedman hypothesise, there are no garden paths (and hence no parsing preferences) which cannot be overcome by the provision of adequate contexts, and if we intend to overcome the preference for right association, then we must define the felicity conditions for each of the alternative attachments of the target constituent.

The Principle of Referential Failure is clearly inapplicable in cases such as (2.2), and is therefore unable to provide any clues as to just what these conditions may be. However, the approach we have taken during the course of this study is about more than just noun phrase reference. It is about the process of establishing just what is, and what isn't, already represented in the hearer's mental model of the domain of discourse. The act of establishing referential success or failure is an example of one such process. Another example was encountered in Chapter 4, when it was found that reading times associated with the modification of a verb by an optional prepositional phrase were sensitive to Given/New considerations.

[2] We shall ignore examples such as

(7.8) While John was eating his toast became cold.

There is a sense in which examples presented like these can be considered to be ungrammatical. They lack the usual linguistic cues which are normally provided: punctuation in the written form, and intonation in the spoken form.

But can the distinction between Given and New provide a satisfactory account of the felicity conditions which must be satisfied by each alternative attachment? It would be wrong to deduce from the results of Experiments 1 and 2 that the action denoted by a verb must be Given if it is to be modified by an optional adverbial (for present purposes we take anything that can modify a verb to be "adverbial"). There exist occasions when a verb can be modified by an adverbial without being Given:

(7.9) J.R. bumped into Cliff and Jamie at the Oil Barons' ball. As usual, they all got into a fight. It started with Cliff insulting J.R., and ended when Jamie grabbed a bottle of champagne and hit J.R. with it.

In this example, "with it" modifies the action "hit J.R.", even though this action is not Given. Nonetheless, it does appear to be the case, as Experiments 1 and 2 demonstrate, that the modification of a verb is on occasion sensitive to Given/New considerations. One might argue, for instance, that in the null context the preferred attachment of "yesterday" in (2.2) is brought about by the HSPM's attempts to minimise the violations concerning the Given/New structure of the sentence. Figure 7.1 illustrates one way in which to interpret the different attachments of "yesterday": only if the processor attaches "yesterday" to "Stephen died.." does it minimise the amount of material presupposed to be Given.

The intended attachment of the target constituent ("yesterday" in 2.2) might therefore be determined on the basis of what is Given and what is New (whether in the null context or in a discourse). A

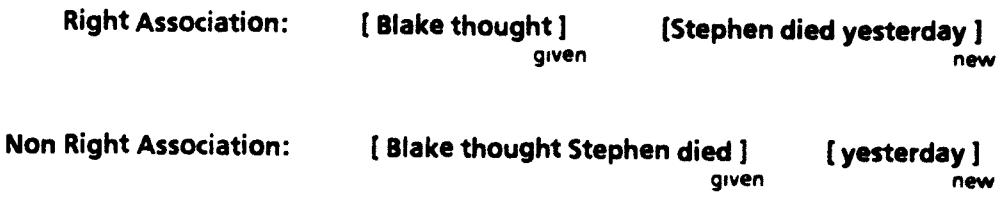


FIGURE 7.1

Given/New structures for the sentence "Blake thought Stephen died yesterday"

drawback with this account is that, unlike the explanation of minimal attachments in the null context, it does not provide an explicit statement of the HSPM's decisions as it processes the sentence from left to right. However, it does suggest that one might be able to develop an account of the Right Association/Late Closure data without recourse to structural considerations alone. The exact form that such an account would take is unclear, because Given/New considerations are not the only factor which might influence the parser's decisions.

For instance, in (2.11) below, the Given/New structure of the sentence does not provide any obvious clues as to why there is a preferred attachment.

(2.11) Joe bought the book that I had been trying to obtain for Susan.

In this example, "for Susan" can attach either to the verb complex within the relative clause, or to "bought". Given that the that-clause can only be interpreted as a relative clause, it must, in the null context, violate a number of presuppositions concerning Given and New: the entire string "... that I had been trying to obtain ..." is presupposed to provide Given information. The proper names "Joe" and "Susan" also violate such presuppositions (unless the reader is already familiar with these two characters). At issue, though, are the additional presuppositions which are violated by each of the alternative attachments of "for Susan". It is unclear why the preferred attachment of "for Susan" to "trying to obtain" should

minimise the number of presuppositions violated.[3]

Materials such as these are complicated by the possibility that several factors may conspire to induce a preferred attachment. For instance, in the case of spoken discourses, intonation may play an important role in influencing the particular attachment chosen by the hearer (e.g. Lehiste, 1973; Lehiste et al., 1976; Berkovits, 1981; Briscoe, 1984). But a preference for attachment over another might also arise as a result of lexical influences. The following example does not appear to exhibit the preference, predicted by Right Association/Late Closure, for attaching "yesterday" to "Stephen died.

(7.10) Blake first heard Stephen died yesterday.

Ford, Bresnan, & Kaplan (1982) propose that lexical preferences are brought about by a preferred rule ordering associated with each verb (see Section 2.3.4). There is an issue as to whether this is in fact the case. It may be that pragmatic information, and not lexico-syntactic information, is responsible for such effects.

Clearly, there are a number of factors which influence the HSPM's decision to modify one verb rather than another. The important point is that these factors are not necessarily structural

[3] It is of course conceivable that certain attachments are preferred because some violations are less "costly" than others, in which case it is not simply the number of violations which must be taken into account.

in origin. It is by no means obvious that the data should only be explained in structural terms. Crain & Steedman's Principle of Parsimony could account for the Right Association/Late Closure data if we assume that each alternative attachment of an optional constituent carries with it different presuppositions concerning the state of the hearer's/reader's beliefs. If the presuppositions implicated by a particular attachment are satisfied, then that attachment will be favoured over one whose associated presuppositions are not satisfied. The problem is to discover the nature of these presuppositions. Some of these presuppositions may be related to notions like Given and New. There is a clear need to experiment (both in theory and in practice) with such notions, and hence determine their applicability to the study of syntactic ambiguity resolution.

7.3 Considerations for Foregrounding

The notion of reference to a discourse model has had an important role to play during the course of this study, even though the study has not been concerned with reference per se. It has been sufficient to assume that there exist processes which can establish what is, and what is not, represented in the hearer's model of the discourse. Just how, and on what basis, these processes operate was not the concern of this study. However, there have been times when it has been necessary to provide a specification of the contextual information to which the resolution of local syntactic ambiguity is sensitive (as in Experiments 1 to 5). For instance, in the case of

(7.11), a "complement-inducing" context was defined as one which introduced just one "woman" into the discourse.

(7.11) The psychologist told the woman that he was having trouble with

A "relative-inducing" context was defined as one which introduced more than one "woman". A referring expression such as "the woman" should not, in the relative-inducing context, provide sufficient information with which to identify the intended "woman". However, such a view is overly simplistic, as according to this definition, the text in (7.12) constitutes a relative-inducing context. But if we follow (7.12) by (7.13), which contains the simple NP "the woman", we see that it clearly <u>is</u> possible to identify which "woman" is intended.

(7.12) a. A psychologist was counseling <u>two</u> <u>women</u>.
b. One of them was in tears.
c. She couldn't stop wailing.
d. The psychologist became impatient.

(7.13) He suddenly stood up and shouted at <u>the</u> <u>woman</u> to leave.

In (7.12) more than one "woman" has been introduced, but the presence of sentences (b) and (c) has led one of these candidate "women" to be "foregrounded" (Chafe, 1972; 1976). To use Chafe's own analogy (Chafe, 1972): within the hearer's (and presumably also speaker's) representation of the discourse is a "stage" on which events "unfold", and onto which, or off from which, the major "characters" move. There are thus at least two kinds of "object" in the

representation at any one moment in time: those that are "on-stage", and those that are "off-stage". If there is more than one candidate to which an expression could be used to refer, then whichever candidate is "on-stage" is chosen.

This is not the place to consider what determines whether a "character" is to move "on or off stage" (see Chafe, 1976). However, the approach to ambiguity resolution we have adopted provides a tool by which such effects can be investigated. In the case of (7.11) - reproduced below - a complement clause analysis should be chosen only if a unique referent can be found for "the woman" (on the assumption that the Principle of Referential Failure does guide the resolution process).

(7.11) The psychologist told the woman that he was having trouble with ...

If more than one "woman" had previously been introduced by a co-text, but subjects were nonetheless to choose the complement analysis, then this would suggest that one of the candidate women had somehow become more salient (i.e. foregrounded). By systematically manipulating the texts which come between the introductory sentence, and the final locally ambiguous target sentence, it should be possible to investigate the determinants of foregrounding.[4]

[4] Sanford & Garrod (1981) describe a paradigm in which some effects on foregrounding were investigated by means of a reading time measure (see also Garrod & Sanford, 1982): "... by measuring the ease of making a coreferential mapping, a number of ... ideas can be examined in the context of multi-sentence processing. ... When an entity is no longer foregrounded, it can be referred to by an

There are, of course, many issues other than foregrounding which must be addressed when considering reference. For instance, one issue concerns the use of general world knowledge as the basis for inferring the existence of a referent (see Sanford & Garrod, 1981, for a review of such issues). Although not the immediate concern of the present study, it is yet another example of the complexity of the processes ultimately responsible for determining whether or not a referring expression is referentially successful.

It would appear, then, that the resolution of local syntactic ambiguity must be under the influence of referential processes which establish not just whether something is represented in the hearer's representation of the discourse, but also how it is represented, or on what basis it can be inferred.

7.4 Summary

The aim of this study has been to investigate the basis on which the HSPM resolves local syntactic ambiguities. In Part I, we considered some existing approaches to the problem. On the one hand is the structural approach, in which syntactic ambiguities are resolved on the basis of syntactic information alone. A number of structural models were presented in Chapter 2. We concluded that

unambiguous pronoun only with difficulty." (Sanford & Garrod, 1981. p. 137). They describe an experiment in which reading times were collected for target sentences containing either a pronoun or a definite noun phrase. Different effects on foregrounding were found depending on the specific manipulations of the text between the target sentence and the sentence which contained the antecedent to the pronoun/noun phrase.

there are just two strategies that the various models have in common: Minimal Attachment (Frazier, 1979; and see Kimball, 1975) and Right Association (Kimball, 1973). The alternative approach to syntactic ambiguity resolution, in which different kinds of information (some of it non-syntactic) are brought to bear during the resolution process, was discussed in Chapter 3. In particular, we considered the interactive model of Crain & Steedman (1985). They hypothesise that there is no parsing preference which cannot be overturned by the provision of adequate contextual information. They propose that the HSPM abides by the Principle of Referential Success, and the more general Principle of Parsimony. This is in direct contrast to the structural accounts, which claim that parsing preferences are syntactic in origin, and cannot be influenced by semantic/pragmatic factors.

In Part II of the thesis, we considered the empirical adequacy of these two sets of claims. Two experiments were described in Chapter 4 which used materials similar to those used in a study by Rayner, Carlson, & Frazier (1983). In that study, it was claimed that semantic/pragmatic information did not influence the parsing preferences, and that these preferences were in fact attributable to the operation of Minimal Attachment. The flaw in Rayner et al.'s argument is that they manipulated only intra-sentential plausibility. But Crain & Steedman's Referential Success hypothesis suggests that extra-sentential factors are critical. Experiments 1 and 2 in Chapter 4 took this consideration into account. The data suggested that semantic/pragmatic factors <u>can</u> influence the HSPM's choice of

analysis.

In Chapter 4 it was pointed out that there is more to sentence complexity than just syntactic reanalysis, and that reading time experiments must take into account the possibility that the reading time data may reflect factors which are unrelated to the process responsible for the resolution of local syntactic ambiguity. In Chapter 5 we returned to the processing of relative clauses, and considered some early data (Wanner, Kaplan, & Shiner, 1974) which had been taken as support for structural models of syntactic ambiguity resolution. In Experiment 3 it was shown that relative clauses do invoke longer reading times than complement clauses once context has been controlled for: a strong effect of referential context was found, but there was also a strong main effect of target construction. In Experiments 4 and 5, we demonstrated that these effects could be eliminated by taking into account the additional processing that restrictive relative clauses require. Experiment 5 demonstrated that the residual differences between relative clauses and complement clauses could be eliminated once the amount of inferencing which is required during the processing of a relative clause was minimised.

Throughout Part II, we considered whether the Minimal Attachment Hypothesis was compatible with the experimental results. There were a number of features of the data which suggested that it was not: in all five experiments strong effects of referential context were found on reading times. It was concluded that such data were inconsistent

with the Minimal Attachment Hypothesis. Moreover, it was concluded that the empirical and computational motivation behind this hypothesis would be put to question if an attempt was made to reconcile the data with the hypothesis. On the other hand, the data were not inconsistent with the Referential Success Hypothesis.

In Part III, we turned to some theoretical considerations concerning Crain & Steedman's original proposals. In Chapter 6, we considered the timing of the HSPM's decisions during the resolution of local syntactic ambiguity. The discussion begun by considering the different kinds of interaction that might be possible between syntax and semantics. We contrasted "weak" (or "selective") interactions with "strong" (or "instructive") interactions. Crain & Steedman advocate a model of language processing which exhibits only weak interactions, in which syntax independently proposes different analyses between which semantics can then choose. They proposed that semantics chooses between the alternative analyses on the basis of their Principle of Referential Success. It was suggested, however, that this principle made inadequate predictions concerning the timing of the HSPM's decisions. An alternative principle, the Principle of Referential Failure, was offered in its place.

We also considered in Chapter 6 the relationship between syntax and semantics, and the implications for this relationship of the various models we have considered during the course of this work. We concluded that a model based on the Principle of Referential Failure would tend towards a model in which the units of autonomy (i.e. the

unit of syntactic representation which is passed up to semantics for interpretation) could not extend to beyond noun phrases. It was suggested that the Minimal Attachment Hypothesis would be motivated if the unit of autonomy did extend to beyond phrases, but that the empirical evidence favour models of language processing in which noun phrases can be interpreted as soon as they are encountered.

In the present chapter, we considered some further implications of the discussions which took place in the preceding chapters. The approach we have developed relies on the distinction between what is, and what is not, already known to the hearer. We saw that this approach makes strong claims about the null context data which we first encountered in Chapter 1. These differ significantly from Crain & Steedman's original suggestions concerning the null context, although their Principle of Parsimony is still adhered to. We considered whether Parsimony could also be extended to account for the Right Association/Late Closure strategy which we first encountered in Part I. Our conclusion, although only tentative, was that Parsimony could be extended to apply to the Right Association/Late Closure data, but that much rests on being able to identify the presuppositions which are implicated by the modification of different kinds of verb.

7.5 Conclusions

With regard to the resolution of local syntactic ambiguities which have elsewhere been explained by Minimal Attachment, we are left with an approach which does not rely on separate mechanisms to

account for the null context data and the contextual data. This approach developed through a consideration of the conversational purpose underlying a speaker's use of different syntactic constructions. Previous studies have taken the view that syntactic processing is only about relations which hold between individual items in a single sentence. But such studies often fail to question the "purpose" behind the speaker's use of a particular construction. There is a sense in which the speaker's intention is to evoke a specific set of processes in the hearer. This is achieved, in part, by way of the syntactic constructions which the speaker chooses to employ. The role of these processes is to establish a relationship between the information that is conveyed by the utterance and the information that is already known to the hearer. Clearly such processes cannot operate on information which is only contained within the single utterance. They must also address information which is external to the utterance. In other words, the processes underlying the comprehension of natural language require more than just single, isolated, sentences over which to operate.

The study has aimed to do more, however, than simply list the criteria by which the processor decides to execute one process rather than another. It has also been concerned with the timing of the processor's decisions. The intention to provide such a dual specification is shared with computational models of natural language processing. It follows that the results reported here, and the theoretical discussions concerning the status of syntax within the language processor, have a contribution to make to computational

models of syntactic ambiguity resolution.

The present results also offer to computational models the sort of psycholinguistic consideration which they are faulted for ignoring. For instance, SHRDLU (Winograd, 1972) used referential information in order to determine whether to continue with, or abandon, the analysis of a noun phrase as simple or complex. Dresher & Hornstein (1976) suggested that

> "The question here ... is why the components interact in just this manner. How do we know that the semantic component can intervene into the parsing of a noun group 'to see whether it is worth continuing'? Winograd cites no empirical evidence that this is the way that parsing works, nor does he show that it captures any generalizations that could not otherwise be captured." (Dresher & Hornstein, 1976, p.354)

Dresher & Hornstein went on to suggest that research in Artificial Intelligence has failed to contribute to what they call "the scientific study of language" simply because researchers in Artificial Intelligence have failed to consider issues such as those mentioned in the excerpt.[5] The present study, on the other hand, <u>has</u> considered these issues.

Winograd had a number of goals in mind when he devised the SHRDLU program. One of these was to create a usable language-understanding system. Another was to explore different linguistic behaviours. The study of such behaviours requires empirical investigation. Studies such as the present one suggest that hearers

[5] See also Dresher & Hornstein (1977).

are sensitive to presuppositions assumed to be true by the speaker. Indeed, it has been noted[6] that the speaker expects the hearer to appreciate which presuppositions are involved in the use of particular forms in particular contexts, and so acts accordingly. So even if the aim of computational linguistics is merely to devise usable language-understanding systems (as opposed to modelling human linguistic behaviour), then it follows that these systems should be sensitive to such presuppositions which motivate their human users. It follows that any study of human language behaviour which attempts to identify the manner in which the hearer makes use of such presuppositions can contribute to research in computational linguistics. Whether or not this will represent a practical contribution in the short term is, in a sense, an empirical question.

But Dresher & Hornstein were not concerned with the practicalities of building usable language-understanding systems. They were instead concerned with the ability (or lack of ability) of theories to be able to "explain why things are the way they are and not some other way" (p.355). In Section 7.1 we discussed an account of the null context data which, like the Minimal Attachment Hypothesis, predicts that simple noun phrases will be preferred to complex noun phrases. Moreover, this account, based on Crain & Steedman's Principle of Parsimony, explains why "things are the way they are" in a way that the Minimal Attachment Hypothesis does not.

[6] Bard, personal communication.

To do this, it questions the significance of the distinction between simple and complex noun phrases: there is more to this distinction than just numbers of NP nodes. Minimal Attachment does not capture any generalizations in the sense advocated by Dresher & Hornstein: it merely describes certain linguistic behaviours.

The Principle of Parsimony also describes certain linguistic behaviours, but at a different more explanatory level of description. The relevance of these different levels of description is as follows: The purely structural vocabulary within which Minimal Attachment is framed does not permit generalisation from the null context condition to the explicit prior discourse condition without the introduction of new terms (such as "reference"). But these new terms would belong to a level of description which is different to the syntactic level to which Minimal Attachment belongs. On the other hand, the vocabulary within which the Principle of Parsimony is framed <u>does</u> permit generalisation from null to non-null context conditions. Of course, the Principle of Referential Failure could be implemented in a number of different ways, and at a number of different levels (depending, for instance, on the degree of Formal autonomy exhibited by the particular model). But the underlying account would reside at the level of description which permits, amongst others, notions such as "reference to discourse models", "candidate referent sets", "restriction", and indeed "context".

In conclusion, whereas previous studies of syntactic ambiguity resolution have ignored the role of context during the interpretation

of natural language, the present study has not. Instead, we have developed a model of the resolution process (as applied to one particular class of ambiguity) which not only accounts for contextually induced parsing preferences, but also explains the preferences in the null context, with which previous studies have been so concerned. Investigations of the human language processor which ignore the role of context are open to the charge that they study something other than natural language processing.

APPENDIX A

The following are the materials used in Experiment 1. In the "null context" conditions, the target sentences were presented without any prior co-text.

1 "VP-Inducing" Co-text

A mechanic walked up to a car with a monkey wrench. He needed to change one of the tyres and also the carburetor. One of them had a faulty valve and the other had a small hole in it.

1 "NP-Inducing" Co-text

A mechanic walked up to a car with a monkey wrench. He needed to change two of the tyres. One of them had a faulty valve and the other had a small hole in it.

1 NP-attached Target Sentence

The mechanic changed the tyre with the faulty valve
but it took a long time.

1 VP-attached Target Sentence

The mechanic changed the tyre with the monkey wrench
but it took a long time.

2 "VP-Inducing" Co-text

A convict was clutching a knife. A warder and the prison governor were trying to take it away from him. One of them held a bunch of keys but the other had both his hands free.

2 "NP-Inducing" Co-text

A convict was clutching a knife. Two warders were trying to take it away from him. One of them held a bunch of keys but the other had both his hands free.

2 NP-attached Target Sentence

The convict suddenly lunged at the warder with the keys before running off.

2 VP-attached Target Sentence

The convict suddenly lunged at the warder with the knife before running off.

3 "VP-Inducing" Co-text

A schoolteacher walked into class carrying a bamboo cane. He decided to beat a boy and a girl for not standing up quickly enough when he came into the room. One of them had a broken leg but the other had just been lazy.

3 "NP-Inducing" Co-text

A schoolteacher walked into class carrying a bamboo cane. He decided to beat two boys for not standing up quickly enough when he came into the room. One of them had a broken leg but the other had just been lazy.

3 NP-attached Target Sentence

The schoolteacher beat the boy with the broken leg but the boy didn't cry out.

3 VP-attached Target Sentence

The schoolteacher beat the boy with the bamboo cane
but the boy didn't cry out.

4 "VP-Inducing" Co-text

John had bought a new stylus for his stereo. He was comparing a record and a cassette. One of them produced a lot of noisy crackle while the other produced a lot of hiss.

4 "NP-Inducing" Co-text

John had bought a new stylus for his stereo. He was comparing two records. One of them produced a lot of noisy crackle while the other produced a lot of hiss.

4 NP-attached Target Sentence

John decided to play the record with the noisy crackle
but it sounded terrible.

4 VP-attached Target Sentence

John decided to play the record with the new stylus
but it sounded terrible.

5 "VP-Inducing" Co-text

A historian was going out to buy a magnifying glass. He had been examining a map and a manuscript. One of them had an appalling tear in it though the other seemed in perfect condition.

5 "NP-Inducing" Co-text

A historian was going out to buy a magnifying glass. He had been examining two maps. One of them had an appalling tear in it though the other seemed in perfect condition.

5 NP-attached Target Sentence

The historian had to study the map with the appalling tear so as to value it.

5 VP-attached Target Sentence

The historian had to study the map with the magnifying glass so as to value it.

6 "VP-Inducing" Co-text

An artist was working with a long brush. On his table were a bottle and a small jug. One of them was full of paint and the other was full of water.

6 "NP-Inducing" Co-text

An artist was working with a long brush. On his table were two bottles. One of them was full of paint and the other was full of water.

6 NP-attached Target Sentence

The artist accidently knocked over the bottle with the paint and swore.

6 VP-attached Target Sentence

The artist accidently knocked over the bottle with the brush and swore.

7 "VP-Inducing" Co-text

A window cleaner was rinsing out his cloth. He had a window and a skylight left to clean. One of them was covered in dirt but the other was relatively clean.

7 "NP-Inducing" Co-text

A window cleaner was rinsing out his cloth. He had two windows left to clean. One of them was covered in dirt but the other was relatively clean.

7 NP-attached Target Sentence

The window cleaner wiped the window with the dirt while singing to himself.

7 VP-attached Target Sentence

The window cleaner wiped the window with the cloth while singing to himself.

8 "VP-Inducing" Co-text

A burglar carrying some dynamite broke into an heiress's house. Once inside he found a safe and a jewelry box. One of them had some diamonds inside whilst the other had several priceless emeralds.

8 "NP-Inducing" Co-text

A burglar carrying some dynamite broke into an heiress's house. Once inside he found two safes. One of them had some diamonds inside whilst the other had several priceless emeralds.

8 NP-attached Target Sentence

The burglar blew open the safe with the diamonds and made off with the loot.

8 VP-attached Target Sentence

The burglar blew open the safe with the dynamite and made off with the loot.

9 "VP-Inducing" Co-text

A workman was setting up a blow torch. He was working on a section of pipe and a valve. One of them was leaking gas but the other seemed ok.

9 "NP-Inducing" Co-text

A workman was setting up a blow torch. He was working on two sections of pipe. One of them was leaking gas but the other seemed ok.

9 NP-attached Target Sentence

The workman cut through the pipe with the gas leak before fitting a new one.

9 VP-attached Target Sentence

The workman cut through the pipe with the blow torch before fitting a new one.

10 "VP-Inducing" Co-text

A man was walking up and down a dark street with an iron crowbar under his coat. He seemed to be watching a shop and a post office suspiciously. One of these had a broken window but the other was completely secure.

10 "NP-Inducing" Co-text

A man was walking up and down a dark street with an iron crowbar under his coat. He seemed to be watching two shops suspiciously. One of these had a broken window but the other was completely secure.

10 NP-attached Target Sentence

The man broke into the shop with the broken window
but nobody saw him.

10 VP-attached Target Sentence

The man broke into the shop with the iron crowbar
but nobody saw him.

11 "VP-Inducing" Co-text

A monkey had been trained to eat using a knife and a fork. It was given a banana and an apple to eat. One of these had a bruise on it but the other was perfect.

11 "NP-Inducing" Co-text

A monkey had been trained to eat using a knife and a fork. It was given two bananas to eat. One of these had a bruise on it but the other was perfect.

11 NP-attached Target Sentence

The monkey ate the banana with the bruise
much to everyone's surprise.

11 VP-attached Target Sentence

The monkey ate the banana with the fork
much to everyone's surprise.

12 "VP-Inducing" Co-text

A caretaker was carrying some nails. He had a door and a locker to repair. One had a large crack in it and the other wouldn't shut properly.

12 "NP-Inducing" Co-text

A caretaker was carrying some nails. He had two doors to repair. One had a large crack in it and the other wouldn't shut properly.

12 NP-attached Target Sentence

The caretaker repaired the door with the crack
and then took a tea break.

12 VP-attached Target Sentence

The caretaker repaired the door with the nails
and then took a tea break.

13 "VP-Inducing" Co-text

A skinhead was walking along a street carrying a large knife. He was stopped by a policeman and a policewoman. One of them had a scar though the other had no such distinguishing feature.

13 "NP-Inducing" Co-text

A skinhead was walking along a street carrying a large knife. He was stopped by two policemen. One of them had a scar though the other had no such distinguishing feature.

13 NP-attached Target Sentence

The skinhead suddenly attacked the policeman with the scar
and then ran off.

13 VP-attached Target Sentence

The skinhead suddenly attacked the policeman with the knife
and then ran off.

14 "VP-Inducing" Co-text

A dressmaker was looking for her scissors. She had with her a piece of material and also a bit of leather. One of these had a pattern on it but the other was completely plain.

14 "NP-Inducing" Co-text

A dressmaker was looking for her scissors. She had with her two pieces of material. One of these had a pattern on it but the other was completely plain.

14 NP-attached Target Sentence

The dressmaker cut the material with the pattern
and then went to have lunch.

14 VP-attached Target Sentence

The dressmaker cut the material with the scissors
and then went to have lunch.

15 "VP-Inducing" Co-text

A fireman carrying a heavy axe was running to the scene of a fire. He was going to have to break through a door and a window. One of these had a very rusty lock and the other was nailed shut.

15 "NP-Inducing" Co-text

A fireman carrying a heavy axe was running to the scene of a fire. He was going to have to break through two doors. One of these had a very rusty lock and the other was nailed shut.

15 NP-attached Target Sentence

The fireman smashed down the door with the rusty lock
but smoke overcame him.

15 VP-attached Target Sentence

The fireman smashed down the door with the heavy axe but smoke overcame him.

16 "VP-Inducing" Co-text

A little girl borrowed a sharp knife from the kitchen. She was about to eat an an orange and a tangerine. One of the fruit had a thick rind while the other's seemed very thin.

16 "NP-Inducing" Co-text

A little girl borrowed a sharp knife from the kitchen. She was about to eat two oranges. One of the fruit had a thick rind while the other's seemed very thin.

16 NP-attached Target Sentence

The little girl tried to cut the orange with the thick rind but to no success.

16 VP-attached Target Sentence

The little girl tried to cut the orange with the sharp knife but to no success.

17 "VP-Inducing" Co-text

A chemist turned on his bunsen burner. He needed to heat up both a solution and some powder. One of them contained some blue crystals and the other some red dye.

17 "NP-Inducing" Co-text

A chemist turned on his bunsen burner. He needed to heat up two solutions. One of them contained some blue crystals and the other some red dye.

17 NP-attached Target Sentence

The chemist heated the solution with the blue crystals but the gas went out.

17 VP-attached Target Sentence

The chemist heated the solution with the bunsen burner but the gas went out.

18 "VP-Inducing" Co-text

A drunk was on the underground wielding a heavy stick. He was acting violently towards a man and a woman. One of them had a stammer but the other's speech was fine.

18 "NP-Inducing" Co-text

A drunk was on the underground wielding a heavy stick. He was acting violently towards two women. One of them had a stammer but the other's speech was fine.

18 NP-attached Target Sentence

The drunk hit the woman with the stammer as she got off the train.

18 VP-attached Target Sentence

The drunk hit the woman with the stick as she got off the train.

19 "VP-Inducing" Co-text

An office executive was trying to get his telephone to work. He wanted to get through to a man and a woman who worked for him. One of them had been shortlisted for promotion while the other hadn't been turning up regularly for work.

19 "NP-Inducing" Co-text

An office executive was trying to get his telephone to work. He wanted to get through to two men who worked for him. One of them had been shortlisted for promotion while the other hadn't been turning up regularly for work.

19 NP-attached Target Sentence

The executive tried to call the man on the shortlist
but couldn't get through.

19 VP-attached Target Sentence

The executive tried to call the man on the telephone
but couldn't get through.

20 "VP-Inducing" Co-text

A university lecturer was travelling to a conference. He had taken with him an article and a book to read. One of them was about a new discovery while the other was a review paper.

20 "NP-Inducing" Co-text

A university lecturer was travelling to a conference. He had taken with him two articles to read. One of them was about a new discovery while the other was a review paper.

20 NP-attached Target Sentence

The lecturer didn't read the article on the discovery
as he'd seen it before.

20 VP-attached Target Sentence

The lecturer didn't read the article on the journey
as he'd seen it before.

21 "VP-Inducing" Co-text

A demolition man had hired a bulldozer for the day. He wasn't sure which of a house and a church to demolish. One of these had an ornate statue whilst the other had some very fine carvings.

21 "NP-Inducing" Co-text

A demolition man had hired a bulldozer for the day. He wasn't sure which of two houses to demolish. One of these had an ornate statue whilst the other had some very fine carvings.

21 NP-attached Target Sentence

The man knocked down the house with the statue
but it wasn't the right one.

21 VP-attached Target Sentence

The man knocked down the house with the bulldozer
but it wasn't the right one.

22 "VP-Inducing" Co-text

A farmer was going out to work on his tractor. He wanted to plough a field and then clean out an old yard. One was covered in clover and the other was covered in mud.

22 "NP-Inducing" Co-text

A farmer was going out to work on his tractor. He wanted to plough two fields. One was covered in clover and the other was covered in mud.

22 NP-attached Target Sentence

The farmer ploughed the field with the clover
but rain forced him to stop.

22 VP-attached Target Sentence

The farmer ploughed the field with the tractor
but rain forced him to stop.

23 "VP-Inducing" Co-text

A doctor walked into a waiting room with a stethoscope around his neck. A woman and a man were waiting to see him. One of them had a temperature and the other had stomach pains.

23 "NP-Inducing" Co-text

A doctor walked into a waiting room with a stethoscope around his neck. Two women were waiting to see him. One of them had a temperature and the other had stomach pains.

23 NP-attached Target Sentence

The doctor examined the woman with the temperature
but couldn't help her.

23 VP-attached Target Sentence

The doctor examined the woman with the stethoscope
but couldn't help her.

24 "VP-Inducing" Co-text

A student was about to take a short train journey. She had chosen a book and a magazine to take with her. One of them had been on a reading list for a course in Contemporary English while she'd only bought the other at the last minute.

24 "NP-Inducing" Co-text

A student was about to take a short train journey. She had chosen two books to take with her. One of them had been on a reading list for a course in Contemporary English while she'd only bought the other at the last minute.

24 NP-attached Target Sentence

The student decided to read the book on the list
in order to pass her exam.

24 VP-attached Target Sentence

The student decided to read the book on the train
in order to pass her exam.

25 "VP-Inducing" Co-text

A woman was looking around for her darning wool. She had a sock and a jumper to repair. One of these had a hole in it but the other was just getting a bit thin.

25 "NP-Inducing" Co-text

A woman was looking around for her darning wool. She had two socks to repair. One of these had a hole in it but the other was just getting a bit thin.

25 NP-attached Target Sentence

The woman repaired the sock with the wool
but pricked her finger in the process.

25 VP-attached Target Sentence

The woman repaired the sock with the hole
but pricked her finger in the process.

26 "VP-Inducing" Co-text

A little boy was playing with a ball. He was very close to both a window and a statue. One of these had a crack in it and the other was very old.

26 "NP-Inducing" Co-text

A little boy was playing with a ball. He was very close to two windows. One of these had a crack in it and the other was very old.

26 NP-attached Target Sentence

The boy managed to break the window with the crack
and so ran home.

26 VP-attached Target Sentence

The boy managed to break the window with the ball
and so ran home.

27 "VP-Inducing" Co-text

A chef was setting up his food mixer. He had been asked to bake a cake and also a pie. He wanted to make one of them with dried fruit and the other with jam.

27 "NP-Inducing" Co-text

A chef was setting up his food mixer. He had been asked to bake two cakes. He wanted to make one of them with dried fruit and the other with jam.

27 NP-attached Target Sentence

The chef started to make the cake with the dried fruit
but soon ran out of eggs.

27 VP-attached Target Sentence

The chef started to make the cake with the food mixer
but soon ran out of eggs.

28 "VP-Inducing" Co-text

A gardener was sharpening his shears. He wanted to cut down a plant and a shrub. One of them had lots of green fly on it and the other had been dead a long time.

28 "NP-Inducing" Co-text

A gardener was sharpening his shears. He wanted to cut down two plants. One of them had lots of green fly on it and the other had been dead a long time.

28 NP-attached Target Sentence

The gardener cut down the plant with the green fly
but was very sad about it.

28 VP-attached Target Sentence

The gardener cut down the plant with the sharp shears
but was very sad about it.

29 "VP-Inducing" Co-text

A detective was sitting at a window with a set of binoculars. He was keeping a man and a woman under surveillance. One of them had a revolver and the other had a grenade.

29 "NP-Inducing" Co-text

A detective was sitting at a window with a set of binoculars. He was keeping two men under surveillance. One of them had a revolver and the other had a grenade.

29 NP-attached Target Sentence

The detective watched the man with the revolver
and then made a phone call.

29 VP-attached Target Sentence

The detective watched the man with the binoculars
and then made a phone call.

30 "VP-Inducing" Co-text

A man had bought a large brush to paint his room with. He was trying to decide which of a door or a window to paint first. One had a large crack in it while the other wouldn't close.

30 "NP-Inducing" Co-text

A man had bought a large brush to paint his room with. He was trying to decide which of two doors to paint first. One had a large crack in it while the other wouldn't close.

30 NP-attached Target Sentence

The man decided to paint the door with the large crack
and do the window later.

30 VP-attached Target Sentence

The man decided to paint the door with the large brush
and do the window later.

APPENDIX B

The following are the materials used in Experiment 2. Only the "2-instrument" co-texts are shown. The "1-instrument" co-texts can be derived by deleting the last noun phrase in the first line of each "2-instrument" co-text. For instance, the first material below starts "A mechanic walked up to a car with a monkey wrench and a pair of pliers.". The "1-instrument" version would be "A mechanic walked up to a car with a monkey wrench."

1 "VP-Inducing" Co-text

A mechanic walked up to a car with a monkey wrench and a pair of pliers. He thought he'd have to change a tyre. On examining the car he found that there was a tyre with a faulty valve and a fuel line with a small hole in it.

1 "NP-Inducing" Co-text

A mechanic walked up to a car with a monkey wrench and a pair of pliers. He thought he'd have to change a tyre. On examining the car he found that there was a tyre with a faulty valve and a tyre with a small hole in it.

1 NP-attached Target Sentence

The mechanic changed the tyre with the faulty valve
but it took a long time.

1 VP-attached Target Sentence

The mechanic changed the tyre with the monkey wrench
but it took a long time.

2 "VP-Inducing" Co-text

A caretaker was walking along a corridor carrying some nails and some glue. He'd been told to repair a door. He saw a door with a large crack in it and a locker which wouldn't shut properly.

2 "NP-Inducing" Co-text

A caretaker was walking along a corridor carrying some nails and some glue. He'd been told to repair a door. He saw a door with a large crack in it and a door which wouldn't shut properly.

2 NP-attached Target Sentence

The caretaker repaired the door with the crack
and then took a tea break.

2 VP-attached Target Sentence

The caretaker repaired the door with the nails
and then took a tea break.

3 "VP-Inducing" Co-text

A doctor walked into a waiting room carrying a stethoscope and rubber gloves. He'd arranged to examine a woman patient. In the waiting room was a woman with a temperature and a man who seemed to have stomach pains.

3 "NP-Inducing" Co-text

A doctor walked into a waiting room carrying a stethoscope and rubber gloves. He'd arranged to examine a woman patient. In the waiting room was a woman with a temperature and a woman who seemed to have stomach pains.

3 NP-attached Target Sentence

The doctor examined the woman with the temperature
but couldn't help her.

3 VP-attached Target Sentence

The doctor examined the woman with the stethoscope
but couldn't help her.

4 "VP-Inducing" Co-text

A woman was looking through some clothes while holding some darning wool and some thread. She needed to repair a sock. She found in the pile of clothes a sock with a hole in it and a jumper which was getting a bit thin.

4 "NP-Inducing" Co-text

A woman was looking through some clothes while holding some darning wool and some thread. She needed to repair a sock. She found in the pile of clothes a sock with a hole in it and a sock which was getting a bit thin.

4 NP-attached Target Sentence

The woman repaired the sock with the wool
but pricked her finger in the process.

4 VP-attached Target Sentence

The woman repaired the sock with the hole
but pricked her finger in the process.

5 "VP-Inducing" Co-text

A little boy was playing in a garden with a ball and a catapult. He was warned that he'd probably break a window. Close to him was a window with a crack in it and a statue that looked very old.

5 "NP-Inducing" Co-text

A little boy was playing in a garden with a ball and a catapult. He was warned that he'd probably break a window. Close to him was a window with a crack in it and a window that looked very old.

5 NP-attached Target Sentence

The boy managed to break the window with the crack
and so ran home.

5 VP-attached Target Sentence

The boy managed to break the window with the ball
and so ran home.

6 "VP-Inducing" Co-text

A chef was working in his kitchen with a food mixer and a whisk.
He was preparing to make a cake. Somebody came into the kitchen
and asked him to bake a cake with dried fruit and a pie with jam.

6 "NP-Inducing" Co-text

A chef was working in his kitchen with a food mixer and a whisk.
He was preparing to make a cake. Somebody came into the kitchen
and asked him to bake a cake with dried fruit and a cake with jam.

6 NP-attached Target Sentence

The chef started to make the cake with the dried fruit
but soon ran out of eggs.

6 VP-attached Target Sentence

The chef started to make the cake with the food mixer
but soon ran out of eggs.

7 "VP-Inducing" Co-text

A gardener was working in a garden with his shears and a small saw.
He decided to cut down a plant. In front of him was a plant with
lots of green fly and a shrub which had been dead a long time.

7 "NP-Inducing" Co-text

A gardener was working in a garden with his shears and a small saw.
He decided to cut down a plant. In front of him was a plant with
lots of green fly and a plant which had been dead a long time.

7 NP-attached Target Sentence

The gardener cut down the plant with the green fly
but was very sad about it.

7 VP-attached Target Sentence

The gardener cut down the plant with the sharp shears but was very sad about it.

8 "VP-Inducing" Co-text

A detective was sitting at a window with a set of binoculars and a telescope. He'd been asked to watch a woman. In the room he was looking into was a woman with a revolver and a man holding a grenade.

8 "NP-Inducing" Co-text

A detective was sitting at a window with a set of binoculars and a telescope. He'd been asked to watch a woman. In the room he was looking into was a woman with a revolver and a woman holding a grenade.

8 NP-attached Target Sentence

The detective watched the woman with the revolver and then made a phone call.

8 VP-attached Target Sentence

The detective watched the woman with the binoculars and then made a phone call.

9 "VP-Inducing" Co-text

A man was decorating a room and had with him a large brush and a roller. He'd decided to paint a door. In the room was a door with a large crack in it and a window which wouldn't close.

9 "NP-Inducing" Co-text

A man was decorating a room and had with him a large brush and a roller. He'd decided to paint a door. In the room was a door with a large crack in it and a door which wouldn't close.

9 NP-attached Target Sentence

The man decided to paint the door with the large crack and do the window later.

9 VP-attached Target Sentence

The man decided to paint the door with the large brush and do the window later.

10 "VP-Inducing" Co-text

A convict escaped from his cell clutching a knife and an iron bar. He was prepared to attach a warder if necessary. Once outside he saw a warder with a bunch of keys and a policeman with a long truncheon.

10 "NP-Inducing" Co-text

A convict escaped from his cell clutching a knife and an iron bar. He was prepared to attach a warder if necessary. Once outside he saw a warder with a bunch of keys and a warder with a long truncheon.

10 NP-attached Target Sentence

The convict suddenly lunged at the warder with the keys before running off.

10 VP-attached Target Sentence

The convict suddenly lunged at the warder with the knife before running off.

11 "VP-Inducing" Co-text

A schoolteacher walked into class carrying a bamboo cane and a walking stick. He sadistically hoped that he would have to beat a boy. On entering he saw that a boy with a broken leg and a girl with glasses didn't stand up for him.

11 "NP-Inducing" Co-text

A schoolteacher walked into class carrying a bamboo cane and a walking stick. He sadistically hoped that he would have to beat a boy. On entering he saw that a boy with a broken leg and a boy with glasses didn't stand up for him.

11 NP-attached Target Sentence

The schoolteacher beat the boy with the broken leg
but the boy didn't cry out.

11 VP-attached Target Sentence

The schoolteacher beat the boy with the bamboo cane
but the boy didn't cry out.

12 "VP-Inducing" Co-text

John had bought a new stylus and new headphones for his stereo. He planned to play a record. He'd been given a record with a lot of noisy crackle and a cassette with a lot of hiss.

12 "NP-Inducing" Co-text

John had bought a new stylus and new headphones for his stereo. He planned to play a record. He'd been given a record with a lot of noisy crackle and a record with a lot of hiss.

12 NP-attached Target Sentence

John decided to play the record with the noisy crackle
but it sounded terrible.

12 VP-attached Target Sentence

John decided to play the record with the new stylus
but it sounded terrible.

13 "VP-Inducing" Co-text

A historian was working in the British Museum with a magnifying glass and a microscope. He'd sat down to study a map. On his desk was a map with an appalling tear and a manuscript which seemed in perfect condition.

13 "NP-Inducing" Co-text

A historian was working in the British Museum with a magnifying glass and a microscope. He'd sat down to study a map. On his desk was a map with an appalling tear and a map which seemed in perfect condition.

13 NP-attached Target Sentence

The historian had to study the map with the appalling tear so as to value it.

13 VP-attached Target Sentence

The historian had to study the map with the magnifying glass so as to value it.

14 "VP-Inducing" Co-text

An artist was working at a table with a long brush and a palette. He had a sudden premonition that he was going to knock over a bottle. On the table was a bottle with some paint and a small jug filled with water.

14 "NP-Inducing" Co-text

An artist was working at a table with a long brush and a palette. He had a sudden premonition that he was going to knock over a bottle. On the table was a bottle with some paint and a bottle filled with water.

14 NP-attached Target Sentence

The artist accidently knocked over the bottle with the paint and swore.

14 VP-attached Target Sentence

The artist accidently knocked over the bottle with the brush and swore.

15 "VP-Inducing" Co-text

A window cleaner was climbing his ladder with a cloth and a sponge. He'd been asked to wipe a window. Once up the ladder he saw that there was a window covered in dirt and a skylight which was relatively clean.

15 "NP-Inducing" Co-text

A window cleaner was climbing his ladder with a cloth and a sponge. He'd been asked to wipe a window. Once up the ladder he saw that there was a window with a lot of dirt on it and a window which was relatively clean.

15 NP-attached Target Sentence

The window cleaner wiped the window with the dirt while singing to himself.

15 VP-attached Target Sentence

The window cleaner wiped the window with the cloth while singing to himself.

16 "VP-Inducing" Co-text

A burglar broke into a bank carrying some dynamite and some gelignite. He planned to blow open a safe. Once inside he saw that there was a safe with a new lock and a strong box with an old lock.

16 "NP-Inducing" Co-text

A burglar broke into a bank carrying some dynamite and some gelignite. He planned to blow open a safe. Once inside he saw that there was a safe with a new lock and a safe with an old lock.

16 NP-attached Target Sentence

The burglar blew open the safe with the new lock
and made off with the loot.

16 VP-attached Target Sentence

The burglar blew open the safe with the dynamite
and made off with the loot.

17 "VP-Inducing" Co-text

A workman climbed down a manhole carrying a blow torch and a hack-saw. He expected to have to cut through a valve. Down the manhole he found a valve with a lot of rust on it and a section of pipe which had been leaking gas.

17 "NP-Inducing" Co-text

A workman climbed down a manhole carrying a blow torch and a saw. He expected to have to cut through a valve. Down the manhole he found a valve with a lot of rust on it and a valve which had been leaking gas.

17 NP-attached Target Sentence

The workman cut through the valve with the rust
before fitting a new one.

17 VP-attached Target Sentence

The workman cut through the valve with the saw
before fitting a new one.

18 "VP-Inducing" Co-text

A man was walking up and down a dark street carrying an iron crowbar and a pickaxe. He hoped to break into a shop. He noticed that there was a shop with a broken window and a post office which was fitted with a burglar alarm.

18 "NP-Inducing" Co-text

A man was walking up and down a dark street carrying an iron crowbar and a pickaxe. He hoped to break into a shop. He noticed that there was a shop with a broken window and a shop which was fitted with a burglar alarm.

18 NP-attached Target Sentence

The man broke into the shop with the broken window
but nobody saw him.

18 VP-attached Target Sentence

The man broke into the shop with the iron crowbar
but nobody saw him.

19 "VP-Inducing" Co-text

A monkey had been trained to eat using a knife and a fork. It was supposed to eat a banana. It was given a banana with a bruise on it and an apple that was perfect.

19 "NP-Inducing" Co-text

A monkey had been trained to eat using a knife and a fork. It was supposed to eat a banana. It was given a banana with a bruise on it and a banana that was perfect.

19 NP-attached Target Sentence

The monkey ate the banana with the bruise
much to everyone's surprise.

19 VP-attached Target Sentence

The monkey ate the banana with the fork
much to everyone's surprise.

20 "VP-Inducing" Co-text

A skinhead was walking along a street carrying a large knife and a chain. He was planning to attack a policeman. He eventually saw a policeman who had a large scar on his face and a policewoman wearing spectacles.

20 "NP-Inducing" Co-text

A skinhead was walking along a street carrying a large knife and a chain. He was planning to attack a policeman. He eventually saw a policeman with a large scar on his face and a policeman wearing spectacles.

20 NP-attached Target Sentence

The skinhead suddenly attacked the policeman with the scar and then ran off.

20 VP-attached Target Sentence

The skinhead suddenly attacked the policeman with the knife and then ran off.

21 "VP-Inducing" Co-text

A dressmaker was working on a dress with a pair of scissors and a small knife. She had to cut some material for the pockets. She had a piece of material with a pattern on it and a piece of leather which was plain.

21 "NP-Inducing" Co-text

A dressmaker was working on a dress with a pair of scissors and a small knife. She had to cut some material for the pockets. She had a piece of material with a pattern on it and a piece of material which was plain.

21 NP-attached Target Sentence

The dressmaker cut the material with the pattern and then went to have lunch.

21 VP-attached Target Sentence

The dressmaker cut the material with the scissors and then went to have lunch.

22 "VP-Inducing" Co-text

A fireman was running to the scene of a fire carrying a heavy axe and a large hammer. He had to smash down a door. When he got to the scene of the fire he found a door with a very rusty lock and a window which was nailed shut.

22 "NP-Inducing" Co-text

A fireman was running to the scene of a fire carrying a heavy axe and a large hammer. He had to smash down a door. When he got to the scene of the fire he found a door with a very rusty lock and a door which was nailed shut.

22 NP-attached Target Sentence

The fireman smashed down the door with the rusty lock but smoke overcame him.

22 VP-attached Target Sentence

The fireman smashed down the door with the heavy axe but smoke overcame him.

23 "VP-Inducing" Co-text

A little girl borrowed from the kitchen a bread knife and a small vegetable knife. She wanted to cut an orange into pieces. She found in the fruit bowl an orange with a very thick rind and a tangerine with a thin rind.

23 "NP-Inducing" Co-text

A little girl borrowed from the kitchen a bread knife and a small vegetable knife. She wanted to cut an orange into pieces. She found in the fruit bowl an orange with a very thick rind and an orange with a thin rind.

23 NP-attached Target Sentence

The little girl tried to cut the orange with the thick rind but to no success.

23 VP-attached Target Sentence

The little girl tried to cut the orange with the bread knife but to no success.

24 "VP-Inducing" Co-text

A chemist was fiddling with a bunsen burner and a small gas ring. He was about to heat up a solution. On his bench was a solution containing blue crystals and a powder containing red dye.

24 "NP-Inducing" Co-text

A chemist was fiddling with a bunsen burner and a small gas ring. He was about to heat up a solution. On his bench was a solution with blue crystals in it and a solution containing red dye.

24 NP-attached Target Sentence

The chemist heated the solution with the blue crystals but the gas went out.

24 VP-attached Target Sentence

The chemist heated the solution with the bunsen burner but the gas went out.

25 "VP-Inducing" Co-text

A drunk was walking along a street wielding an empty bottle and a heavy stick. He felt like smashing a window. In front of him he saw a window with stained glass and a glass door with bars covering it.

25 "NP-Inducing" Co-text

A drunk was walking along a street wielding an empty bottle and a heavy stick. He felt like smashing a window. In front of him he saw a window with stained glass and a window with bars covering it.

25 NP-attached Target Sentence

The drunk hit the woman with the stammer
as she got off the train.

25 VP-attached Target Sentence

The drunk hit the woman with the stick
as she got off the train.

26 "VP-Inducing" Co-text

A vet went into an animal cage armed with a dart gun and an ordinary syringe. He had to tranquilise a lion. Inside the cage was a lion with a big mane and a puma that didn't have any mane at all.

26 "NP-Inducing" Co-text

A vet went into an animal cage armed with a dart gun and an ordinary syringe. He had to tranquilise a lion. Inside the cage was a lion with a big mane and a lion that didn't have any mane at all.

26 NP-attached Target Sentence

The vet tranquilised the lion with the big mane
and then examined it.

26 VP-attached Target Sentence

The vet tranquilised the lion with the dart gun
and then examined it.

27 "VP-Inducing" Co-text

A tribesman was running through the forrest carrying a long spear and a sharp knife. He intended to kill a lion. On arriving at a clearing he found a lion with very sharp teeth and a tiger with strange coloured paws.

27 "NP-Inducing" Co-text

A tribesman was running through the forrest carrying a long spear and a sharp knife. He intended to kill a lion. On arriving at a clearing he found a lion with very sharp teeth and a lion with strange coloured paws.

27 NP-attached Target Sentence

The tribesman killed the lion with the sharp teeth
and carried it back home.

27 VP-attached Target Sentence

The tribesman killed the lion with the long spear
and carried it back home.

28 "VP-Inducing" Co-text

A demolition man was on his way to work with a bulldozer and some explosives. He'd been told to go to a particular address to demolish a house. Once at the address he found a house with an ornate statue in its garden and a church covered in fine carvings.

28 "NP-Inducing" Co-text

A demolition man was on his way to work with a bulldozer and some explosives. He'd been told to go to a particular address to demolish a house. Once at the address he found a house with an ornate statue in its garden and a house covered in fine carvings.

28 NP-attached Target Sentence

The man demolished the house with the statue
but it wasn't the right one.

28 VP-attached Target Sentence

The man demolished the house with the bulldozer
but it wasn't the right one.

29 "VP-Inducing" Co-text

A furniture repairer was setting to work with some paint stripper and a blow torch. He'd been asked to strip a cabinet. In his workshop was a cabinet with brass hinges and a chest with aluminium hinges.

29 "NP-Inducing" Co-text

A furniture repairer was setting to work with some paint stripper and a blow torch. He'd been asked to strip a cabinet. In his workshop was a cabinet with brass hinges and a cabinet with aluminium hinges.

29 NP-attached Target Sentence

The repairer stripped the cabinet with the brass hinges and then waxed it.

29 VP-attached Target Sentence

The repairer stripped the cabinet with the paint stripper and then waxed it.

30 "VP-Inducing" Co-text

A secretary was sitting at her desk on which was a manual typewriter and a wordprocessor. She was told to type up a report. She was handed a report with diagrams all over it and a letter written in different colours.

30 "NP-Inducing" Co-text

A secretary was sitting at her desk on which was a manual typewriter and a wordprocessor. She was told to type up a report. She was handed a report covered with diagrams and a report written in different colours.

30 NP-attached Target Sentence

The secretary typed up the report with the diagrams and then went to lunch.

30 VP-attached Target Sentence

The secretary typed up the report with the typewriter and then went to lunch.

31 "VP-Inducing" Co-text

A burglar was trying to break into a house and had a credit card and a small penknife in his hand. He wanted to open a door. He found a door with a faulty lock and a window with a cracked frame.

31 "NP-Inducing" Co-text

A burglar was trying to break into a house and had a credit card and a small penknife in his hand. He wanted to open a door. He found a door with a faulty lock and a door with a cracked frame.

31 NP-attached Target Sentence

The burglar opened the door with the faulty lock and quickly slipped inside.

31 VP-attached Target Sentence

The burglar opened the door with the credit card and quickly slipped inside.

32 "VP-Inducing" Co-text

A cleaning lady started on her morning rounds carrying a brush and a vacuum cleaner. She would have to clean a corridor that morning. In the building there was a corridor with lots of plants along it and a lounge with lots of pictures on the walls.

32 "NP-Inducing" Co-text

A cleaning lady started on her morning rounds carrying a brush and a vacuum cleaner. She would have to clean a corridor that morning. In the building there was a corridor with lots of plants along it and a corridor with lots of pictures on the walls.

32 NP-attached Target Sentence

The cleaning lady cleaned the corridor with the plants and then lit a cigarette.

32 VP-attached Target Sentence

The cleaning lady cleaned the corridor with the brush and then lit a cigarette.

APPENDIX C

The following are the materials used in Experiment 5.

1 Relative-Inducing Co-text: Minimal Bridging

An electrician was explaining why a fuse had burnt out to two girls living in a flat together. The electrician was pleasantly surprised by how much one of them knew about electrical systems but was also disappointed by how little the other understood.

1 Complement-Inducing Co-text: Minimal Bridging

An electrician was explaining why a fuse had burnt out to a boy and a girl living in a flat together. The electrician was pleasantly surprised by how much one of them knew about electrical systems but was also disappointed by how little the other understood.

1 Complement-Inducing Co-text: Bridging

An electrician was explaining why a fuse had burnt out to a boy and a girl living in a flat together. The electrician couldn't believe how much one of them knew about electrical systems but was also disappointed by how little the other understood.

1 Relative-Inducing Co-text: Bridging

An electrician was explaining why a fuse had burnt out to two girls living in a flat together. The electrician couldn't believe how much one of them knew about electrical systems but was also disappointed by how little the other understood.

1 Complement Target Sentence

The electrician told the girl that he was surprised by
the amount she knew.

1 Relative Target Sentence

The electrician told the girl that he was surprised by
to help her flatmate.

2 Relative-Inducing Co-text: Minimal Bridging

An antique dealer was trying to buy a painting belonging to two elderly sisters. He was arguing over the price with one of them even though he'd already agreed a price with the other.

2 Complement-Inducing Co-text: Minimal Bridging

An antique dealer was trying to buy a painting belonging to an elderly brother and sister. He was arguing over the price with one of them even though he'd already agreed a price with the other.

2 Complement-Inducing Co-text: Bridging

An antique dealer was trying to buy a painting belonging to an elderly brother and sister. He couldn't come to any agreement over the price with one of them even though he'd already agreed a price with the other.

2 Relative-Inducing Co-text: Bridging

An antique dealer was trying to buy a painting belonging to two elderly sisters. He couldn't come to any agreement over the price with one of them even though he'd already agreed a price with the other.

2 Complement Target Sentence

The dealer told the woman that he was arguing with
them for no good reason.

2 Relative Target Sentence

The dealer told the woman that he was arguing with
to be reasonable.

3 Relative-Inducing Co-text: Minimal Bridging

A teacher was talking to the fathers of two children in his class. He was shocked that one of them lacked all interest in education but was cheered up by the other's positive attitude.

3 Complement-Inducing Co-text: Minimal Bridging

A teacher was talking to the parents of a child in his class. He was shocked that one of them lacked all interest in education but was cheered up by the other's positive attitude.

3 Complement-Inducing Co-text: Bridging

A teacher was talking to the parents of a child in his class. He found it hard to believe that one of them lacked all interest in education but was cheered up by the other's positive attitude.

3 Relative-Inducing Co-text: Bridging

A teacher was talking to the fathers of two children in his class. He found it hard to believe that one of them lacked all interest in education but was cheered up by the other's positive attitude.

3 Complement Target Sentence

The teacher told the man that he was shocked by
his negative attitude.

3 Relative Target Sentence

The teacher told the man that he was shocked by
to change his attitude.

4 Relative-Inducing Co-text: Minimal Bridging

A musician was talking to two guitarists about a group he wanted to form. He was very impressed by one of them but didn't think much of the other.

4 Complement-Inducing Co-text: Minimal Bridging

A musician was talking to a guitarist and a drummer about a group he wanted to form. He was very impressed by one of them but didn't think much of the other.

4 Complement-Inducing Co-text: Bridging

A musician was talking to a guitarist and a drummer about a group he wanted to form. He liked the music written by one of them but didn't think much of the other.

4 Relative-Inducing Co-text: Bridging

A musician was talking to two guitarists about a group he wanted to form. He liked the music written by one of them but didn't think much of the other.

4 Complement Target Sentence

The musician told the guitarist that he was impressed by his skilled playing.

4 Relative Target Sentence

The musician told the guitarist that he was impressed by to meet him again.

5 Relative-Inducing Co-text: Minimal Bridging

A detective suspected two women of having committed a murder. He had his doubts about the alibi given by one of them and had definite proof of the guilt of the other.

5 Complement-Inducing Co-text: Minimal Bridging

A detective suspected a man and a woman of having committed a murder. He had his doubts about the alibi given by one of them and had definite proof of the guilt of the other.

5 Complement-Inducing Co-text: Bridging

A detective suspected a man and a woman of having committed a murder. He was suspicious of the alibi given by one of them and had definite proof of the guilt of the other.

5 Relative-Inducing Co-text: Bridging

A detective suspected two women of having committed a murder. He was suspicious of the alibi given by one of them and had definite proof of the guilt of the other.

5 Complement Target Sentence

The detective told the woman that he had his doubts about her story.

5 Relative Target Sentence

The detective told the woman that he had his doubts about to confess.

6 Relative-Inducing Co-text: Minimal Bridging

A photographer had just finished working with two models. He was sure he'd taken bad pictures of one of them but was happy with those he'd taken of the other.

6 Complement-Inducing Co-text: Minimal Bridging

A photographer had just finished working with an athlete and a model. He was sure he'd taken bad pictures of one of them but was happy with those he'd taken of the other.

6 Complement-Inducing Co-text: Bridging

A photographer had just finished working with an athlete and a model. He was sure he'd over exposed the pictures of one of them but was happy with those he'd taken of the other.

6 Relative-Inducing Co-text: Bridging

A photographer had just finished working with two models. He was sure he'd over exposed the pictures of one of them but was happy with those he'd taken of the other.

6 Complement Target Sentence

The photographer told the model that he'd taken bad pictures of her that day.

6 Relative Target Sentence

The photographer told the model that he'd taken bad pictures of to pose again.

7 Relative-Inducing Co-text: Minimal Bridging

A fireman was telling two women how serious the situation had been when their house had caught fire. He had had to risk his life to rescue one of them while the other had stayed outside.

7 Complement-Inducing Co-text: Minimal Bridging

A fireman was telling a man and a woman how serious the situation had been when their house had caught fire. He had had to risk his life to rescue one of them while the other had stayed outside.

7 Complement-Inducing Co-text: Bridging

A fireman was telling a man and a woman how serious the situation had been when their house had caught fire. He had had to fight his way through the flames to rescue one of them while the other had stayed outside.

7 Relative-Inducing Co-text: Bridging

A fireman was telling two women how serious the situation had been when their house had caught fire. He had had to fight his way through the flames to rescue one of them while the other had stayed outside.

7 Complement Target Sentence

The fireman told the woman that he had risked his life for many people before.

7 Relative Target Sentence

The fireman told the woman that he had risked his life for to stop smoking.

8 Relative-Inducing Co-text: Minimal Bridging

A piano teacher had been giving lessons to two boys. She was impressed with the playing of one of them but thought the other quite ordinary.

8 Complement-Inducing Co-text: Minimal Bridging

A piano teacher had been giving lessons to a girl and a boy. She was impressed with the playing of one of them but thought the other quite ordinary.

8 Complement-Inducing Co-text: Bridging

A piano teacher had been giving lessons to a girl and a boy. She thought one of them very talented but the other quite ordinary.

8 Relative-Inducing Co-text: Bridging

A piano teacher had been giving lessons to two boys. She thought one of them very talented but the other quite ordinary.

8 Complement Target Sentence

The piano teacher told the boy that she was impressed with his piano playing.

8 Relative Target Sentence

The piano teacher told the boy that she was impressed with to practise a lot.

9 Relative-Inducing Co-text: Minimal Bridging

An auctioneer was talking to two men. He had just sold a picture for one of them but had never dealt with the other one before.

9 Complement-Inducing Co-text: Minimal Bridging

An auctioneer was talking to a man and a woman. He had just sold a picture for one of them but had never dealt with the other one before.

9 Complement-Inducing Co-text: Bridging

An auctioneer was talking to a man and a woman. He had just auctioned a painting for one of them but had never dealt with the other one before.

9 Relative-Inducing Co-text: Bridging

An auctioneer was talking to two men. He had just auctioned a painting for one of them but had never dealt with the other one before.

9 Complement Target Sentence

The auctioneer told the man that he had sold a picture for two hundred pounds.

9 Relative Target Sentence

The auctioneer told the man that he had sold a picture for to collect his money.

10 Relative-Inducing Co-text: Minimal Bridging

A lawyer was asked to defend two men. He was going to go into court with only one of them but not the other.

10 Complement-Inducing Co-text: Minimal Bridging

A lawyer was asked to defend a woman and a man. He was going to go into court with only one of them but not the other.

10 Complement-Inducing Co-text: Bridging

A lawyer was asked to defend a woman and a man. He was going to represent only one of them but not the other.

10 Relative-Inducing Co-text: Bridging

A lawyer was asked to defend two men. He was going to represent only one of them but not the other.

10 Complement Target Sentence

The lawyer told the man that he was going into court with him for the trial.

10 Relative Target Sentence

The lawyer told the man that he was going into court with to plead innocent.

11 Relative-Inducing Co-text: Minimal Bridging

A social worker visited the two fathers of some children in her care. She was horrified by how aggressive one of them was towards children even though the other one seemed very kind.

11 Complement-Inducing Co-text: Minimal Bridging

A social worker visited the mother and father of some children in her care. She was horrified by how aggressive one of them was towards children even though the other one seemed very kind.

11 Complement-Inducing Co-text: Bridging

A social worker visited the mother and father of some children in her care. She just couldn't believe how aggressive one of them was towards children even though the other one seemed very kind.

11 Relative-Inducing Co-text: Bridging

A social worker visited the two fathers of some children in her care. She just couldn't believe how aggressive one of them was towards children even though the other one seemed very kind.

11 Complement Target Sentence

The social worker told the father that she was horrified by his aggressiveness.

11 Relative Target Sentence

The social worker told the father that she was horrified by to change his ways.

12 Relative-Inducing Co-text: Minimal Bridging

A policeman was talking to two women claiming to be CIA agents. He had his doubts about one of them but did not suspect the other.

12 Complement-Inducing Co-text: Minimal Bridging

A policeman was talking to a man and a woman claiming to be CIA agents. He had his doubts about one of them but did not suspect the other.

12 Complement-Inducing Co-text: Bridging

A policeman was talking to a man and a woman claiming to be CIA agents. He suspected that one of them was lying but did not suspect the other.

12 Relative-Inducing Co-text: Bridging

A policeman was talking to two women claiming to be CIA agents. He suspected that one of them was lying but did not suspect the other.

12 Complement Target Sentence

The policeman told the woman that he had his doubts about her true identity.

12 Relative Target Sentence

The policeman told the woman that he had his doubts about to give him proof.

13 Relative-Inducing Co-text: Minimal Bridging

A doctor was treating two sisters. He could find no cure for one of them even though he could easily cure the other.

13 Complement-Inducing Co-text: Minimal Bridging

A doctor was treating a brother and sister. He could find no cure for one of them even though he could easily cure the other.

13 Complement-Inducing Co-text: Bridging

A doctor was treating a brother and sister. He diagnosed one of them as having an incurable disease though he could easily cure the other.

13 Relative-Inducing Co-text: Bridging

A doctor was treating two sisters. He diagnosed one of them as having an incurable disease though he could easily cure the other.

13 Complement Target Sentence

The doctor told the girl that he could find no cure for her at all.

13 Relative Target Sentence

The doctor told the girl that he could find no cure for to be brave.

14 Relative-Inducing Co-text: Minimal Bridging

A girl was drinking alone at a bar when two blokes came up and started talking to her. She'd had a big row with one of them but had always got on well with the other.

14 Complement-Inducing Co-text: Minimal Bridging

A girl was drinking alone at a bar when a bloke and his girlfriend came up and started talking to her. She'd had a big row with one of them but had always got on well with the other.

14 Complement-Inducing Co-text: Bridging

A girl was drinking alone at a bar when a bloke and his girlfriend came up and started talking to her. She'd had a bitter argument with one of them but had always got on well with the other.

14 Relative-Inducing Co-text: Bridging

A girl was drinking alone at a bar when two blokes came up and started talking to her. She'd had a bitter argument with one of them but had always got on well with the other.

14 Complement Target Sentence

The girl told the man that she'd had a row with him once too often.

14 Relative Target Sentence

The girl told the man that she'd had a row with to leave her alone.

15 Relative-Inducing Co-text: Minimal Bridging

A couple were visiting two young boys in hospital. They had brought a present for one of them but they hadn't brought anything for the other, less ill, child.

15 Complement-Inducing Co-text: Minimal Bridging

A couple were visiting a young girl and boy in hospital. They had brought a present for one of them but they hadn't brought anything for the other, less ill, child.

15 Complement-Inducing Co-text: Bridging

A couple were visiting a young girl and boy in hospital. They had bought a toy for one of them but they hadn't brought anything for the other, less ill, child.

15 Relative-Inducing Co-text: Bridging u A couple were visiting two young boys in hospital. They had bought a toy for one of them but they hadn't brought anything for the other, less ill, child.

15 Complement Target Sentence

The couple told the boy that they had brought a present for them both.

15 Relative Target Sentence

The couple told the boy that they had brought a present for to share it.

16 Relative-Inducing Co-text: Minimal Bridging

A teacher was talking to two boys about their exam results. She was worried about one of them for having done badly although she was pleased with the other one's results.

16 Complement-Inducing Co-text: Minimal Bridging

A teacher was talking to a girl and a boy about their exam results. She was worried about one of them for having done badly although she was pleased with the other one's results.

16 Complement-Inducing Co-text: Bridging

A teacher was talking to a girl and a boy about their exam results. She was troubled by how badly one of them had done although she was pleased with the other one's results.

16 Relative-Inducing Co-text: Bridging

A teacher was talking to two boys about their exam results. She was troubled by how badly one of them had done although she was pleased with the other one's results.

16 Complement Target Sentence

16 Complement Target Sentence

The teacher told the boy that she was worried about his poor performance.

16 Relative Target Sentence

The teacher told the boy that she was worried about to work even harder.

17 Relative-Inducing Co-text: Minimal Bridging

A lecturer was giving a tutorial to two sisters. He was impressed with one of them though he thought the other pretty stupid.

17 Complement-Inducing Co-text: Minimal Bridging

A lecturer was giving a tutorial to a brother and sister. He was impressed with one of them though he thought the other pretty stupid.

17 Complement-Inducing Co-text: Bridging

A lecturer was giving a tutorial to a brother and sister. He found one of them to be very intelligent though he thought the other pretty stupid.

17 Relative-Inducing Co-text: Bridging

A lecturer was giving a tutorial to two sisters. He found one of them to be very intelligent though he thought the other pretty stupid.

17 Complement Target Sentence

The lecturer told the girl that he was impressed with her good work.

17 Relative Target Sentence

The lecturer told the girl that he was impressed with to keep it up.

18 Relative-Inducing Co-text: Minimal Bridging

A company executive was about to interview two women for a job. He had received a letter of application from one of them but the other had simply phoned.

18 Complement-Inducing Co-text: Minimal Bridging A company executive was about to interview a man and a woman for a job. He had received a letter of application from one of them but the other had simply phoned.

18 Complement-Inducing Co-text: Bridging

A company executive was about to interview a man and a woman for a job. He had received a postal application from one of them but the other had simply phoned.

18 Relative-Inducing Co-text: Bridging

A company executive was about to interview two women. He had received a postal application from one of them but the other had simply phoned.

18 Complement Target Sentence

The executive told the woman that he had received a letter from her that day.

18 Relative Target Sentence

The executive told the woman that he had received a letter from to sit down.

19 Relative-Inducing Co-text: Minimal Bridging

A doctor was talking to two men. He had made out a prescription for one of them but not for the other.

19 Complement-Inducing Co-text: Minimal Bridging

A doctor was talking to a man and a woman. He had made out a prescription for one of them but not for the other.

19 Complement-Inducing Co-text: Bridging

A doctor was talking to a man and a woman. He had prescribed some pills for one of them but not for the other.

19 Relative-Inducing Co-text: Bridging

A doctor was talking to two men. He had prescribed some pills for one of them but not for the other.

19 Complement Target Sentence

The doctor told the man that he had made out a prescription for some pills.

19 Relative Target Sentence

The doctor told the man that he had made out a prescription for to go to bed.

20 Relative-Inducing Co-text: Minimal Bridging

A baker was talking to two women. He had baked a cake for one of them and scones for the other.

20 Complement-Inducing Co-text: Minimal Bridging

A baker was talking to an woman and a man. He had baked a cake for one of them and scones for the other.

20 Complement-Inducing Co-text: Bridging

A baker was talking to an woman and a man. He had made a gateaux for one of them and scones for the other.

20 Relative-Inducing Co-text: Bridging

A baker was talking to two women. He had made a gateaux for one of them and scones for the other.

20 Complement Target Sentence

The baker told the woman that he had baked a cake for himself as well.

20 Relative Target Sentence

The baker told the woman that he had baked a cake for to pay him later.

21 Relative-Inducing Co-text: Minimal Bridging

A politician was talking to two boys. He disagreed with the views of one of them but thought the other to be very sensible.

21 Complement-Inducing Co-text: Minimal Bridging

A politician was talking to a boy and a girl. He disagreed with the views of one of them but thought the other to be very sensible.

21 Complement-Inducing Co-text: Bridging

A politician was talking to a boy and a girl. He was arguing with one of them but thought the other to be very sensible.

21 Relative-Inducing Co-text: Bridging

A politician was talking to two boys. He was arguing with one of them but thought the other to be very sensible.

21 Complement Target Sentence

The politician told the boy that he disagreed with
him over his ideals.

21 Relative Target Sentence

The politician told the boy that he disagreed with
to try and see sense.

22 Relative-Inducing Co-text: Minimal Bridging

A film director was at a party talking to two actors. He had heard of one of them before, but knew nothing about the other.

22 Complement-Inducing Co-text: Minimal Bridging

A film director was at a party talking to an actor and an actress. He had heard of one of them before, but knew nothing about the other.

22 Complement-Inducing Co-text: Bridging

A film director was at a party talking to an actor and an actress. He'd been told about one of them before, but knew nothing about the other.

22 Relative-Inducing Co-text: Bridging

A film director was at a party talking to two actors. He'd been told about one of them before, but knew nothing about the other.

22 Complement Target Sentence

The director told the actor that he had heard of him from someone else.

22 Relative Target Sentence

The director told the actor that he had heard of to audition for his film.

23 Relative-Inducing Co-text: Minimal Bridging

A hospital patient was having a discussion with two nurses. He was complaining about the way he had been treated by one of them and hoped the other would do something about it.

23 Complement-Inducing Co-text: Minimal Bridging

A hospital patient was having a discussion with a doctor and a nurse. He was complaining about the way he had been treated by one of them and hoped the other would do something about it.

23 Complement-Inducing Co-text: Bridging

A hospital patient was having a discussion with a doctor and a nurse. He was angry at the way he had been treated by one of them and hoped the other would do something about it.

23 Relative-Inducing Co-text: Bridging

A hospital patient was having a discussion with two nurses. He was angry at the way he had been treated by one of them and hoped the other would do something about it.

23 Complement Target Sentence

The patient told the nurse that he was complaining about her treatment of him.

23 Relative Target Sentence

The patient told the nurse that he was complaining about to treat him better.

24 Relative-Inducing Co-text: Minimal Bridging

A burglar was in a pub talking to two men. He'd been in prison with one of them but only vaguely knew the other.

24 Complement-Inducing Co-text: Minimal Bridging

A burglar was in a pub talking to a man and a woman. He'd been in prison with one of them but only vaguely knew the other.

24 Complement-Inducing Co-text: Bridging

A burglar was in a pub talking to a man and a woman. He'd served a sentence with one of them but only vaguely knew the other.

24 Relative-Inducing Co-text: Bridging

A burglar was in a pub talking to two men. He'd served a sentence with one of them but only vaguely knew the other.

24 Complement Target Sentence

The burglar told the man that he'd been in prison with some nice people.

24 Relative Target Sentence

The burglar told the man that he'd been in prison with to keep in touch.

25 Relative-Inducing Co-text: Minimal Bridging

A ski instructor was talking to two girls. He was very happy with the ability of one of them but he was worried about the other.

25 Complement-Inducing Co-text: Minimal Bridging

A ski instructor was talking to a girl and a boy. He was very happy with the ability of one of them but he was worried about the other.

25 Complement-Inducing Co-text: Bridging

A ski instructor was talking to a girl and a boy. He was very pleased with the ability one of them showed but he was worried about the other.

25 Relative-Inducing Co-text: Bridging

A ski instructor was talking to two girls. He was very pleased with the ability one of them showed but he was worried about the other.

25 Complement Target Sentence

The instructor told the girl that he was happy with her overall skiing ability.

25 Relative Target Sentence

The instructor told the girl that he was happy with to try something harder.

26 Relative-Inducing Co-text: Minimal Bridging

A duchess was at a garden party talking to two men. She was terribly bored by one of them though she quite liked the other.

26 Complement-Inducing Co-text: Minimal Bridging

A duchess was at a garden party talking to a man and a woman. She was terribly bored by one of them though she quite liked the other.

26 Complement-Inducing Co-text: Bridging

A duchess was at a garden party talking to a man and a woman. She found one of them to be terribly dull though she quite liked the other.

26 Relative-Inducing Co-text: Bridging

A duchess was at a garden party talking to two men. She found one of them to be terribly dull though she quite liked the other.

26 Complement Target Sentence

The duchess told the man that she was bored by his dull conversation.

26 Relative Target Sentence

The duchess told the man that she was bored by to go somewhere else.

27 Relative-Inducing Co-text: Minimal Bridging

A dentist was at a sherry party talking to two women. He'd been at school with one of them but had never met the other one before.

27 Complement-Inducing Co-text: Minimal Bridging

A dentist was at a sherry party talking to a man and a woman. He'd been at school with one of them but had never met the other one before.

27 Complement-Inducing Co-text: Bridging

A dentist was at a sherry party talking to a man and a woman. He had been classmates with one of them but had never met the other one before.

27 Relative-Inducing Co-text: Bridging

A dentist was at a sherry party talking to two women. He had been classmates with one of them but had never met the other one before.

27 Complement Target Sentence

The dentist told the woman that he had been at school with the other for years.

27 Relative Target Sentence

The dentist told the woman that he had been at school with to ring him sometime.

28 Relative-Inducing Co-text: Minimal Bridging

A little boy was being bullied by two older girls. The little boy was afraid of one of them but wasn't too worried about the other.

28 Complement-Inducing Co-text: Minimal Bridging

A little boy was being bullied by an older boy and his sister. The little boy was afraid of one of them but wasn't too worried about the other.

28 Complement-Inducing Co-text: Bridging

A little boy was being bullied by an older boy and his sister. The little boy was worried by the threats one of them was making but wasn't too worried about the other.

28 Relative-Inducing Co-text: Bridging

A little boy was being bullied by two older girls. The little boy was worried by the threats one of them was making but wasn't too worried about the other.

28 Complement Target Sentence

The little boy told the girl that he was afraid of the two of them.

28 Relative Target Sentence

The little boy told the girl that he was afraid of to leave him alone.

29 Relative-Inducing Co-text: Minimal Bridging

A woman was talking to two brothers. She was annoyed with one of them but not with the other.

29 Complement-Inducing Co-text: Minimal Bridging

A woman was talking to a girl and her brother. She was annoyed with one of them but not with the other.

29 Complement-Inducing Co-text: Bridging

A woman was talking to a girl and her brother. She had got cross with one of them but not with the other.

29 Relative-Inducing Co-text: Bridging

A woman was talking to two brothers. She had got cross with one of them but not with the other.

29 Complement Target Sentence

The woman told the boy that she was annoyed with
his behaviour.

29 Relative Target Sentence

The woman told the boy that she was annoyed with
to go upstairs.

30 Relative-Inducing Co-text: Minimal Bridging

An art teacher told two girls about an exhibition at a local museum. He wanted to go with one of them but not the other.

30 Complement-Inducing Co-text: Minimal Bridging

An art teacher told a girl and a boy about an exhibition at a local museum. He wanted to go with one of them but not the other.

30 Complement-Inducing Co-text: Bridging

An art teacher told a girl and a boy about an exhibition at a local museum. He asked one of them to go with him but not the other.

30 Relative-Inducing Co-text: Bridging

An art teacher told two girls about an exhibition at a local museum. He asked one of them to go with him but not the other.

30 Complement Target Sentence

The art teacher told the girl that he wanted to go with her to the exhibition.

30 Relative Target Sentence

The art teacher told the girl that he wanted to go with to meet him later.

31 Relative-Inducing Co-text: Minimal Bridging

A psychologist was counselling two wives. He was worried about one of the pair but wasn't concerned about the other.

31 Complement-Inducing Co-text: Minimal Bridging

A psychologist was counselling a man and his wife. He was worried about one of the pair but wasn't concerned about the other.

31 Complement-Inducing Co-text: Bridging

A psychologist was counselling a man and his wife. He was particularly concerned with the problems of one of the pair but wasn't concerned about the other.

31 Relative-Inducing Co-text: Bridging

A psychologist was counselling two wives. He was particularly concerned with the problems of one of the pair but wasn't concerned about the other.

31 Complement Target Sentence

The psychologist told the woman that he was worried about her marital problems.

31 Relative Target Sentence

The psychologist told the woman that he was worried about to see him again.

32 Relative-Inducing Co-text: Minimal Bridging

A girl was talking to two of her teachers. She had been having trouble with one of them for a long time whilst the other had always been good to her.

32 Complement-Inducing Co-text: Minimal Bridging

A girl was talking to her mother and her teacher. She had been having trouble with one of them for a long time whilst the other had always been good to her.

32 Complement-Inducing Co-text: Bridging

A girl was talking to her mother and her teacher. She was complaining because one of them had been illtreating her for a long time whilst the other had always been good to her.

32 Relative-Inducing Co-text: Bridging

A girl was talking to two of her teachers. She was complaining because one of them had been illtreating her for a long time whilst the other had always been good to her.

32 Complement Target Sentence

The girl persuaded the teacher that she was having trouble with her home life.

32 Relative Target Sentence

The girl persuaded the teacher that she was having trouble with to see reason.

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